POLITECNICO DI MILANO



FEATURE-BASED CHARACTERIZATION OF LOUDSPEAKERS

DIPARTIMENTO DI ELETTRONICA, INFORMAZIONE E BIOINGEGNERIA

MSc. TELECOMMUNICATION ENGINEERING
SOUND ENGINEERING AND MUSICAL ACOUSTICS

Supervisor: Prof. Giuseppe Bertuccio
External Supervisor: Ing. Lucio Bianchi

Author:
Alessio Sopranzi
Student number: 898531

A.Y. 2019-2020

Abstract

In this thesis we present a methodology relying on a feature-based description of loudspeakers to find relationships between their objective characterization and sound quality reproduction assessment. Listening tests are a well-known but costly solutions to get reliable results in the latter sense. Measurements offer instead a quick and significant portrait of transducers especially if combined with common speaker parameters. We investigated the problem of discovering well-generalizable links between the two fields and predicting sensory data. Experimental activities included the design and conduction of formal listening tests for single driver monophonic reproduction. Proposed metrics related to linear and non-linear lumped parameters and a low-level characterization of harmonic distortion and on-axis frequency curves in terms of statistical spectral descriptors. An explicit regression methodology was tested. A merging procedure of near-field and far-field responses of an extended dataset was performed to gain a further insight about the problem. Finally, a correlation analysis of subjective ratings with common drivers' linear electro-mechanical parameters was carried out. Predictive modelling attempt was inadequate due to the restricted specimen of speakers and listening test results employed. Partial conclusions are due to the lack of possibility of gathering further data too. A comparison with an existed validated model allowed further insights about drivers' equalization and preference ranking. Inferences about the role of electro-mechanical parameters in perceived sound quality need to be counter-validated with supplementary work. Our methodology is yet not enough informative to a full replacement of a subjective assessment; the expectation is providing a new strong experimental background in the field as regards experimental choices and the optimal number of degree of freedoms for such a complex task.

Estratto in lingua italiana

In questa tesi proponiamo una metodologia basata su una descrizione tramite feature di altoparlanti per cercare relazioni tra una loro caratterizzazione oggettiva e una valutazione della qualità del suono riprodotto. I test d'ascolto costituiscono una soluzione ben nota ma costosa per ottenere risultati affidabili per il secondo scopo. Le misure offrono invece una rappresentazione valida e veloce di questi trasduttori soprattutto se unite a tipici parametri degli altoparlanti. Abbiamo investigato il problema di trovare un collegamento tra i due mondi e predire dati sensoriali. Le attività sperimentali hanno riguardato il design e la conduzione di test formali in uno scenario di riproduzione monofonica con singoli driver. Le metriche proposte riguardano parametri concentrati lineari e non lineari e una caratterizzazione a basso livello di curve di distorsione armonica e risposta in frequenza in asse in termini di descrittori spettrali statistici. È stato fatto un tentativo di costruire un modello regressivo. Inoltre, misure in campo vicino e in campo lontano sono state unite per ottenere un dataset più esteso di altoparlanti per speculare una ulteriore percezione del problema. In ultimo è stata cercata una correlazione dei voti soggettivi con i classici parametri elettromeccanici dei trasduttori. Il tentativo di modellazione predittiva è stato inadeguato a causa del campione limitato di dispositivi e risultati di test d'ascolto impiegati. Le parziali conclusioni sono dovute anche all'impossibilità di collezionare altri dati. Un confronto con un modello validato esistente ha permesso ulteriori approfondimenti sull'equalizzazione degli altoparlanti e una gerarchia di preferenze. I risultati riguardo il ruolo dei parametri elettromeccanici nella qualità del suono percepito devono essere contro-validati con un lavoro supplementare. La nostra metodologia non è ancora abbastanza informativa per sostituire completamente una valutazione soggettiva; ci aspettiamo di fornire un background sperimentale solido sul campo per quanto riguarda le scelte sperimentali e il numero ottimale di gradi di libertà per un problema così complesso.

Contents

Abstract iii

Estratto in lingua italiana v

List of figures xi

List of tables xv

Acknowledgments xix

1. Introduction 1

Outline 4

2. Background and State-of-the-art 5

2.1 Electrodynamic loudspeakers 6

2.1.1 Basic historical perspective and main elements 6

2.1.2 Basic principle of Electro-Mechanical transduction 8

2.2 Loudspeaker modelling 8

2.2.1 Signal-based modelling 11

2.2.2 Lumped parameter characterization 18

2.3 Evaluation of sound quality in speaker reproduction 26

2.3.1 Frequency response and sound quality perception 26

2.3.2 Distortion and sound quality perception 28

2.3.3 Timbre and sound quality perception 30

2.4 Correlation between subjective and objective data 31

2.5 Conclusions and remarks 36

3. Methodology and Implementation 39

3.1 Criteria for listening test design 40

3.1.1 Rationale behind the choice of test method 42

3.1.2 Rationale behind the choice of independent variables 45

3.1.3 The design of the experiments (DoE) 50

3.2 Objective Measurements and Features 52

3.2.1 Spectral descriptors analysis 56

3.3 Merging Far field and Near field data 61

3.3.1 Near-field and Far-field measurements 61

3.3.2 The merging procedure 65

3.4 Setting up experiments 68

3.5 Criteria for relating sensory and objective data 74

3.5.1 Statistical validation method of listening test data 74

3.5.2 Regression and Correlation analysis 75

3.6 Conclusions and remarks 81

4. Evaluation 83

4.1 Experimental conditions 83

4.1.1 Test environment and logistics 84

4.1.2 Hardware setup and playback settings 85

4.2 Objective Data results 88

4.3 Subjective Data Results 97

4.4 Predictive model and correlation analysis Results 105

4.5 Conclusions and remarks 117

5. Conclusions and Future works 123

Future work 125

Appendix A Documentation of the programme material 129

Appendix B Technical Specifications 133

B.1 Drivers 133

B.2 Microphones 137

B.3 Power Amplifiers 138

B.4 A/D and D/A Converter 139

Appendix C Feature Extraction results 143

Appendix D Electromotive force 147

Bibliography 149

List of figures

Figure 2.1 – C.W.Rice and E.W.Kellogg. Loudspeaker. US Patent1,707,570. Apr. 1929. Picture from [118] 6

Figure 2.2 – Schematic example of a woofer diagram. Picture from [3] 7

Figure 2.3 – Loudspeaker "Grey model". Picture adapted from [6] 9

Figure 2.4 – Loudspeaker's cone pistonic and non-pistonic range - left; piston Mode below 1kHz right; First Bending Resonance at a frequency ~1.4kHz. Picture from [6] 10

Figure 2.5 – Example of a measured full-frequency response. Picture from [34] 12

Figure 2.6 – Sound pressure spectrum generated by a two-tone stimulus. A high distance between the fundamental components $f2 \gg f1$, nicely separates harmonic and intermodulation components. Picture from [17] 14

Figure 2.7 – Static and dynamic nonlinearities modelling in transducers. Left; static memory-less polynomial nonlinear system. Right; convolution Volterra integrals of different orders replacing polynomials. Picture from [12] 15

Figure 2.8 – Simplified low-frequency Thiele-Small circuit for drivers (impedance analogy), free air, infinite baffle 20

Figure 2.9 – Electro-mechanical equivalent circuit of the loudspeaker. Picture from [29]. 22

Figure 2.10 – Lumped linear loudspeaker parameter model (electromechanics behavior). Picture from [10]. 22

Figure 2.11 – Magnitude of electrical impedance versus frequency f. Picture from [26]. 22

Figure 2.12 – Nonlinearities shapes; Force factor Bl(x), stiffness $Kmsx$ and voice-coil inductance Lex versus displacement for a 6-inch well-designed woofer intended for high quality consumer application. Pictures from [17]. 23

Figure 2.13 – Electrical equivalent circuit considering motor and suspension nonlinearities. Picture from [30]. 24

Figure 3.1 – A revision of the filter model describing all the psychological, and inner psychomotor processes undertaken by subjects during a test. Picture from [43]..... 41
Figure 3.2 – Sound wheel of reliable, precise non-overlapping attributes for reproduced sound. Inner ring: Main groups, middle ring: Categories, outer ring: Attributes. Picture from [68]..... 46
Figure 3.3 – Construction rules for n x n Latin Square (left) and Balanced Latin Square (right) designs..... 52
Figure 3.4 – On-axis pressure amplitude and harmonic distortion. Picture from [83]..... 55
Figure 3.5 - Loudspeaker, measurement microphone, and nearest reflecting surface define an ellipsoid within which free field measurements can be performed using time-selective techniques. Picture from [74]..... 63
Figure 3.6 - Example of a windowed impulse response of a woofer. Picture from Klippel dB-Lab 210.610..... 64
Figure 3.7 - Graphic illustration near field and far field measurements combining. Top; Near field and far field measurement before magnitude compensation. Bottom; overlap region after rescaling. Pictures from [74]..... 65
Figure 3.8 - Equal-loudness contours of the ISO 226 standard (solid curve) and A-, B-, C-weighting curves superimposed (dotted lines). Picture from [55]..... 70
Figure 3.9 - Quantization effect in distribution of data due to horizontal lines crossing sliders. Pictures from [56]..... 72
Figure 3.10 - 9-point hedonic scale for preference rating. Picture from [55]..... 72
Figure 3.11 - Example of a rating (timbral balance case) with the GUI used by listeners; .. 73
Figure 3.12 - Three-way structure of data (left), three-way sensory data to be related to a regular two-way objective data matrix (right). The vertical dimension must be the same. Picture from [98]..... 77

Figure 3.13 - Relating X to Y data for the purpose of prediction and improved understanding. Picture from [98] 77
Figure 4.1 - Picture from listening room: a subject's performing the test with blinded drivers 85
Figure 4.2 - Block diagram of the stimulus playback architecture for listening tests. Exemplificative signal flow in case of a single speaker. Picture made with free-source software INKSCAPE 0.92 [119] 86
Figure 4.3 - Beyerdynamic MM1 microphone mounted at one end of the rotor (left), Drivers mounted in the sealed boxes without black curtains (center), Control location to manage experiments and hardware rack (left) 88
Figure 4.4 – Woofers All SPL on-axis. Picture exported from REW. 89
Figure 4.5 – Tweeters All SPL on-axis. Picture exported from REW. 89
Figure 4.6 – Hertz C165 Cento Harmonic distortion plot. Picture exported from REW. 90
Figure 4.7 – Audison TH 6.5 II Sax Harmonic distortion plot. Picture exported from REW. 90
Figure 4.8 – Hertz ESK EV 165.5 Harmonic distortion plot. Picture exported from REW. . 91
Figure 4.9 – Hertz MP 165.3 Mille Pro Harmonic distortion plot. Picture exported from REW. 91
Figure 4.10 – Hertz C26 Harmonic distortion plot. Picture exported from REW. 92
Figure 4.11 – TH 1.5 II Violino Harmonic distortion plot. Picture exported from REW. 92
Figure 4.12 – ET 26.5 Harmonic distortion plot. Picture exported from REW. 93
Figure 4.13 – Hertz MP 25.3 Mille Pro Harmonic distortion plot. Picture exported from REW. 93
Figure 4.14 – Example (B035) of input curves before merging. Picture exported from Klippel dB-Lab Analyzer 95

Figure 4.15 – Example (B035) of input curves after merging. Picture exported from Klippel dB-Lab Analyzer. 95
Figure 4.16 – Mean loudness ratings (woofers), 95% confidence interval 97
Figure 4.17 – Mean Timbral Balance ratings (woofers), 95% confidence interval 98
Figure 4.18 – Mean Preference ratings (woofers), 95% confidence interval 98
Figure 4.19 – Mean Loudness ratings (tweeters), 95% confidence interval 99
Figure 4.20 – Mean Timbral Balance ratings (tweeters), 95% confidence interval 99
Figure 4.21 – Mean Preference ratings (tweeters), 95% confidence interval 100
Figure 4.22 – Boxplots for Woofers ratings 101
Figure 4.23 – Boxplots for Tweeter ratings 102
Figure 4.24 – Woofer ratings one-way ANOVA; Loudness (top), Timbral Balance (center), Preference (bottom) 103
Figure 4.25 – Woofer ratings one-way ANOVA; Loudness (top), Timbral Balance (center), Preference (bottom) 103
Figure 4.26 – RMSE metric on testing 106
Figure 4.27 – R2 metric on testing 106
Figure 4.28 – Adjusted R2 metric on testing 107
Figure 4.29 – Woofers, Loudness Score plot 108
Figure 4.30 – Woofers, Loudness Loadings plot 109
Figure 4.31 – Woofers' measured SPL on-axis and average response (green) 112

List of tables

Table 3.1 Tracks involved in the test; samples came from easily accessible sources and represent common choices. 49
Table 4.1 – Merging Results 96
Table 4.2 – Correlation of single listeners with group mean scoring 104
Table 4.3 - Example of Olive's features value for woofers involved in the listening sessions 110
Table 4.4 – Heuristic algorithm results 111
Table 4.5 – Loudspeaker predicted Preference ranking Case 1 and Case 2 113
Table 4.6 – Loudspeaker preference ranking; predicted vs listening test results 113
Table 4.7 – Min and Max Preference Range and feature values for Case 2 114
Table 4.8 – Min and Max values for lumped parameters in the experiment for both 35 and 4 woofer dataset 115
Table 4.9 – Electro-mechanical parameters and attribute ratings correlation 115
Table B.0.1 - Experimental setup: woofers' specifications 133
Table B.0.2 - Experimental setup: tweeters' specifications 135
Table B.0.3 - Experimental setup: measurement room microphone's specifications 137
Table B.0.4 - Experimental setup: listening room microphone's specifications 137
Table B.0.5 - Experimental setup: measurement room power amplifier 138
Table B.0.6 - Experimental setup: listening room power amplifier 138
Table B.0.7 - Experimental setup: 30 Inputs/30 Outputs channels, 4 mic/instrument Preamps, 24-Bit/192 kHz resolution Digital Audio USB interface 139
Table C.0.1 – Woofer Feature Extraction results 143
Table C.0.2 – Tweeters' Feature Extraction results 145

Acknowledgments

I would like to thank Elettromedia s.r.l to have provided me with all the facilities to do this work especially regarding measurement and listening rooms. A special thank goes to my thesis advisor Ing. Lucio Bianchi who was always open to any question about my research or writing and he guided me in the right direction whenever I needed letting this paper be my own work.

1. Introduction

To express a predictable judgment of perceived sound quality is not a simple task. It is difficult to determine objectively what sounds “good”, “bad” or “crystalline”, and such quantification is an old question of which researchers have the ball for a long time. The task has been faced in different ways, especially in the low-bit rate coding for transmission systems field and many standards are still a reference for the academic community. In the context of loudspeakers, companies’ production units often need the most objective comparison possible in terms of this high-level characterization so that perceptual evaluation is carried out through listening tests. The task is high-interdisciplinary and encompasses knowledge from signal processing, room acoustics, electroacoustics, audio recording and production, not to mention psychology and statistics. Assessors determine their listening experience by inner processes involving a quantification of auditory attributes sensorial strength and an audio quality overall impression. Typical tests consist of few stimuli reproduced on speakers which the interviewee rates according to different test methods. This huge field needs a careful experimental planning to avoid poor-quality results. However, large-scale tests are often a high-cost choice, especially in terms of time and people involved. Common objective characterization of loudspeakers relies instead on easy-accessible physical measurements in the electrical, mechanical, and acoustic domains crossed by the reproduced audio signal. Many ways of modelling or simulating both linear and different non-linear behaviors of these systems exists too. A full assessment of transducers sound quality relies on both areas, and the goal would be filling the gap in between the two worlds, identifying subjective characteristics of sound in the most possible objective way. The final intention is finding generalizable and compliant results with repeated

informative tests. Exploration aims specifically at relating a subset of attributes to some scalar numbers acting as a concise and representative descriptor of measurable and modellable physical phenomena. This non-redundant low-level representation of data hopefully simplify the analysis still providing a consistent insight about relevant properties of the systems. Often, audio-based features are employed in many different machine learning tasks especially in the Music Information Retrieval (MIR) field and are computed by segmenting highly time-variant input signals on steady-state subframes. Such values incorporate different inspection standpoints (temporal, spectral, perceptual) to which their formula apply. In the context of this work, a feature-based characterization of loudspeaker results from specific operations in raw input measurable speaker data, combined with lumped parameters accounting for other phenomena not englobed in the former signal-based characterization. These are precisely the figures which we want to be most informative as possible for establishing a correlation with subjective data. Replacing *completely* listening tests with a repeatable, easy-accessible and accurate objective characterization of audio signal quality would be the holy grail of research; reliable results would represent a precious cost-effective alternative to carrying out tests when it comes to take specific company decisions. Many developments attempts have been ongoing for many years and an option is estimating sound quality through use of predictive modelling techniques. This typically happened with loudness models or audio perceptual models for telecommunication applications where the aim is predicting results of speech quality tests or quality assessment of audio coding. To simulate listeners' response, they quantify directly in general the quality of system output by processing both a reference and degraded signal, using a psychoacoustic model of human auditory system. The difference between the two is evaluated by a cognitive model to provide an objective score of the degraded one. Such models come easily in the form of relatively simple software whose task is analyze input sound files. When employed for loudspeaker assessment, this approach does not allow to separate the influence of the listening environment and to easily correlate the quality results with common technical parameters used for design. It is a general well-known statement in literature that

both predicting and correlation analysis of subjective scores with objective measurements cannot be done regardless the listening environment in which devices sit and without considering enough resolution in measurements. The present-day scenario is that no predictive model can include all aspects of audio perception, and it will be some time before listening test can be eliminated completely still making them an essential component of perceptual audio evaluation. Successful research studies in last 40 years essentially demonstrated graphical correlation between a set of measurements about direct, early reflected, and diffuse sound; different philosophies about the most relevant measurements about perception of sound quality exist. The only example of validated model considers only a prediction of overall impression of relative preference score for loudspeakers evaluated in a multi-comparison listening test scenario. The interesting point is the existence of a quantitative relationship between average preference scores and a linear combination of valuable features accounting for some statistical properties of anechoic frequency response and sound power measurements. This feature-based approach in the context of predictive modelling is an idea we are interested much. A new different approach investigates the correlation between a set of loudspeaker parameters and listening preference but is limited to the adoption of the sole linear ones. At the present time it still results in an over-simplified analysis. Besides this, such numbers have never been considered for becoming potential predictors. A new fast and employable feature-based methodology, as much independently as possible from environment employed for measures, would be to fill these holes.

This thesis embraces both the audio sensory science field and a model-based objective characterization of electro-dynamical transducers. Our attention is devoted to investigating the most relevant parameters to infer correlations between physical properties of a device and listening perception. To produce a speaker objective characterization strictly related to sound quality both linear and nonlinear loudspeaker parameters are involved. The expected outcome is finding an informative listening test methodology whose results, in conjunction with our low-level characterization allows a replacement of sensory data on different auditory attributes

by instrumental measurement and simple calculations, or, to prove some unknown correlations for at least a prior direction and knowledge on listening test design.

Outline

Manuscript is organized as follows. In Chapter 2 we introduce the all the necessary theory behind our experiments and a state-of-the-art techniques review to address the problem with emphasis on the main scopes and limitations behind them according to our specific research direction. In Chapter 3 the procedure chosen for achieving our task and the way our datasets are collected are introduced along with all technical and implementation-oriented problems. A meticulous design of the experiments is described along with the problem of feature extraction. In Chapter 4 an appropriate analysis of the results will be carried out, with respect to the nature of collected data. All possible experimental outcomes are discussed to prove the effectiveness of our methodology. This discussion aims at highlighting every possible quantitative correlation between the results provided by measurements and test ratings. Finally, we draw some conclusions and propose possible future works. Appendix A and B show additional documentation about programme material involved in the listening tests and technical specifications of all hardware involved in the work. Appendix C simply provides a further mathematical clarification of electro-dynamical loudspeaker working principle.

2. Background and State-of-the-art

This Chapter is devoted to a state-of-the-art review about sound quality evaluation objective techniques based on speakers' measurements and their capability of predicting or well-correlating with listeners' sound quality ratings. The options provided by the literature consist mainly in graphical attempts to find trend on measurements especially concerning frequency response when juxtaposed to subjective ratings to explain some ranking or correlation in the corresponding objective data. Existing predictive models can be coarsely categorized as based on loudspeaker sound power measurements, in-room measurements and perceptual models revealing various underlying dimensions of listening impression. All objective techniques dealing with human perception, despite their accuracy, suffer unavoidably from how and where devices are measured. A recent attempt to generalize a quantification of perceived sound quality in a wide variety of environments has been done through virtualized listening tests but still showing, however, lots of limitations. A special attention in this work will be dedicated to the most relevant example of a validate predictive regression model existing in literature which is however restricted to the only estimation of preference ratings. Finally, another research direction investigates the correlation between a set of electromechanical lumped parameters of transducers and attributes of sound quality.

The rest of this chapter is organized as follows. In Section 2.1 we present a preliminary overview on electro-mechanical dynamical speakers as regards their transduction principle, main components, and classification. The essential mathematical background for their modelling according to their physical behavior is presented in Section 2.2. In Section 2.3 we highlight the complex problem of objectively establishing what *sounds good* in terms of speaker

measurements such as frequency response and distortion. Subjective timbre perception in speaker reproduction is addressed too. Finally, in Section 2.4 we review the state-of-the-art approaches to correlate objective to subjective data along with their strengths and limitations.

2.1 Electrodynamic loudspeakers

2.1.1 Basic historical perspective and main elements

Moving-coil electrodynamic loudspeakers are the most common type found in both consumer and professional audio applications. The device was patented in 1929 by Edward W. Kellogg and Chester W. Rice (Figure 2.1) and, surprisingly enough, in one century, little changes in the construction idea of these transducers occurred: a real difference is the absence of the permanent magnet in original patent due to the lack, at that time, of reliable materials to realize it [1]. The main components of these drivers are illustrated in Figure 2.2 instead.

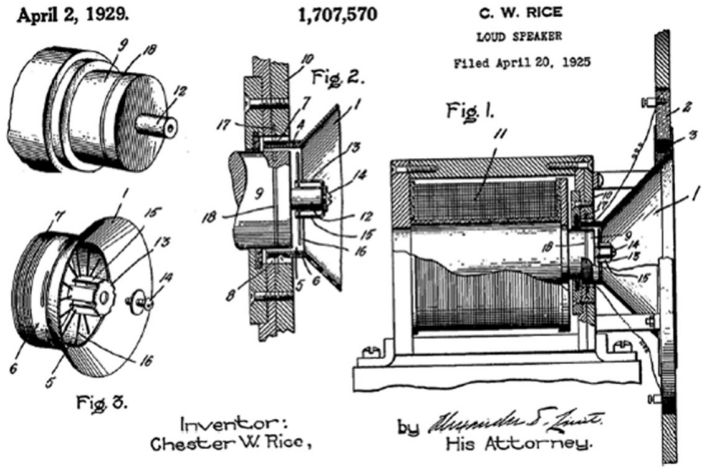


Figure 2.1 – C.W.Rice and E.W.Kellogg. Loudspeaker. US Patent1,707,570. Apr. 1929. Picture from [118]

We can think a speaker diaphragm as a stiff curved cone with a dust cap in the center and a coil wound attached to its top. This coil is in the gap of a magnetic path, comprising a pole piece, a pole plate and a permanent magnet held producing constant magnetic field. The cone is supported at the rim and near the voice coil by a surround and a spider respectively so that

it can move freely only in an axial direction [2]. Loudspeakers can be classified in different ways. A first arrangement is according to the frequency band of sound being reproduced. Drivers made for high frequencies are called “tweeters”, those for middle ones are called “mid-range drivers” while those dedicated to the low-end of the audible range are the “woofers”, sometimes very low frequencies being treated by “sub-woofers” alone. If more driver nits are present in the same loudspeaker, the incoming signal is separated into different frequency bands by a properly designed filter network called “crossover”.

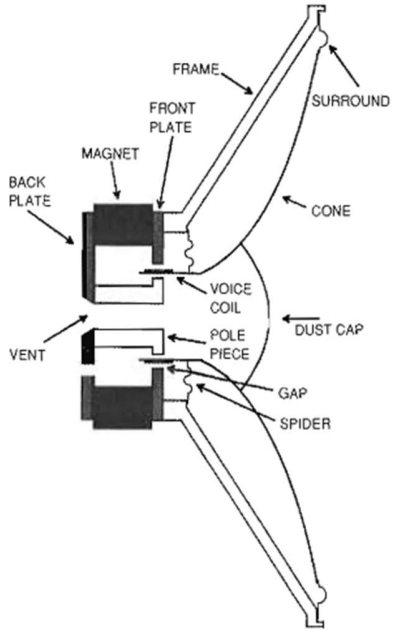


Figure 2.2 – Schematic example of a woofer diagram. Picture from [3]

This is really needed in many applications as the three decades wide (20Hz – 20kHz) audio bandwidth has conflicting requirements at his extremes: low frequencies needs a lot of air movement so that large membranes and heavy moving elements are desirable, while, on the other side, lighter and faster ones better fit higher frequencies reproduction. We could also categorize transducers according to the cone shape or acoustic load; in this work, we implicitly refer to direct radiators, in which membrane is immediately facing outside air.

2.1.2 Basic principle of Electro-Mechanical transduction

Three separate but interrelated systems operate in unanimity in an electro-dynamic driver: the motor system, the diaphragm, and the suspension system. The first is represented by the aforesaid moving voice coil suspended in the constant magnetic field. When an alternating current passes through the wire, due to Lorentz force law, the coil is forced to move rapidly back and forth translating then to the diaphragm. In a first approximation, a perpendicularly suspended voice coil, is subject to the following force:

$$F = Bl \cdot i \tag{2.1}$$

A further mathematical insight about the meaning equation above is given in Appendix D. The proportionality coefficient Bl is called "force factor" and represents the coupling between mechanical and electrical domain when this alternating current i runs through the wire thanks to its interaction with the air-gap flux of the magnet. In the electrical domain, the force factor defines a back electromotive force acting as an excitation voltage $v_{EMF} = Blu$ created by the voice-coil movement with a velocity $u = \frac{dx}{dt}$ in the same magnetic field. Such kind of interactions was understood almost one century ago. Voice coil is coupled to a suitable acoustic radiator vibrating correspondingly with it and this movement pushes air molecules at its surface. In the end an electrical audio signal is converted to acoustic waves, i.e. microscopic propagating air pressure variations that our brain recognize as sound [5].

2.2 Loudspeaker modelling

Different kinds of loudspeaker modelling exist, each of them providing a partial description of the system specific to its abstraction level while preserving relevant features according to the scope of the model itself. However, all maintain the general structure shown in Figure 2.3; a model is essentially a "black box" in which some input variables produce an output. In case of a speaker, the most typical input is a voltage signal applied to its terminals, while common

outputs are considered according to the purpose of the model, we are building, i.e. diaphragm velocity, diaphragm displacement, acoustic pressure etc. A model consists of a structure, operators, parameters and instantaneous-state variables, such as voice coil displacement $x(t)$, temperature $T(t)$ and current $i(t)$ which are time-variant and stimulus-dependent. All of these are measurable in electrical, mechanical, and acoustic domain. As far as structure is concerned, it can be categorized as linear or nonlinear, deterministic, or stochastic, static, or dynamic and lumped or distributed. It provides a general description of the system physical mechanisms to be modelled and is representative of the transducer principle.

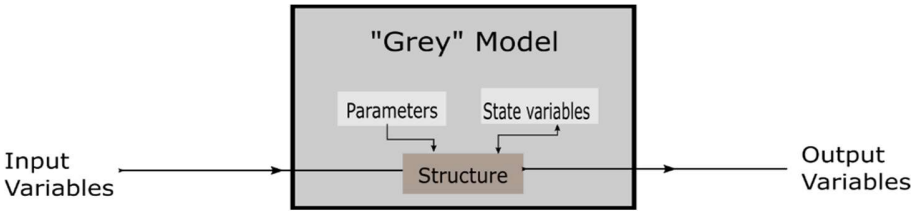


Figure 2.3 – Loudspeaker "Grey model". Picture adapted from [6]

Example of model structures in loudspeakers can be Finite Element Models (FEM), Boundary Element Models (BEM), Lumped Parameter Models and Differential Equations. The free parameters are descriptive of the properties of one individual unit: they should be independent on both stimulus and measurement conditions and may be constant or functions of one or more variables. Examples are materials, geometry, linear and nonlinear parameters [6]. Two operating conditions are also distinguishable in speakers: small signal and large signal domain. The former is correctly representable by a linear model[1] while the latter needs a nonlinear one. The most important state variable in discriminating this is $x(t)$. Roughly speaking (value depends on specific speaker), in small signal working range maximum excursion may be of $\sim 10^{-3}$m while in the other case even about ten times higher. Still, if we increase it, loudspeaker

[1] For a linear system superposition principle is satisfied as well as stability, causality. Input-output relation is of convolving nature in time domain and is completely characterized by an invertible transfer function in frequency domain.

behavior can go from weakly nonlinear to strongly non-linear, possibly affected also by other time-variant phenomena. A high-level driven speaker generates inherent artefacts for the non-linear nature of electro-dynamic transducers such as amplitude-dependent distortion and additional spectral components not present in the exciting stimulus [7]. A key assumption for a linear characterization is that all parameters are constant with respect to x, while contrarily large signal modelling considers such dependence. Above all, the parameters that vary with voice-coil displacement are the electric inductance L_e, the force factor Bl and stiffness K_{ms}. Details on these aspects are reminded to Subsection 2.2.1.2. Last worth-mentioning aspect is the difference between lumped or distributed parameters for modelling.

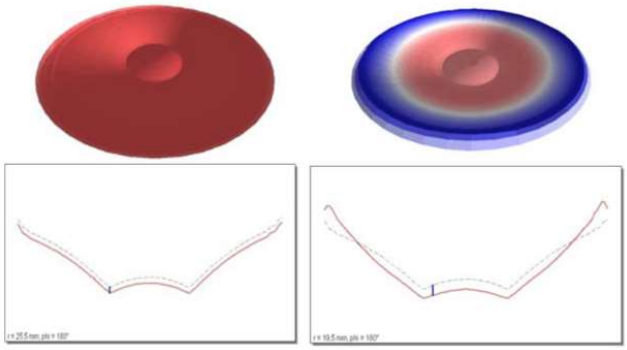


Figure 2.4 – Loudspeaker's cone pistonic and non-pistonic range - left; piston Mode below 1kHz right; First Bending Resonance at a frequency ~1.4kHz. Picture from [6]

Qualitatively speaking, typically a lumped model is theoretically valid only for frequencies around the fundamental resonance determined by mechanical mass and compliance. In this situation, the cone is assumed to be in a "pistonic" range, that is, sufficiently stiff to move as a rigid body in sync with voice coil. At higher frequencies, cone starts to flex and bend due to its own resonance modes (usually denoted as "breakup behavior") producing irregularities in the frequency-response [2]. In this case, part of the cone is out of phase with the other and distortion at the output is guaranteed. This appears along with the above-mentioned nonlinear behavior at large amplitudes due to the x-dependent characteristics of the speaker.

2.2.1 Signal-based modelling

A system can be formally defined in a general mathematical sense by a function that maps an excitation signal $x(t)$ to an output $y(t)$:

$$y(t) = F[x(t)] \tag{2.2}$$

The $F[\cdot]$ operator may be linear or nonlinear, while has memory if function of previous output values and/or input signals. This is what confers to the system its dynamical properties [8].

2.2.1.1 Frequency response

A dynamical loudspeaker transduces a voltage $u(t)$ to sound pressure $p(t)$ which in this context are exactly the input and output respectively of our audio system. The acoustical pressure is a useful measurable quantity to assess signal quality. An (amplitude) frequency response is the curve obtained by plotting in dB (y-axis) an output pressure recorded by a microphone versus frequency (x-axis) as a response to an electric signal. In frequency domain, with input voltage signal $U(j\omega) = FT\{u(t)\}$ and pressure $P(j\omega) = FT\{p(t)\}$ spectra, the model can be described by a transfer function:

$$H(j\omega) = \frac{P(j\omega)}{U(j\omega)} \tag{2.3}$$

Along with the phase information this figure of merit provides a full *linear* characterization of our speaker. This ratio is specific to the location used for measurement. Typically, what is shown by manufacturers is the on-axis response where the microphone is placed on a reference axis in the far-field. Such response accounts for just the first sound we hear. In acoustics, a far-field-near-field dichotomy can be discriminated according whether the wavelength of reproduced sound is large (far-field) or small (near-field) if compared to the size of our device. The former kind of response is usually measured under free field condition, an idealized sound propagation in which only the direct sound is detected, and data is free of any room reflections. Usually an anechoic room artificially reproduce this proper environment. The other extreme

is a reverberant field, where sound is assumed to arrive from all directions with equiprobability and with the same magnitude. A "listening window" or a "first reflections" frequency responses also exist, obtained by spatially averaging sound from several measurements positions around the speaker or bouncing back from room boundaries, respectively. A sound power response characterizes instead the total radiated sound power at different frequencies and is measured at many angles around a sphere both in anechoic and reverberant environments.

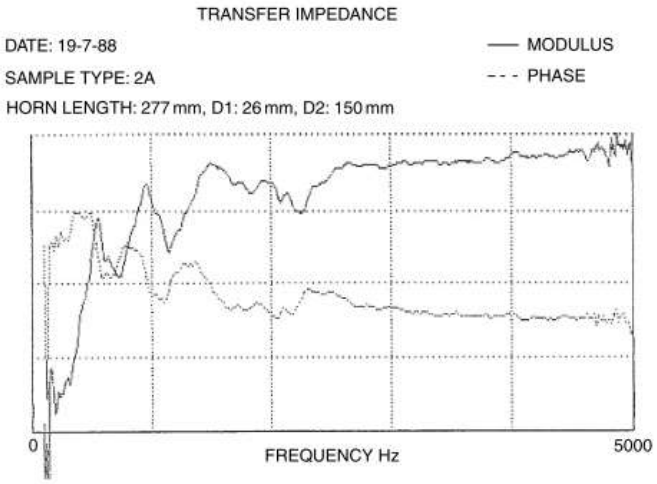


Figure 2.5 – Example of a measured full-frequency response. Picture from [34]

If displacement $x(t)$ is known, then low-frequency sound pressure can be predicted with knowledge of the membrane area by near-field sound pressure measurements [9] for which direct sound is much higher than the diffuse and ambient noise contributions. Low-frequency sound pressure response in far-field can also be predicted precisely [10]. Assuming a radiation in a free field half space, the acoustical pressure is given by the following formula:

$$p(t) = \rho_0 \frac{S_D}{2\pi d} \cdot \frac{d^2 x(t)}{dt^2} \tag{2.4}$$

where S_D is the effective surface of the loudspeaker's diaphragm, ρ_0 is the air density (1.184 kg/m^3 @25°C), and d the distance of measurement point to the diaphragm [11]. Recalling that $X(j\omega) = H_x(j\omega)U(j\omega)$, the magnitude of far-field response transfer function is:

$$|P(\omega)| = \omega^2 |H_x(j\omega)||U(j\omega)| \cdot \rho_0 \frac{S_D}{2\pi d} \tag{2.5}$$

where $H_x(j\omega)$ is the mechanical transfer function defined as the ratio of the displacement signal spectrum and the input voltage signal spectrum, the former easily measurable by a laser.

2.2.1.2 Distortion and nonlinearities in electrodynamic speakers

With reference to the previous general mathematical framework, we can distinguish between linear and nonlinear distortion occurring in speakers. In the linear case the system produces a signal at the output with a change in amplitude or phase with no new frequencies added while the latter case occurs whenever new frequency components that were not present in the input appears in the output signal. In the context of loudspeakers, non-linear distortion is usually a prevalent disturbance factor due to its effect on perceived sound. We can categorize nonlinear systems into exhibiting two non-linearities kinds. Memoryless ones, called *static nonlinearities*, occur with instantaneous dependence of output signal on the input and $F[\ \cdot\ ]$ is representable by a 1D function, while *dynamical nonlinearities* are nonlinearities with memory.

A static model is an idealized approximation for speaker nonlinearities. Lack of memory implies that linear and nonlinear characteristics of system are not frequency-dependent [12]. In this case all input-output curves can be expanded in a Taylor-series to desired accuracy:

$$F[x(t)] = \sum_n a_n x^n(t) \tag{2.6}$$

where the nth power of x corresponds to the nth order nonlinearity. Artificial nonlinearities functions commonly found in electroacoustic systems are shown in [11]. If a single tone is applied as an input, output will be a periodic signal with the same fundamental frequency plus harmonics added. This holds for a polynomial nonlinearity of arbitrary order [8], [13]. Motor and suspension non-linearities of the speaker will follow such a model (Section 2.2.1.2). A very common frequency-dependent distortion metric can be defined as the ratio of the total level of harmonics to the level of corresponding fundamental tone at the output:

$$THD = \frac{\sqrt{\sum_{k=2}^{N+1} P_k^2}}{P_1} \tag{2.7}$$

where N is the total number harmonics, P_1 is the amplitude of the fundamental frequency in the pressure spectrum output and $P_k = P(j\omega_k)$ is the kth harmonic amplitude. Sometimes it is expressed without root (power ratio) or in percentage. Such a curve depends also on different motor and input voltage signals [13]. If the input signal is a two-tone or multi-tone stimulus, in addition to harmonics, the combination of tones with frequencies equal to sums or differences of them are generated. These are the called *intermodulation products* altogether representing the intermodulation distortion (IMD) as shown in Figure 2.6.

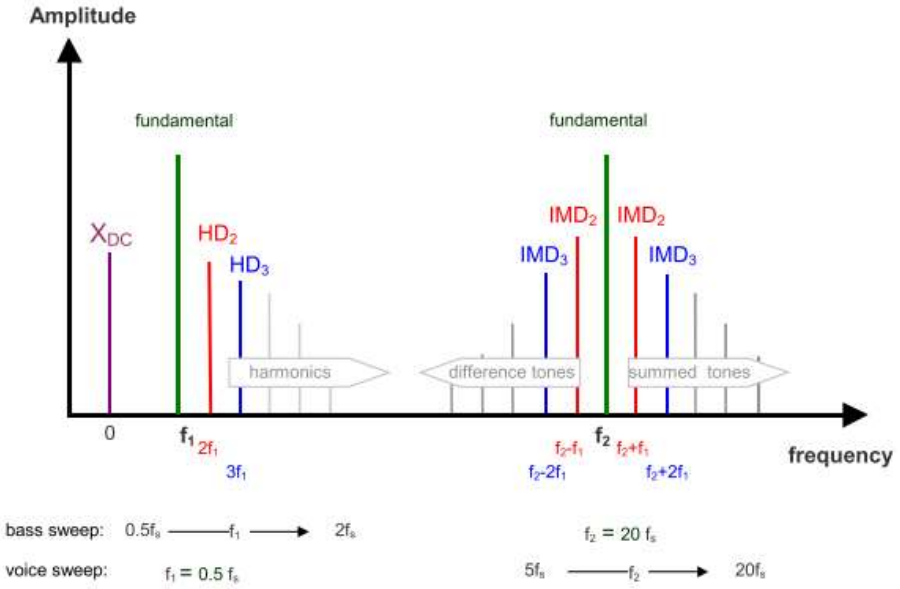


Figure 2.6 – Sound pressure spectrum generated by a two-tone stimulus. A high distance between the fundamental components ($f_2 \gg f_1$), nicely separates harmonic and intermodulation components. Picture from [17]

However, electro-dynamical transducers exhibit much more complex nonlinear behavior that cannot be described merely by algebraic polynomials and simple 1D input-output relationships [12]. A complete analysis would require sophisticated mathematical tools such as Volterra series [14], particularly useful in case of weak nonlinearities.

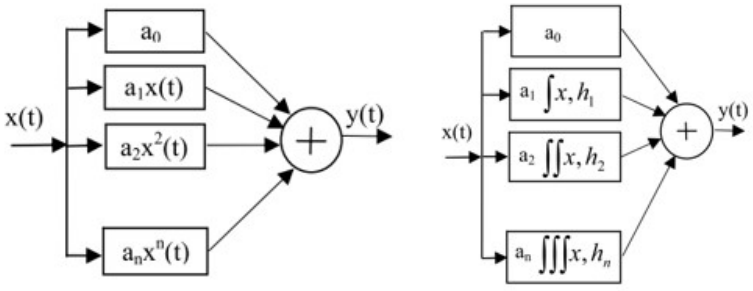


Figure 2.7 – Static and dynamic nonlinearities modelling in transducers. Left; static memory-less polynomial nonlinear system. Right; convolution Volterra integrals of different orders replacing polynomials. Picture from [12]

This is a nonparametric modelling consisting in infinite series, where the nth power is replaced by an n-dimensional convolution integral with a nth multidimensional impulse response called *kernel*, the 1st order one coinciding with the classical impulse response. The frequency domain counterpart is a multidimensional n^{th} order frequency response where simple one-tone distortion measurements lie in a line [13], [11]. For very strong non-linear systems Volterra modelling is still non-valid. Perceptually relevant aspects of distortion deserve more interest here and are addressed in Section 2.3.1.

2.2.1.2.1 Causes and symptoms

Significant elements in non-linear distortion generation reside at the moving coil speaker unit. Most dominant contributions are motor and suspension nonlinearities. The driving mechanism itself leads to two non-linear distortion sources both related to effects in the iron magnetically permeable structure: a magnetic attraction force between coil and field structure and a distortion due to non-uniform magnetic field [15]. Design choices for compensating for such electro-magnetic phenomena are well-illustrated in [3], [17]. The voice-coil self-inductance L_e is in principle nonlinear in both $x(t)$ and $i(t)$; we will always implicitly neglect here the second. The force factor x-dependance is non-trascurable at large displacements, as away from the rest position, a voice coil leaving the magnetic gap steps away from the maximum magnetic field region thereby producing less force for the same current. Generally, this excursion

nonlinearity is recognized to be stronger than a current one. Moreover, for high cone excursions driver suspensions respond stronger than predicted by a simple Hooke's law, getting stiffer the more it is stretched away from rest position. Real suspension nonlinear behavior is more complicated and should also include a hysteresis, a "creep-effect" time-varying behavior (leading to a frequency-dependent compliance) and a temperature dependence as well [8]. The traditional method of assessing large signal performance relies on measurement of special *symptoms* of monitored variable such as $p(t)$, $i(t)$ or $x(t)$. Such symptoms are *not* system parameters as they highly depend on stimulus, and are generated only if nonlinearities are properly activated [6]. Distortion measurements results highly depend on the properties of chosen stimulus; different inputs may produce different symptoms in the same system and this is a substantial difference to the small signal domain where the input/output relationship is fully characterizable by a linear transfer function (or impulse response). Symptoms do not completely describe the large signal performance but must be intended only as a manifestation of a specific cause. Harmonics and Intermodulation products are nothing but an example. $Bl(x)$ and $K_{ms}(x)$ lead to such distortions in sound production. More specifically, as regards force factor, notable harmonic distortion is generated only at low frequencies where cone displacement, current and velocity are larger. It is a great source of intermodulation distortion, still amplified at low frequencies and when the two-tones are close apart [16]. This nonlinearity causes both nonlinear damping and parametric excitation (see Section 2.2.2.3). As displacement is a low pass filtered signal, its multiplication with $K_{ms}(x)$ leads to a stiff restoring force producing both distortion symptoms. A voice coil non-linearity produces relatively low harmonic distortion instead. Just from these considerations, it is yet evident that harmonic distortion is not uniquely distinguishable Bl and K_{ms} nonlinearities although being a more sensitive symptom for the latter. For a deeper analysis reader is referred to [17].

2.2.1.2.2 Main approaches in assessing transducers distortion and non-linearities

Assessing nonlinearities in transducers is complex because of both the multidimensional nature of the task itself and the intricacy of the human hearing response to a contaminated signal. Historically we can divide three main approaches in doing that: identification models, measurement models and perceptual models [18]. As regards physical measurements methods are concerned, they aim at obtaining symptoms of nonlinearities through metrics such as THD, traditional IMD methods, such as CCIF [19] and SMPTE [20], TIMD or through multitone testing with different kind of high order weightings. Employment of coherence function is also a possibility [13]. All these provide an "integral" assessment of nonlinear distortion and convey different information. Moreover, measuring, graphing, and interpreting nonlinear distortion as it relates to subjective perception is a well-known "dilemma" in literature: much more information than is required to assess a linear system is needed but a comprehensible graphical and numerical presentation is a hard task [12]. Other methods exploit noise-like excitation signals for measurement of nonlinear transducers [21], [22]. When it comes to correlation with subjective sound quality impression, although objectively better than one-or two-tone measurements no standards exist for these kinds of measurements [8]. The perceptual approach, based on the simulation of psychoacoustic effects, is another worth-mentioning one. Originally, it had wide application in the context of objective evaluation of codecs quality for low-bit compression of musical and speech signals and nowadays some standards [23],[24] still survive. However, nonlinearity nature in loudspeakers is different with respect to these compression schemes. In the formers, signal is distorted by nonlinear physical effects making the radiated pressure dependent on the voice coil, while in the latter is impaired by quantization noise which is not level dependent. References [12], [25] provide a complete historical survey.

2.2.2 Lumped parameter characterization

With lumped parameter modelling, each driver element belonging to different domains is ideally concentrated in one single point neglecting physical dimensions of the system and all can be described by a physical constitutive law. There are some basic assumptions behind:

- Linearity
- Low frequency
- Cone is assumed as a perfectly flat plane rigid piston radiating in an infinite baffle

An approximative criterion to describe its range of validity is the Helmholtz number [6]:

$$He = \frac{2\pi a}{\lambda} = \frac{2\pi f a}{c} = ka < 1 \tag{2.8}$$

where a is the radius of the speaker cone and λ is the wavelength of the sound wave reproduced. The effect of an infinitely rigid piston can be *lumped* to one single point, behaving similarly to a pure spherical source at low frequencies from an impedance standpoint [1]. Two aspects can be stressed about modelling speakers with equivalent circuits; firstly, analogies between different domains are exploited, secondly, electro-mechanically speaking, dynamical loudspeaker drivers are reciprocal transducers. To tackle mechanical domain as if its elements were electrical, the most natural way is the "impedance analogy" where the across and through Kirchhoff electrical variables (u, i) have a biunivocal correspondence to mechanical force and velocity (F, v) respectively. This will be implicit from now on for bipoles. Mechanical friction is modelled as a resistance, a compliance with a condenser and a mass with an inductor. Analogy stems from similarities of physical laws between the domains; if one writes second Newton's, Hooke's and viscous friction laws, the mathematical structure of equations is the same of condenser, inductance, and resistor constitutive laws, respectively. The transduction is modelled with a gyrator whose coupling constant equals Bl [26]. An "admittance analogy" also exists where correspondence of variables is dually swapped. A more detailed discussion can be found in Chapter 3 of Beranek and Mellow [2].

2.2.2.1 Linear lumped parameters

In the low-frequency behavior, diaphragm moves as a rigid body with no acoustic modes so that loudspeaker can be characterized with some lumped electro-mechanical parameters:

- $R_e = |Z_e(f = 0)|$: DC electrical resistance of voice coil
- Bl: force factor
- $C_{ms} = \frac{1}{K_{ms}}$: mechanical compliance (stiffness) of driver suspensions
- $M_{ms} = M_{md} + M_{ar}$: mechanical mass of diaphragm assembly (air load + voice coil)
- R_{ms}: mechanical resistance
- S_D: surface area of diaphragm

We already know that actual values of these parameters may vary according to driving level, temperature, etc. and above all displacement. If a signal is applied to loudspeaker (voice coil) terminals, speaker electric behavior is summarized by a resistance R_e and some amount of inductance L_e, that can be neglected in a first approximation for very low frequencies. The system mechanical part is basically a lumped resonator with two suspension elements acting as a spring (compliance) along with mechanical losses explained by a single value of mechanical resistance. The force factor models the motor and we can "see" the amount of magnetic induction at our disposal through it. In the acoustical domain, the air moved by diaphragm could be represented as a load where drivers are designed to spend as much energy as possible in terms of propagating waves. Different kinds of radiation impedances modelling at different frequencies are not explored in the context of this work: S_D is the only parameter accounting for the coupling between mechanical and acoustical domains. The assumption behind is replacing all speaker radiating parts with a rigid piston of this area of same voice-coil velocity and a surface generating the equivalent sound pressure in far field as the real radiator [27]. Theoretically, this is a complex function of frequency $S_D(f)$ that, at low frequencies can be

approximated with a real value, usually derived from the geometry of the radiating surface by reading the diameter d in the middle of the surround area:

$$S_D = |S_D(f_s)| \approx \frac{\pi}{4}\left(\frac{d_i + d_o}{2}\right)^2 = \frac{\pi}{4}d \tag{2.9}$$

Above formula typically holds when the surround is wider than 10 % of the outer diameter d_o and it is particularly suitable for woofers [28]. Putting all elements together we obtain the complete circuit for a driver alone, a rigid baffle acting in free air, without any enclosure: Air load is included directly in the mechanical mass and acoustic resistance in the rightmost part of the circuit which is null here so that volume velocity in the medium $U_0 = S_D v$ coincides with the one directly impressed by diaphragm motion.

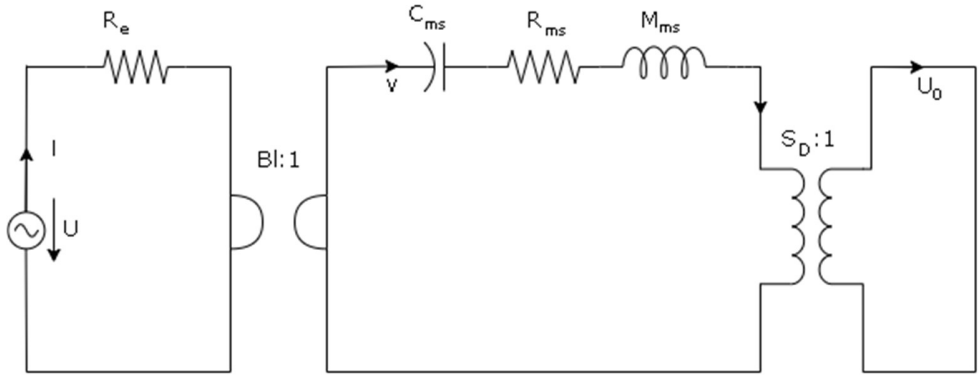


Figure 2.8 – Simplified low-frequency Thiele-Small circuit for drivers (impedance analogy), free air, infinite baffle

The above circuit name is in honor of two men that about 50 years ago devoted their research to the characterize loudspeaker low-frequency response starting from driver and enclosure specifications with the aim of finding convenient design choices on this basis. Hence, some "Thiele-Small parameters" exist too and are commonly found in any specification sheet. Ones of our interest will be discussed henceforth. Many of those numbers are strictly defined only at the resonant frequency, but the approach is generally applicable in the pistonic motion range. At low frequencies, the natural resonance frequency of mechanical oscillator is:

$$f_s = \frac{1}{2\pi\sqrt{M_{ms} \cdot C_{ms}}}(Hz) \tag{2.10}$$

At this frequency, energy stored is maximum resulting in maximum cone velocity. As in any harmonic oscillator, a quality factor Q can be defined as the ratio of the reactive to the resistive part quantifying how fast energy is vanished due to losses when the oscillator moves freely. In speakers two subtracting energy mechanisms different in their physical nature occur: a mechanical and an electrical resistance, the former living inside the moving assembly while the latter acting as an electromagnetic brake in the motor. Therefore, they produce both a mechanical and an electrical Q:

$$Q_{ms} = \sqrt{\frac{M_{ms}}{C_{ms}}}\frac{1}{R_{ms}} = \frac{2\pi f_s M_{ms}}{R_{ms}} \tag{2.11}$$

$$Q_{es} = \sqrt{\frac{M_{ms}}{C_{ms}}}\frac{R_e}{(Bl)^2} = \frac{2\pi f_s M_{ms} R_e}{(Bl)^2} \tag{2.12}$$

A total Q can be derived from the previous two as they were two parallel impedances:

$$Q_{ts} = \frac{Q_{es}Q_{ms}}{Q_{es} + Q_{ms}} \tag{2.13}$$

The equivalent box volume of suspension, is instead given by:

$$V_{as} = \rho_0 c^2 S_D^2 C_{ms} \tag{2.14}$$

This parameter represents the air volume which, when acting upon a piston of area S_D, is equivalent to the compliance of driver's suspension as if the piston was not radiating in an infinite baffle but with some mechanical stiffness acting. Lastly, as voice coil does not operate in free-air, special linear models can be applied to describe losses in the iron material [17] known as "eddy currents", which make the inductance to decrease at higher frequencies thus not modellable with an ideal inductor. The lumped equivalent circuit transforms as shown in Figure 2.9. To find a suitable impedance curve covering the full audio band we need an inductance model which fits best the frequency dependency of $Z_L(j\omega, x)$ with a minimal number of free parameters, hence the so-called LR2 [29] can be adopted (Figure 2.10).

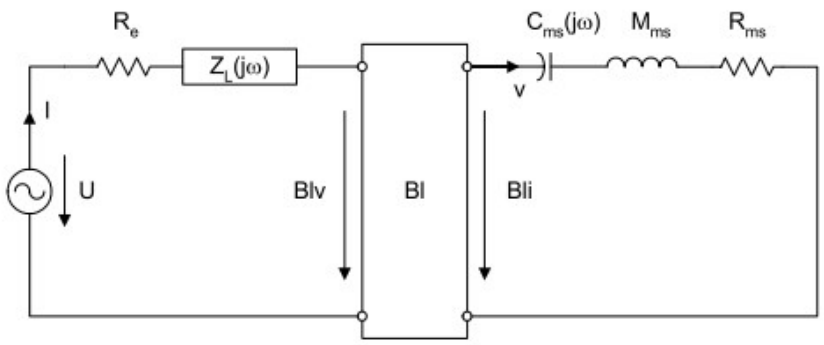


Figure 2.9 – Electro-mechanical equivalent circuit of the loudspeaker. Picture from [29].

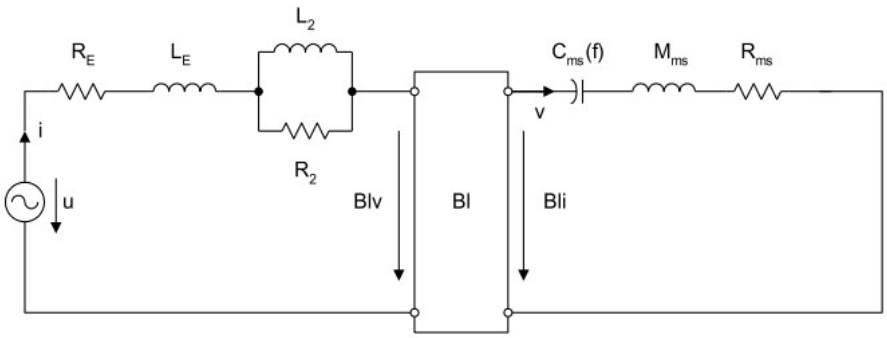


Figure 2.10 – Lumped linear loudspeaker parameter model (electromechanics behavior). Picture from [10].

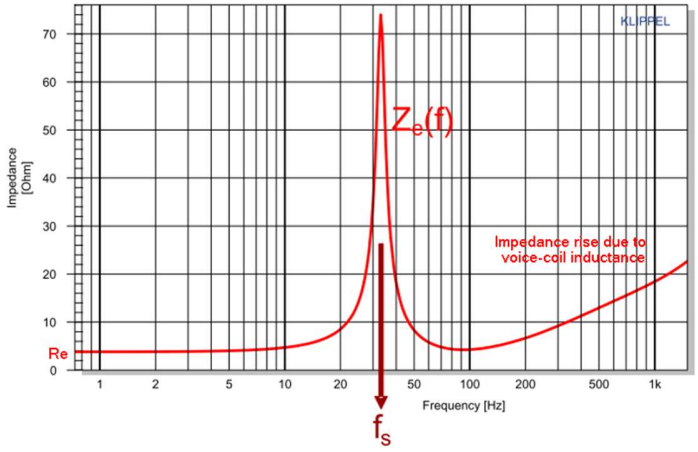


Figure 2.11 – Magnitude of electrical impedance versus frequency f. Picture from [26].

Three further numbers L_e, L_2 and R_2 are included in the modelling; this lossy inductance is nothing but the residual electrical impedance after removing the DC resistance R_e and the

effect of the back EMF Blv [26]. Note that the R_e is not included in $Z_L(j\omega)$ so that it is measurable at electrical terminals by blocking coil movement and subtracting it. Neglecting losses, electrical impedance acts circuitally in series with the driving amplifier output impedance, is made of R_e, L_e plus the *motional electric impedance*, that is, the equivalent impedance seen by the electrical side of the circuit if effects of the other two domains are "brought back" [2]. The electrical impedance is a transfer function defined as the ratio of voltage and current spectra.

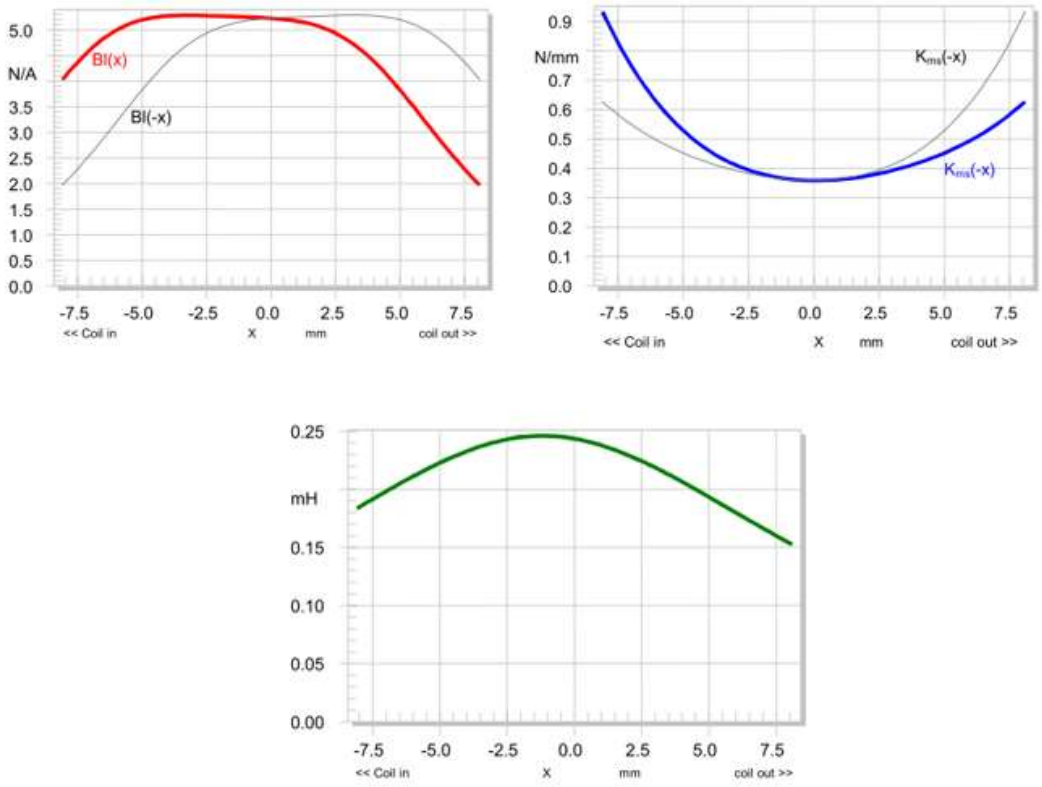


Figure 2.12 – Nonlinearities shapes; Force factor Bl(x), stiffness $K_{ms}(x)$ and voice-coil inductance $L_e(x)$ versus displacement for a 6-inch well-designed woofer intended for high quality consumer application. Pictures from [17].

By plotting its frequency response (in log-scale) a resonance can be noticed (Figure 2.11). Despite appearing as an electrical resonance, it accounts also parametrically for the mechanical one, its amplitude not diverging due to the presence of friction. At very low frequencies, the

resistive behavior is dominant while a prevalent inductive phenomena L_e at higher frequencies explains the increasing behavior.

2.2.2.2 Non-linear lumped parameters

With large signal modelling we assume the dependence of drivers' parameters on instantaneous voice coil displacement as shown in Figure 2.12. Usually high displacement is required to cause significant variation of these parameters, whose impact on modelling is discussed henceforth. An electro-mechanical lumped model can account for dominant nonlinearities including $L_e(x, i = 0)$, $Bl(x)$ and $K_{ms}(x)$. Because of these dependencies, the magnetic flux variation translates in a x-dependent back induced voltage in the electrical circuit due to the magnetic flux $\phi_x = L(x)i$ derivative and in an additional reluctance force $F_m \sim i^2(t)$ driving mechanical system. Low-frequency behavior with non-linear signal dependent elements modifies as follows:

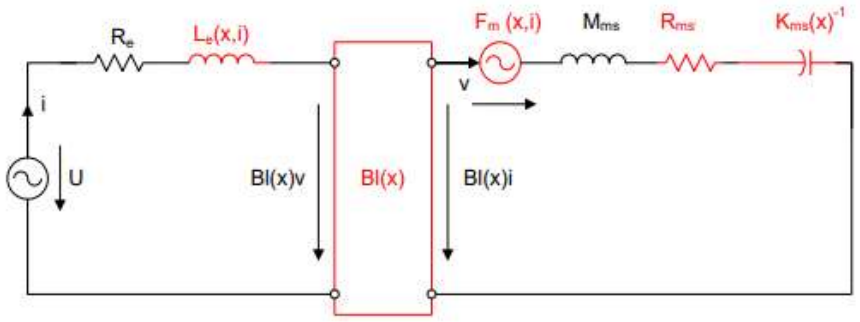


Figure 2.13 – Electrical equivalent circuit considering motor and suspension nonlinearities. Picture from [30].

Here, considering the LR-2 model for large signal modelling, we will consider only the first parameter x-dependence, compliantly with analyzer software measurement performances [30] while leaving $R_2(x = 0)$ and $L_2(x = 0)$ only for the linear model. The voice coil inductance, Force factor and stiffness nonlinearity thus can be modelled as:

$$L_e(x) = \sum_{i=0}^{N} L_{xi} x^i \tag{2.15}$$

$$Bl(x) = \sum_{i=0}^{N} Bl_i x^i \tag{2.16}$$

$$K_{ms}(x) = \sum_{i=0}^{N} K_{ms_i} x^i \tag{2.17}$$

Static nonlinear functions have no frequency dependency and can be represented as graphs or power series expansion exactly as described in paragraph 2.2.1.2 and linear models may be easily expanded in this way. The parameters corresponding to $i = 0$ represents the linear parameters $L_e(x = 0)$, $Bl(x = 0)$, $K_{ms}(x = 0)$ evaluated at rest position. Note that, as a loudspeaker operates, it is the overall nonlinearity due to this parametric excitation of both circuit sides to be dynamic and more complex than a simple relationship described by a Taylor series. Circuitally speaking it manifests as an interdependence of components values on some variables of the circuit; for instance, if one substitutes in the leftmost loop of Figure 2.13 the effect of the gyrator through its port with a controlled generator its gain will depend on the integral of the current flowing. This is an example of what makes such kind of non-linearity different, for instance, from ones occurring by combining many static nonlinear characteristics of components in a circuit. From a mathematical standpoint each static nonlinearity is multiplied by either displacement or itself in the system equation.

2.2.2.3 Dynamical representation

Dynamical behavior can also be described with a system of nonlinear coupled ordinary differential equations modelling transduction in a biunivocal correspondence with the electrical circuit examined so far. Current $i(t)$ and displacement $x(t)$ are the time-dependent unknowns of the system with a known electrical excitation signal $u(t)$, i.e. a voltage supply:

$$u(t) = R_e i(t) + \frac{d(L_e(x,t) \cdot i(t))}{dt} + Bl(x)\frac{dx(t)}{dt} \tag{2.18}$$

$$F(x,t) = Bl(x)i(t) = M_{ms}\frac{d^2x(t)}{dt^2} + R_{ms}\frac{dx(t)}{dt} + K_{ms}(x)x(t) + \frac{dL_e(x)}{dx}\frac{i^2(t)}{2} \tag{2.19}$$

By subtracting the two above equations and dividing all for R_e one can write:

$$K_{ms}(x)x(t) + \left[R_{ms} + \frac{Bl(x)}{R_e}\right]\frac{dx(t)}{dt} + M_{ms}\frac{d^2x(t)}{dt^2} = \frac{Bl(x)}{R_e}\left(u(t) - \frac{d(L_e(x,t)\cdot i(t))}{dt}\right) + \frac{dL_e(x)}{dx}\frac{i^2(t)}{2} \tag{2.20}$$

Left side shows the sum of restoring force, inertia and damping force corresponding to mechanical stiffness, resistance and moving mass connected in series. $Bl(x)$ causes both a nonlinear electrical damping and electro-dynamical excitation variation of the on the right side. An additional reluctance force due to $L_e(x)$ drives the mechanical system directly [30].

In case of linear lumped parameter modelling, previous equation simplifies in a 2nd order nonhomogeneous differential equation with $\frac{d(L_e(x)\cdot i(t))}{dt} = L\frac{di(t)}{dt}$ and $\frac{dL_e(x)}{dx}\frac{i^2(t)}{2} = 0$. In this case, motion can be seen as the classical driven damped harmonic oscillation, whose general solution (transient) exponentially decreases with a damping β independent on the forcing function:

$$\beta = \frac{R_{ms}}{2\sqrt{K_{ms}M_{ms}}} = \pi R_{ms} f_s \sqrt{\frac{C_{ms}}{M_{ms}}} = \frac{\pi f_s}{Q_{ms}} \tag{2.21}$$

This number also determines the amplitude of the oscillation near resonance which in turns is inversely proportional to M_{ms} [31].

2.3 Evaluation of sound quality in speaker reproduction

Evaluation of sound quality tackles the problem of finding the most meaningful measurements regarding the impact on listening impression. These come generally from electrical, mechanical and acoustical signals whereas listening tests provide sensation judgments in a certain listening environment, the two things competing to provide such estimation [6].

2.3.1 Frequency response and sound quality perception

What researchers claimed for a long time is that total sound-power output is the most reliable indicator of sound quality just for naturally being closer to the predominantly reverberant sound field experienced by listeners in normal rooms. Up to the late 80s there was, above all,

a non-agreement in loudspeaker amplitude response measurements when it comes to finding correlation between subjective ratings. The only common choice has been for a long time a low resolution 1/3-octave steady-state measurements. In his large study Toole [32] stressed the importance of far field measurements should in free field condition with enough frequency resolution to reveal resonance effects. In fact, all conventional amplitude-responses show irregularities within loudspeaker in terms of peaks and dips describable by a central frequency and an associated Q with their height and width. Such inherent audible departures from flatness persist at all off-axis angles [34]. The 1/3-octave in-room data, although retaining spatial resolution, could not distinguish between low and medium-high Q the former being much more audible at low amplitudes [33]. To assess their audibility, he recommended a minimum frequency resolution of 1/20- octave over the entire audio-band. Spatial smoothing of measures in the front hemisphere highlighted that effects of acoustical interference can be removed although indicators of other causes of irregularities could persist. Moreover, sound-power data alone are not capable of revealing finer details of our devices but only allow to recognize the worst speakers leaving ambiguities in ranking better ones, which is the real challenge. At the other extreme, pure anechoic on-axis response neglect information about reflected sound field. Toole noted anyway that trusting the axial response alone one probably successfully identifies a device in one of the top two categories for his study On-axis response is fundamental as it conveys much of the basic information that is revealed in other amplitude measurements still preserving much of the character that persist after off-axis spatial averaging and sound-power measurements (see Fig. 7,8,12 of [32]). A monotonical increase in the smoothness of frequency response as a function of mean fidelity ratings for all listeners was also noticed. A flat, high-resolution wide-band on-axis response seems an incontestably essential objective attribute together with extended bass. A flat amplitude response on reference axis and a flat sound-power response are also mutually exclusive; the same excellence criterion is not applicable to all measurement options. Those finding were confirmed also later in Olive's study [35] where a descending-order ranking of speakers was possible based on

preference ratings when correlated with objective measurements; highest-rated speakers were the ones with the flattest, smoothest and most extended frequency response curves while the lowest rated ones coinciding either to a lack of bass or irregularities in frequency response. In relating objective and subjective data, one must also consider the objective influences of the listening process aspects related to psychoacoustical processing and Toole's perceptual data in his work were basically limited to overall evaluations of fidelity ratings. Gabrielsonn [36], [37] investigated the relation between loudspeaker's frequency response ratings in seven subjective scales. A good reproduction was found to be characterized by high values of clarity, fullness, spaciousness, absence of extraneous sounds and by a proper balance between brightness and dullness as well as softness and sharpness. Free field, reverberation room and listening room frequency response measurements are fairly or very dissimilar among themselves for the same device if related to these perceptual dimensions. Anyway, through a frequency response analysis combined with physical characteristics of the music program, Gabrielsonn still advocates that, listening room responses reflect the stimulus to listener's ears more truthfully than others unavoidably interacting with the other variables in the reproduction system.

2.3.2 Distortion and sound quality perception

In essence, a linear distortion is a smooth frequency-response deviation for the ideal flat curve. Obviously, listening environment have some influence on audibility: test results indicated as significant effects for the perception of diffraction/reflection types of linear distortion the level, delay and absolute playback level of the sound [38]. Olive and Toole's reports [39] suggested that lower amounts of added linear distortion could be perceived if delayed in time, but the combined effect with playback level has stronger implications for which a simple frequency response graph is not enough in revealing that. Perceptual effects of linear distortion have also been addressed in [40] mimicking different linear filtering possibly introduced by transducers so that relationships between quality and frequency response irregularities can be quantified. Perceptual aspects of linear distortion translated into spectral ripples and spectral tilts in the

frequency response especially correlating with listeners' judges about "naturalness" of sound as regards such filtered signals used for the tests. When it comes to nonlinear distortions subjective perceptibility, from most of the research of the past 40 years, seems clear that conventional measurement methods such as THD alone not reveal a complete information about *perceived* distortion. Nonetheless, despite not characterizing nonlinearity in its entirety and complexity, such curves provide useful information about loudspeaker under test anyway: relationship between harmonics of even and odd orders tells about the symmetry (or the lack of it) in loudspeaker x-dependent parameters or flux modulation distortion affects the THD curve at high frequencies [13]. One reason for being THD a poor indicator is that our hearing system inevitably deals with some basic psychoacoustical aspects such as masking, not revealable through a single tone. If loudspeaker is fed with multiple input frequencies, from two upwards, sum and difference of tones become very unpleasant artefacts instead. The question, however, is a little subtler. Generally, the presence of nth order linearity exhibits the presence of at least a nth order harmonic which yield IMD in any signal more complex than the simplest pure tone. This is detrimental to the sound as IM products occur at non-harmonic frequencies and they possess the "harsh frequencies", coinciding with the difference products with frequencies lower than fundamental tones. These are poorly masked because of the non-symmetry of the auditory systems' masking curves [41]: because of stronger masking of higher frequency components, less masked tones may confusingly result more annoying only for their being naturally more audible. A recent study revealed that distortion products become typically less significant as the frequency of the signal increases because of the combined effect upward-frequency masking phenomenon and reduced excursion of speaker diaphragm [42]. The proposed method assesses the audible degradation for low-frequency speakers (<500Hz) deriving custom masking curves for specific frequencies in an anechoic half plane environment from listening tests in a previous study. Then a comparison on a critical-band basis of the FFT-transformed signal reproduced by the speaker to auditory-masking level curves was made. The result was an overall audibility distortion metric by combining audibility values for each

band. Nevertheless, such high-precision test is strictly valid to the designated reference position especially when applied to loudspeaker reproducing music signals. Distortion audibility of the same driver for other distances was not addressed as well as room effects although products and masking processes vary in level depending on acoustic attenuation.

2.3.3 Timbre and sound quality perception

Timbre is a non-spatial characteristic on which listeners' process of assessing perceived quality of sound is based. It is deemed to be an important intrinsic perceptual dimension affecting such process and related to the spectral content of sound. Roughly speaking, it is what differentiates musical sounds from one another duration, pitch, and loudness being equal. Furthermore, the frequency-dependent loudness characteristics of the ear change the perceived timbre of the stimuli at different levels [43]. About this, an attempt in quantifying such aspect was made graphically by Staffeldt in his work [44]. He transformed listening room sound pressure responses in a special plot with interesting property of its underlying area telling how much this frequency range in the spectrum is contributing to total loudness and tonal balance of the speaker. A decade later, he confirmed through his mathematical modelling [45] that timbre is one of the most important perceptual attributes of sound reproduction and a very helpful feature in distinguishing two sounds. This is highlighted also by multidimensional view of sound perception. In 1977, Grey [46] uncovered such dimensions through multidimensional scaling (MDS) of timbral similarities between several time-variant musical instrument tones. A 3D solution for interpretability was needed: first dimension is related to the spectral energy distribution, while the other two to the presence of high-frequency energy in the initial attack segment and synchronicity in the attack and decay of upper harmonics respectively. However, MDS methods do not guarantee the correspondence of one dimension to perceptual attributes and to excite all timbral differences between the speakers, a signal should include, the attack, the stationary part, and the decay part of single instrumental sounds. If, in an oversimplified way we think at sound as divided into a transient and a stationary part, by neglecting the

former, it would be even almost impossible to recognize tones played by different instruments even if spectra of the stationary parts of the tones are quite different [47]. A recent study focused on listeners' perceived differences in transient distortion investigating relations between loudspeaker electromechanical parameters and subjective scores [31]. Electrodynamic loudspeaker dynamic behavior was described in terms of lumped parameter modelling through second order nonhomogeneous linear differential equations allowing also for a quantitative mathematical interpretation for the above-mentioned two parts. Results showed that the $\frac{Bl}{M_{ms}}$ ratio of a loudspeaker is what influences the transient process affecting the playback of signals with a short attack and decay time. However, both Bl and M_{ms} seems to be well-related with attack and dynamic range but the ratio of the two not. Perhaps, a ratio of two linear lumped parameters does not simply lead enough information in assessing that.

2.4 Correlation between subjective and objective data

The problem of no direct correlation between objective parameters and perceptible sound quality, which constitutes the main purpose of our work, sinks its root early in the past. A great historical survey on the most relevant perspectives up to late 80s is given by Toole (see Section 1 of [48]). However, our interest is not only finding such correlation but investigate a predictive modelling attempt to calculate subjective ratings.

Many of earlier studies were conducted with the aforesaid underlying assumption that the 1/3-octave radiated sound power responses best characterize what listeners perceive in a room. In 1973, Rosemberg reported good correlation between such measurements in a reverberation chamber and listening tests [47]. The study was conducted with loudspeakers for normally furnished domestic spaces and he recognized frequency response, transient response, distortion, and directivity to be the most relevant physical parameter. Anyway, the author did not specify an exact model to predict his data, and his measurement method "may be improved". Still in the same times, Staffeldt performed paired comparison listening tests with trained listeners

achieving an objective rank order of loudspeakers [44]. The key idea is the assumption that some psychological dimensions are extractable from experimental data. Through a heuristic analysis, subjective data were summed up in two dimensions concerning treble and bass reproduction by grouping original subjective qualities used in listening tests. Correlations were demonstrated between the two and steady-state 1/3-octave sound power response measured in the listening room while anechoic sound pressure response specifications appeared not so much informative. After this very first attempt, in 1984 the author proposed a model for predicting a 2D graphic representation of perceived timbre [45]. A subjective frequency response can be drawn starting from a plane-wave level combining free-field and diffuse field contributions from the speaker. Such mathematical demonstration enforces Staffeldt's idea that, if two sound-reproducing systems produce the same 1/3 octave steady-state sound pressure level frequency spectrum at the ear canal entrance also the subjectively perceived frequency response is the same. Unfortunately, the study lacked a testing and verification for such a model. An in-room measurements approach was proposed also in 1991 by Gabrielsonn [37]. His overall evaluation of the 2-way and 3-way reproduction systems in terms of fidelity may be considered as a weighted combination of judgments in the different perceptual scales similarly to a multiple regression equation in statistics. Relation between frequency response and fidelity ratings can accordingly be expected to reflect the above-mentioned relations in proportion to how important a perceptual dimension for achieving good fidelity itself is. This is indicated by correlations with fidelity ratings from a previous study [36]. The general hypothesis is that other factors assumed constant, fidelity is favored by a frequency response like that indicated for clarity and spaciousness. For the two dimensions in each listening session Gabrielsonn looked for a correspondence between the two highest and lowest rated loudspeakers and corresponding free field, reverberation room and in-listening room measured frequency responses. Still, only the results obtained for the last one is in good agreement with the aforesaid hypothesis. An interesting attempt in explaining listening impression from objective measurements was made in 1990 by Klippel. He managed to explain subjective tests

results by merging relevant physical properties of the signal at the listening place and modelling key aspects of psychoacoustical processing [49]. The novelty his approach was the introduction of a perceptual-based model for *predicting* perceptual dimensions and overall sound quality. A prior statistical analysis revealed clearness, sharpness, general and low bass emphasis, feeling of space, bass clearness and brightness to be the most important factors. According to Klippel's hybrid model, the most relevant "box parameters" to account for all of them are both free-field anechoic on-axis response and the sound power response for the reverberant component of sound. Still, a 1/3 octave spectral resolution seemed enough to him. Using a loudness model, he estimated differences in loudness density between the reference and measured spectra for the programme material reproduced across each 1/3-octave center frequency. This "discoloration spectrum" information can be transformed into few basic relevant features for modelling listening impression correspondingly to the 7 above-mentioned dimensions whose subjective magnitude was predicted and weighted based on high correlations with overall quality ratings. An objective measure of "naturalness" of sound was specified in terms of a *linear equation* combining 3 orthogonal dimensions such as discoloration defects (DV), brightness defects (DH) and defects in the feeling of space (DR). Four years before Toole proposed a 2-year study [32] in which he performed 34 listening window measurements in an anechoic chamber for both front and rear hemisphere of the speaker containing all the most common forms of amplitude-response measurements. Recognizable *visual* characteristics that closely mirror fidelity ratings are present in objective data. A reliable and high-resolution *ranking* of loudspeaker performance was possible in terms of this set of comprehensive anechoic measurements. He was also able to synthetize a typical room response between 200-300 Hz and 10 kHz accounting for early-reflected and reverberant sounds by spatially averaging certain sets of free-field response data (see section 7.2 of the work). Similarly Dr. Sean Olive made acoustic measurements of loudspeakers to determine the extent to which they correlate with listening test results [35]. Loudspeakers were measured at 70 different angles around horizontal and vertical orbits to fully characterize on- and off-axis sound and remove

resonances effects and colorations in the reproduction. Anechoic measurements were analyzed in descending order for each device based on subjective preference rating and similar visual correlations between listeners' preferences and this set of graphs supported Toole's earlier findings. To accurately predict loudspeakers' sound quality, many anechoic responses, the first reflection frequency response curve, and the sound power response are needed.

Besides all the important graphic results discussed so far, the only tested model with published results was Klippel's hybrid approach. Nonetheless, no one offered a real predictive model based on objective measurements to predict listeners' up to Olive in 2004 [51], still being, to our knowledge, the only accurate and verified work of this kind existing in literature. He tested a predicting approach of loudspeaker's sound quality with 1/20-octave spatially averaged anechoic measurements; motivation was that, for over 35 years, Consumer Reports magazine (CU) recommended loudspeakers to customers based on an accuracy score related to the 1/3-octave 'flat sound power' criterion. By showing very low agreement between listeners' preference and those scores he was also able to demonstrate that such model does not accurately predict listeners' loudspeaker ratings. His work was so decisive that, after 2006, CU no longer publishes reviews based on this obsolete model. Moreover, such ratings were based on anechoic chamber measurements and they are not perceptually relevant assessment from listening tests and the fact that a low-resolution sound power information is not a desirable features was not absolutely not new even at that time [48], [35]. However, Olive developed a pioneering multivariate regression model to predicts preference ratings whose central idea is their relationship with "the mean amplitude deviation in its frequency response measured around horizontal and vertical radiating orbits". Thus, a decrease in deviations should correspond to an increase in the loudspeaker's preference rating. His model only considers linear distortions related to frequency response and the interesting thing is that all the independent variables in the model are graphical extracted features from 7 different kinds of amplitude response measurements statistically quantifying such deviations. Predictive power of each variable was inferred by looking at its correlation with preference ratings from listening

tests reported in Part I of his study [50]. Furthermore, he almost demystified. the question whether predictive models based on 1/3-octave measurements provide accurate predictions of preference compared to 1/20-octave measurements. He compared three models exploiting an anechoic predicted in-room response and two in-room smoothed 1/20- or 1/3-octave curves. Two important discoveries can be summarized in this way. Models such in-room and power ones not including direct sound as an independent variable produced less accurate prediction of sound quality and the accuracy of any model strictly depends on the frequency resolution of the measurements with no doubt. Even sound power models could give good predictions if they are based on measurement data with adequate frequency resolution but still not as good as ones containing direct sound modelled directly in a feature-based way. Olive's model lacked however variables that account for nonlinear distortion.

Traditional research, including Olive's work, follows a classical approach of quantifying subjective perception based on a set of technical measurements differently accounting for the influence of the direct, early reflected, and reverberant sounds on listeners' impression. Apart from of Klippel's work parenthesis, sophisticated perceptual-based models were devoted only to subjective quality assessment of low-bit rate audio codecs. The general idea behind this modelling approach was briefly stated in Introduction and in Section 2.3.1. In the context of loudspeakers, these models still include psychoacoustic effects associated to complex frequency-dependent radiation properties and its interaction with room. Recently a study aimed at generalizing quantification of loudspeakers overall perceived sound quality in a wide variety of environments independently on technical measurements [52]. In this context, reference signals were binaural recordings of musical fragments played over devices of which one wants to assess the quality. Recordings were collected with a Head and Torso Simulator (HATS), using the best quality loudspeakers available, in the ideal listening spot in the best quality environments. The reference signal with the highest subjective quality was compared to the degraded loudspeaker output to quantify and predict loudspeakers' perceived sound quality in normal listening. Nevertheless, this does not allow to separate completely the environment

and correlate quality results with technical design parameters. Main limitations also concern the need of further model validation, highly-dependence of model performance on HATS implementation and the non-negligible dropping of accuracy of the model with new subjective data. Last year, another interesting research direction as regards loudspeaker measurements and subjective tests appeared [31]. The correlation between a set of speakers' electromechanical lumped parameters and some attributes of sound quality was studied. Critical factors affecting transient distortion in relation to subjective scores has already been discussed in Section 2.3.1. However, when it comes into ranking different sample speakers according to Preference, Bass clearness and attributes related to subjective perception of dynamic characteristics, the adoption of the sole linear parameters still results in an over-simplified analysis.

2.5 Conclusions and remarks

A significant assessment of speakers' quality cannot be done regardless of all issues analyzed in this Chapter. Subjective results are intrinsically contaminated by many factors like human perception itself. The suggestion that in-room measurements at listeners' ears give the closest representation of perception was a logical reasoning for a long time, although our auditory system processes direct sound, early and late reflections separately [32]. A complete sound quality discernment embraces a combination of them all. Highly preferred loudspeakers showed a flat and smoot frequency response maintained uniformly both on-axis and off-axis while the shape and slope of the average direct sound plays a more important role in loudspeaker's perceived spectral balance rather than sound power. Frequency response curves alone largely predicted listening tests outcomes but any amplitude-response of the ideal kind requires additional information [51]. Assessing distortion meaningfully and unambiguously in relation to sound quality is not easy. As will be discussed in Section 3.2this problem resulted in several significant metrics along with their underlying limitations. A complete dynamic response of complex nonlinear systems cannot be extrapolated from its reaction to simple testing signals,

such as a sweeping tone and multitone distortion testing exhibits in general better correlation between objective data and distortion subjective assessment. However, simple metrics like THD are still useful for objectively compare loudspeakers of the same type. Three main classes of approaches aimed at correlate objective data with subjective ratings have been presented along with two newest ones. Klippel's contribution was historically the first example of a correctly tested work where a simple linear equation modelled such a sophisticated process. However, still 30 years ago light was shed on an important limitation: accuracy and reliability of a predictive model strongly depend on the subjective data on which it is based and on the "actual listening conditions". The most important state-of-art contribution remains Olive's work offering at the present time, a suitable feature-based regression model for predicting preference ratings as a good compromise of many considerations highlighted here. Correlation of sound quality with only small signal linear parameters seems an interesting point although speakers with similar small signal characteristics may sound different at high amplitudes [28]. Our direction of thinking goes to finding the most proper features to be employed for this work, derived from both measurements, and lumped parameters world. With the ambition stated in the introduction an attempt of a complete low-level feature-based characterization of speakers for building a predictive model and conduct a reliable correlation analysis, to which following chapters are dedicated, is highly motivated.

3. Methodology and Implementation

This chapter is devoted to the rationale behind all the main issues of our work. Three macroscopic problems are addressed on this wise. Primarily, we want to define a proper subjective test methodology for audio sensory evaluation to collect accurate data. Repeatable and reliable scientific outcomes must be ensured to represent a solid ground truth for predictive modelling and correlation analysis. Then, a significant set of parameters to relate with gathered ratings is identified. Finally, an approach to "mimic" test results with artificial intelligence from objective data is explored with an assessment of the most influential features.

An audio sensory evaluation method "*evokes* a response from assessors, which can be *measured* such that data can be statistically *analyzed* and *interpreted*" [43]. Thus, a careful experimental planning reduces the effect of disturbing variables potentially leading to noisy results. The process concerns the selection of speakers along with the way in which they are presented to listeners, the test method itself, the program material, the listening panel, and attributes to be evaluated. Listeners identifies changes in such attributes recording responses to reproduced audio excerpts. A statistical design of variables allows to control stimuli as much objectively as possible. Managed variables also include listeners location, playback level and procedures such as loudness normalization between both speakers and tracks. When it comes to a physical characterization of speakers, beyond a linear parameter-based description, we also desire to accommodate nonlinear behavior of devices altogether with perceptual aspects discussed in Chapter 2. Aim is finding the most representative set of numbers to compactly describe such phenomena. Application of standardized audio spectral features customized to our speakers' frequency response is discussed. By exploiting three-way structure of data, we can reduce

through an assessor-wise reasoning subjective results into a matrix to relate scores to two-way objective data with regression techniques. A merging procedure for an assessment refinement and a comparison with previous literature findings is also described. This involved the processing of further woofer measurements. An investigation on the linear parameters' role as regards correlation with perception deserves attention too.

In Section 3.1 we propose a methodology for devising a suitable listening test. Besides presenting stimuli to assessors, we must decide *how* to measure response as discussed in paragraph 3.1.1. This must be done in conjunction with all the test independent variables to fit into the method's framework, which is the content of paragraph 3.1.2. Some uncontrolled variables such as the order of presentation of stimuli must be monitored to prevent further inaccuracy in the data as addressed in Section 3.1.3. Section 3.2.1 deals with technicalities behind merging true far-field on-axis measurements with predicted frequency response from linear behavior. Furthermore, Section 3.2 discuss the construction of a meaningful objective dataset while Section 3.2.1 addresses more implementation-oriented issues. A way to correlate sensory and external data is described in Section 3.5. In Section 3.6 we draw some conclusions.

3.1 Criteria for listening test design

Audio quality estimation involves a combination of both perceptual and affective judgements where subjects rate the sensorial strength of specific auditory attributes or a global impression of sound. These impressions can be, in turn, affected by acoustical and even non-acoustical factors leading to systematic errors in the results, that, for a given experimental condition, are called *biases*. No listening test is completely bias-free due to issues related to all forms of scaling subjective impressions and such errors can be encountered at several stages of an audio test preparation processes. This is due to a tricky internal mapping process about how listeners translate judgments into some form of external response, occurring roughly speaking, on one of two levels, according to what is known in literature as "filter model" (Figure 3.1).

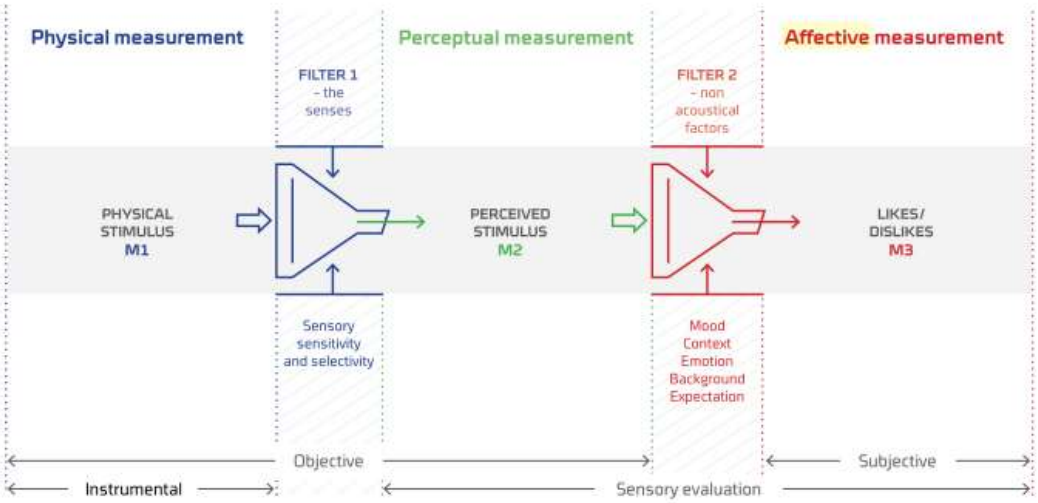


Figure 3.1 – A revision of the filter model describing all the psychological, and inner psycho-motor processes undertaken by subjects during a test. Picture from [43].

A great systematic literature-based survey of the key ones is presented in [56]. Discussion will be limited here to the risks related to specific choices for our experiment. Getting rid of them by statistical postprocessing is not easy and acting in advance for preventing a reduction in data consistency is always recommended. The initial listening test condition is defined by a set of variables including speakers themselves to handle experiment in the most cause-effect possible manner. Those are what "decide" if the outcome is a true representation of subjects' impression and can be generally divided in "dependent" and "independent" in the statistical sense, the formers being essentially assessors' answers while the latter the ones adjusted by us as researchers to which this section is mainly devoted. Here we will implicitly refer to simple rating-based tests methods where score for each system on a given scale is provided, in contrast to ranking-based and multiple/paired-comparison tests. Lastly, it is worth specifying that, differently from existing standard recommendations [57], [58], [59], the choice and the design of our test does not have the purpose of detecting impairments in the reproduced stimuli but aims at inferring sound quality and preference with special attention to the mapping between the perceptual/affective and physical domains.

3.1.1 Rationale behind the choice of test method

A wide variety of listening tests families can be framed into previous definition and literature is plenty of examples of well-established practices to assist the selection procedure depending on the test purposes. ITU.BS1283 standard [59] offers an interesting systematic guide for the task. Response format and variables are correctly combined collocating our choices in the general context provided by literature. Coarsely speaking, tests can be subdivided into:

- Discrimination methods
- Integrative methods (Preference and Affective methods, Audio quality methods)
- Descriptive Analysis (DA) methods (Consensus Vocabulary (CV), Individual Vocabulary (IV) and Associative methods)

The three categories have not to be intended strictly, and our method employs different aspects of each. A conventional quantitative descriptive analysis methodology (QDA) is used as a baseline sensory method for experiments. This falls under the CV discrimination methods driven with the goal of obtaining a *magnitude* estimate of speakers' perceptual characteristics terms of proper attributes characterizing the sound quality nature. The core is a common and agreed set of them to be used, either developed through a development (elicitation) process or selected from existing lexicons such in our case. This contrasts completely with IV methods in which each assessor employs his/her own descriptors set, potentially differing in definitions, terminology, and number. Whilst being less time expensive compared to more traditional methodologies they need advanced multivariate analysis techniques that are beyond the scope of this work. Associative methods instead, are a different category in which assessors are totally free to report what they perceive. In this framework, a large study conducted in the latest 5 years explored some rapid audio sensory profiling methods as potential alternatives to conventional QDA protocol [60]. Parallel tests using the same stimuli set have been conducted for assessing their time-, cost-efficiency and reliability. In particular, verbal methods such as Check-All-That-Apply (CATA) [61], [62] and Flash Profile [63] have been introduced for the

first time in the audio field. Although producing robust and repeatable results we did not consider them in their "pure" forms. Last choice was not suitable in our context, because it potentially generates dimensionality inconsistencies in the dataset to be related with objective measures due to the total freedom given to each assessor. CATA tests are multiple-choice questionnaire on attributes belonging to a pre-selected list on different tracks and speakers. Subjects are presented with a checkbox with which they are asked to *select* all the attributes they think apply most. This is faster than QDA, due to the ease of providing such binary answers in comparison to a scale rating operation. A disadvantage of CATA-like questions is the requirement of many assessors, typically around 50-100, not available for our experiments, because most of the a-posteriori statistical data analysis relies on the frequency of association of a word across devices under test. Moreover, such yes/no response format does not allow for a direct measurement of the evaluated sensory attributes intensity, fitting evaluation of coarser rather than subtle, product differences, thus potentially disguising discrimination between products with similar profiles [64]. A relatively new Rate-that-all-apply (RATA) [64], [65] approach appeared in the sensory evaluation scenario as a CATA generalization where subjects can also assess attributes' intensity on a proper scale in a sequential fashion. This hybrid technique is particularly interesting if some attributes are excited only in particular samples but suffers from the same issue of a big test panel size. Moreover, compared to a canonical DA, data structure is more challenging to analyze due to different ratings for each product. To our present knowledge, this possibility has never been explored in the audio field especially when to be relative with an objective dataset. It is also true that the real advantage of these methods is circumventing the common laborious steps of attribute and scaling alignment of a classical QDA experiment. This involves different phases including selection and training of listeners, attributes elicitation along with scale definitions and endpoints commonly requiring several hours of additional sessions, a potential bottleneck in this work. We retain the idea of using pre-selected attributes and a direct measurement of intensity per single attribute without asking for their "applicability" on separate scales. With respect to an *integrative* sound quality

quantification we offered an estimation of the overall preference of sound. Such assessment is usually done with a single dependent variable. Audio quality integrative methods are frequently employed in the telecommunication field for estimation of small impairments [58], or basic audio quality such as in the case of the famous MUSHRA [57] test. Listening Only Tests (LOT) methodologies described in the recommendation ITU-TP.800 [66] can be used more widely than needed as they are not exclusive to that field. Within all these, one has generally three paradigms for presenting assessors with attributes and samples. Instead of presenting all attributes to be rated on all systems at once and comparatively, we resort to a single stimulus - single attribute choice, where one attribute is rated for one sound at a time for each trial and samples are assessed in a sequential (or "monadic")order. This allowed to partially avoid a well-known such source of variance in ratings due to contextual effects. Models exploiting objective-based measurements to predict ratings, generally need to ensure that each speaker is evaluated against each other under the same "contexts" [50] as a device tends to be rated higher if judged against more weaker ones. Our method, however, does not involve any comparison but an absolute-judgments monadic test implicitly accepts a trade-off: contraction bias, i.e., the conservative tendency in using scale by avoiding extremes is a risk. Moreover, a centering bias may cause the scores to float, making the scale relative rather than absolute contaminating the projection of judgments in the form of an erroneous shift of scores [56]. Choice of proper scales and usage is discussed in Section 3.2.1. Subjective assessment of single drivers such as woofers and tweeters will be analyzed separately due to the intrinsic different nature of the systems under consideration. To not overburden listeners and avoid concentration weakening, test was designed not to exceed 30-40 minutes for a single listener, with no breaks. Independent variables have been chosen accordingly in combination with the other aspects discussed in Section 3.1.2 and 3.1.3. A suitable test room (see Section 4.1) was available for all the configuration and administration of the test. Due to listening room accessibility restrictions and the panel availability no more than two separate listening sessions for both of drivers were carried out.

3.1.2 Rationale behind the choice of independent variables

Independent known variables are discussed here, inasmuch controlled by us, valid for both the sessions. They influence the objectivity of a listening test; the goal is to ensure that these are truly responsible for data variation instead of any unknow factor and this separation naturally manifests in the Analysis of Variance (ANOVA) model of Section 3.5.1. This includes the choice of proper attributes, sound material, speakers and subjects involved.

3.1.2.1 Choice of suitable attributes

In a sensory evaluation scenario, an attribute is a label associated with sound perception, and with complex stimuli like music, several ones are expected to be excited. It thus becomes important to identify the most relevant ones in conjunction with excerpts exciting speakers. Our profiling method strongly relies on them and a deep comprehension of their meaning plays an important role. Therefore, a precise description must be formulated especially if sound quality is based on the sensory strength on proper scales for each. Good discrimination power among stimuli, unambiguity to all listeners and uncorrelation between descriptors are desirable properties. To this extent, in the CV context, we exploited an experimenter defined attribute list from an existing lexicon, i.e. a standardized profiling vocabulary to describe product sensory properties in a wide range of applications. A hierarchical visual representation for reproduced sound called *sound wheel* (Figure 3.2) has been developed recently in the audio field [68]. Often, a small number (< 10) is acceptable to discriminate the products in detail. We restricted our choice only to loudness and timbral balance to not exceed time limits. A relationship between the loudspeakers' preference and spectral ratings was uncovered in quantitative manner [50] in correlation with objective measurements. Perceived spectral balance is an important underlying factor and preferred speakers tend to have little or no tilts in the response across bandwidth, and as mean spectral rating deviates from the ideal condition preference ratings monotonically decrease. A third preference attribute was added also in our small attribute list. Complete definitions can be found in Appendix A. An evaluation of all

descriptors in absolute sense protects us against the so-called “halo bias”, describing the relative influence of other products when other attributes are evaluated for the same stimulus, which is common, for instance, with wrong anchors choices in comparison-type tests [56], [50].

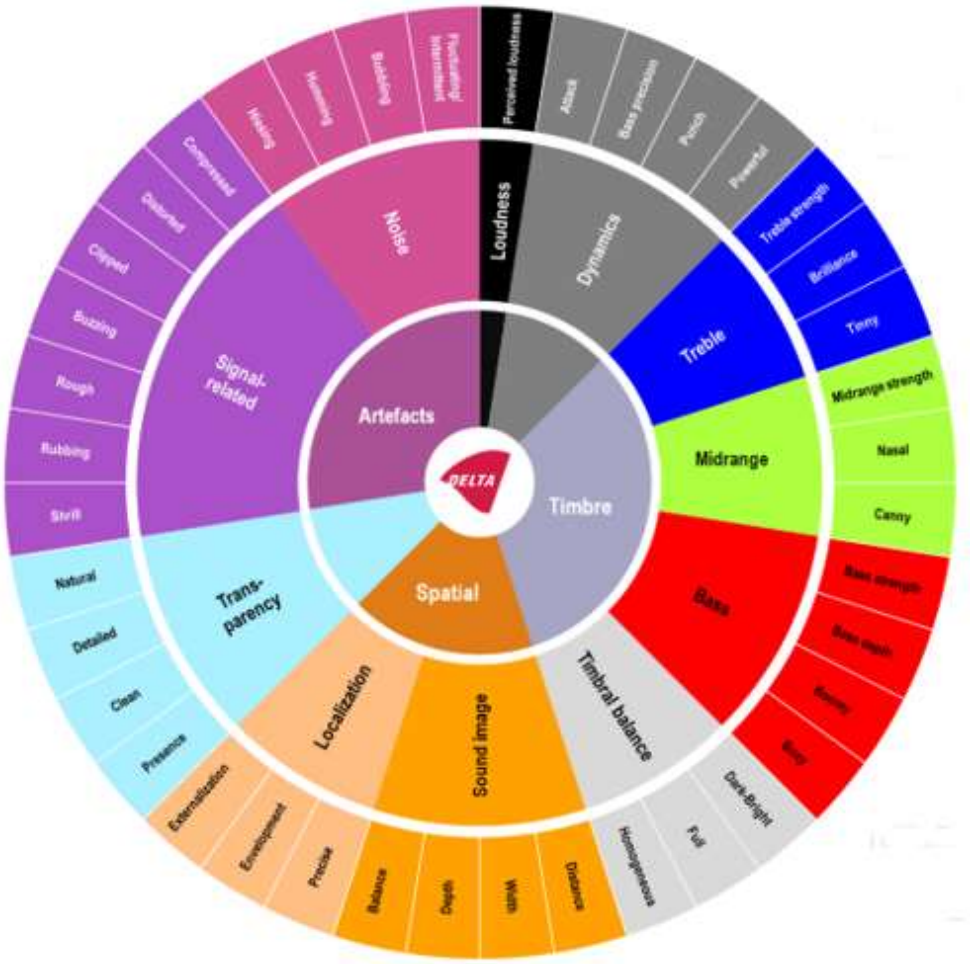


Figure 3.2 – Sound wheel of reliable, precise non-overlapping attributes for reproduced sound. Inner ring: Main groups, middle ring: Categories, outer ring: Attributes. Picture from [68].

The issue that naturally comes next, is designing scales with which subjects assign numbers for converting sensations into a concrete magnitude rating. These must be unequivocal and possess enough resolution, especially if a screen-based interface is employed like our case so a careful attaching of anchor labels is needed as well. Usage of standardized ones is recommended when possible. Sound wheels terms can fit a bi-polar scale structure in which ends are labeled

with an adjective and its opposite as anchors, referred in literature as "semantic differential". Analysis of such data is straightforward with common univariate techniques. We opted for a different Preference scale instead. It is finally yet important that assessors have instructions, scale labels, attributes, and their definitions available in their native language to reflect every subtle definitions nuances. Their meaning is very language- and culture-specific; a special problem relates to labels indicating positions on the scale. Examples of such cultural differences are discussed in [56] along with biasing implications. Complete discussion is postponed to the Graphical User Interface (GUI) implementation aspects in Section 3.2.1.

3.1.2.2 Test panel selection

An adequate level of expertise for assessors is a central problem in sensory evaluation. Their performance is usually defined with terms like *untrained*, *naïve,* or *expert* regularly employed in the audio literature. A distinctive definition of these categories does not exist, and a wide range of descriptions are found. ISO standardized a set of terms (see Table 1 of [113]). We can generally consider as *naïve* people with undefined skill level. Such consumer-representative panel is typically employed for affective methods, selection occurring for demographic requirements or random sampling. An *expert* listener is anyone with prior competence in the field and high degree of sensory sensitivity to guarantee discrimination between stimuli. A small expert panel provide ratings more analytically so is commonly used for descriptive-based testing. As a rule of thumb, 10-20 people are enough for the task. The former panel instead usually requires higher size. This is a matter of choice for any given experiment: depending on the test method itself, there may be specific requirements [43]. Typically methods like ours require expertise because of the intricate nature attributes involved along with the complex pre-selection, post-selection and training procedures to monitor expertise level [55]. The group from which subjects could be recruited was limited as well as the time scheduled for the tests and an intensive training was not possible. We believed that well-designed written instructions (see Appendix A) delivered before performing the test were sufficient to familiarize with

attributes' meaning due to their easiness and small number. To support instructions, we provided further verbal explanations prior the beginning. A user-interface acquaintance procedure was also allowed to make them familiar with scaling perception. Assessors were considered in this way well-aligned in terms of the definition of such words. The same mixed panel performed the two separated tests and it was created by using 13 Elettromedia s.r.l. employers with previous experience in the audio field and listening tests specifically plus 2 naïve high-school students. A 2 year work by Olive [35] studied differences between 12 trained listeners and 256 people with no formal training or experience in controlled tests. A significant finding was that nominally untrained listeners agreed with trained ones in rating preference as regards a ranking of the loudspeakers involved in the test, although, as a group, trained listeners gave lower ratings, as they may be harder to please. An ANOVA test on the individual ratings to infer differences in performance between the panels explained *why* they are usually employed. Trained subjects performed better than any of the untrained groups by a pretty large margin. Experience leads to significant gains in performance, so that fewer listeners are required to achieve same statistical power (i.e. extrapolation to a larger untrained population). However, both categories recognize and prefer high-end loudspeakers such accuracy characterizable, again, with a set of comprehensive anechoic measurements.

3.1.2.3 Choice of stimuli

Monophonic tests were intended in this work for focusing on subjective characterization of single drivers. A selection of the devices was made so that they were slightly different in terms of spectral aspects. Technical specifications are detailed in Appendix B. Tests are intrinsically paradoxical in this sense, due to what has been coined by Toole as "circle of confusion" [69]. Subjects form sound opinions by listening through loudspeakers without knowing how tracks are supposed to sound; valid recordings are used to assess loudspeakers, which are used in turn to make recordings and so on. The only way to escape the loop is to combine that process with objective loudspeaker measurements. However, our work aims at finding a link between

the two faces of the coin and not relying on tests in absolute sense. If specific frequency areas must be tested, spectral characteristics are one of the primary factors determining the timbre of instruments [55] and a representative selection of them must be included. Common categories of musical tracks such as orchestra, pop, blues etc. reveal significant differences in relation to the frequency-dependent average power distribution [70]. It is also important to consider these categories in relation to the subjects' background to avoid some bias because of unfamiliarity with a certain genre. We selected 3 excerpts taken from the tracks listed in Table 3.1 in a digital 44.1 kHz wav stereo file whose channels have been summed for the reproduction ensuring low background noise and avoiding clipping artefacts. Their length must also evoke a consistent response, but at the same time not leading to differences in perception between the same sample. Usually judgments are biased toward the most recently auditioned part leading to a wrong audio quality estimation. This *recency effect* relates to the psychological dominance of short-term memory over long-term one [56].

Table 3.1 Tracks involved in the test; samples came from easily accessible sources and represent common choices.

Program code	**Artist, Track and Album**
DP	Daft Punk, Get Lucky, Random Access Memories (2013)
EC	Eric Clapton, So Tired, Back Home (2005)
NJ	Norah Jones, Sunrise, Feels Like Home (2004)

Our music material includes an example of a rock piece, a pop song with a female voice in evidence and a basic pop/funk commercial track. After a process of expert listening, we believed such sounds to be critical for exciting the interested attributes and reveal differences among drivers while stressing them. All the excerpts guaranteed enough bandwidth, midrange clarity and high dynamic range between quiet and loud parts. Concurrently, instruments playing are easily recognizable which is an interesting feature to reveal spectral balance defects. NJ is an example of a well-portrayed female voice with an excellent timbre to test higher frequencies. DP is a well-engineered example of mastering. Bass is a full, warm, and clean

backbone to the track, retaining crispness of other instruments and dynamics are compressed enough for a clear-sounding mix in which melodic contributions are pushed forward. This is especially useful on smaller drivers. EC is a high-standard production choice with a differentiated bassline from the lead guitar to "touch" a stimulating low-end of the audio band.

3.1.3 The design of the experiments (DoE)

There are some independent variables not directly involved in the research question such as loudspeakers' position in the room, loudness equalization, etc. While these are held at a constant level and not part of the statistical planning (see Section 4.1.1), some others need. The DoE, sometimes called *statistical experimental design*, deals with organizing all of them along with the conditions to prevent test results to be affected by systematic disturbing factors. This is done by converting as much as possible all unknown variables to randomized ones, which unpredictably influence scores but still in a statistically modellable way, the residual variance inherently mirroring in the model. Administration involves the correct presentation order of stimuli, number of presentations per session and number of subjects and replications needed to achieve a certain resolution in statistical analysis. All other factors[2] being already fixed, the first aspect in the list deserves attention. In this case DoE concerns the "allocation of stimuli" design (in combination to "treatment design"). Stimuli presentation order hide bias effects too as perceived quality of a speaker highly hinge on others being evaluated before it; if a low-quality speaker is rated right after a high-quality one, the former scoring results lower than the one eventually obtained if preceded by a device of comparable quality. Moreover, a wrong presentation lead to *sequential contraction bias* [56], i.e. listeners expect what is about to come next with a fixed order. Especially in a monadic scenario, these effects become relevant. Sequential dependencies are eliminated by using either truly random presentation orders or

[2] In statistical terms, any independent explanatory variable manipulated by the experimenter is also called "factor", whose possible different values are called "levels" and combination of the two are called "treatments".

by applying a specific scheme. However, the global level of context (specific experimental setting etc.) is still present even with the second choice [71]. About treatment design, within the full factorial framework of this work, all possible combinations of independent variables are included, which is optimal from a merely statistical point of view. Allocation of stimuli design relates instead to how these are presented to each assessor. In a within-subjects design alternative considered here, all assessors in each session are presented with all combinations specified by the treatment design, i.e. the maximum number of speaker x track x attribute possible. Obvious limitation is again time; duration of the test really matters in this context, and previous choices also accounted for this fact. Presentation order was not completely randomized. A sequence can be chosen by *blocking*[3] each single attribute together for all samples meanwhile focusing listener attention on response variable [43]. Many different types of blocking designs exist, such as Latin square, whose name comes from an old riddle where letters must are arranged in a square matrix each symbol appearing only once in each row and column ensures that each letter is evenly followed by another letter [72]. However, we are yet not completely protected against order effects; to totally avoid contraction bias, it would be perfect to guarantee also that loudspeaker A was not always evaluated after loudspeaker B, loudspeaker C after D. This is commonly obtained with Williams squares (Balanced Latin squares design) [73], a particular arrangement in which presentation order and subjects are the two blocking restrictions. Algorithmic rules for construction are outlined in Figure 3.3. Though, we could not take advantage of this solution in its pure form. Each of the 15 assessors completed 1 uniquely ordered repetition of the design, that is a specific sequential order of all the 36 combinations of 4 speakers x 3 tracks x 3 attributes to be rated. This is a non-square structure being the number of total subjects different from the number of total treatments. Our half-way solution complied with a $n = 36$ Latin Square design where the exemplificative

[3] Blocking generally refers to arranging variables in similar groups while losing the possibility of testing interactions among them. Non-testable effects can be predetermined, and this is where design rules come into play.

integer numbers of Figure 3.3 corresponds in our case to a uniquely coded triad related to the i[th] speaker evaluated for the j[th] track on the k[th] attribute combination ($i = 1,2,3,4$, $j = 1,2,3$ and $k = 1,2,3$). Starting from first row (1,1,1), (1,1,2), (1,1,3), (1,2,1) etc. all subsequent rows in the square are circular shifted by one position towards right.

Latin Square						
1	2	⋯	⋯	n-1	n	
2	3	⋯	⋯	n	1	
⋮	⋮	⋱	⋱	⋮	⋮	
n-1	n	⋯	⋯	n-3	n-2	
n	1	⋯	⋯	n-2	n-1	

Balanced Latin Square						
1	2	n	3	n-1	4	⋯
2	3	1	4	n	5	⋯
3	4	2	5	1	6	⋯
⋮	⋮	⋮	⋯	⋯	⋮	⋮
n-1	n	n-2	1	n-3	2	⋯
n	1	n-1	2	n-3	3	⋯

Figure 3.3 – Construction rules for n x n Latin Square (left) and Balanced Latin Square (right) designs

Then, 15 rows related to the presentation of stimuli for each of the subjects were sampled in the vertical direction with at approximately uniform spacing from the full 36x36 matrix including first and last row. Every subject did not start from the same speaker, and for the subset of listeners who did, nobody rated the same attribute on the same track as a first stimulus in the sequence. This was implemented in a lookup table form in the GUI back-end. This deterministic solution still ensures that each treatment appears once and only once after each of the other ones and should guarantee a high level of independence in the assessor's rating from the actual sequence. No repetitions of the test happened.

3.2 Objective Measurements and Features

All-important outcomes highlighted in Section 2.3 and 2.4, such as Toole's work [32], [48] must not be intended as amplitude response being the only important measurement. Through this single dimension we can simply have the best insight about speaker behavior in another. To quote the author himself, "given *both* the amplitude response information and additional data on the important linear and nonlinear distortions, a designer is in the position to recognize

performance of the highest caliber with little uncertainty". Typical measurements categories concern linear input-output relationships determined by a logarithmic sine sweep technique [80] such as frequency responses plus harmonic distortion estimates or mechanical measures of cone displacement for voltage-excited drivers. Moreover, both linear and non-linear model parameters can be measured/simulated through a Klippel measurement system with different setups. Such measurements were possible for all drivers thanks to Elettromedia s.r.l. facilities. Construction of objective dataset includes features derived by them or measured alone.

Firstly, we included all linear lumped parameters discussed in Section 2.2.2. In addition to them, we considered two easily manufacturer-available numbers:

$$L_{nom} = L_m + 10\log_{10}\left(\frac{Z_{nom}}{R_e}\right) \tag{3.1}$$

$$L_m = 112.2 + 10\log_{10}(\eta_0) \tag{3.2}$$

being the nominal and effective sensitivity respectively, where η_0 is the reference efficiency (2π radiation using R_e). The former is the SPL in dB developed on the nominal impedance of the loudspeaker Z_{nom}, typically 4Ω, 8Ω or 16Ω, for a given input, i.e. a convenient reference loosely describing speaker impedance over much of the audio band. However, linear parameters are senseless if the frequency range where the device response is concentrated the most goes beyond their validity range. This is what happens with tweeters which lacked all this characterization. Four figures were considered instead such as Z_{nom} as well, low and high frequency response limits f_{low}, f_{high} and Sensitivity expressed in dB SPL, measurable with a microphone 1m away from speaker whose level is tuned so that amplifier delivers 1W of power.

Despite appealing linear parameters alone are meaningless in large signal domain too. They only show a small part of the big picture and are interpretable as mean values of the corresponding parameter x-dependent curves weighted by displacement probability density function (pdf) [10]. In this scenario, measuring speaker symptoms is an easy task but explaining the transition from effect to cause is what interests the most. This is the context

in which our feature choices fall with nonlinearities description through polynomial functions described by Eq.2.15, Eq.2.16 and Eq. 2.17 truncated up to the 4th order. Merit et al. showed [16] that such order is enough to reach ideal conformity and satisfactory accuracy. This leads to a total of 15 numerical coefficients, that is L_{xi}, Bl_i and K_{ms_i} with $i = 0, \dots, N = 4$. Such parameters can be dynamically measured (or simulated with FEA) and consent a high-precision prediction of nonlinear symptoms revealing dominant sources of distortion, problem for which harmonic analysis is not exemplificative at all. Large signal modelling is non-meaningful for tweeters due to their lower excursion range and dataset lacked those parameters as well. Relationships between nonlinearities shapes and different distortion components order also exist. For instance, a symmetrical nonlinearity generates high 3rd-order distortion components with no DC-displacement or, as long as nonlinear system is stable, an asymmetrical Bl curve is likely to excite 2nd-order order harmonic and IMD [17]. Specific metrics quantify symmetry of these curves providing an information directly linked to loudspeaker assembly topology. Asymmetry of $K_{ms}(x)$ is assessed with a single number exploiting negative and positive peak voice-coil displacement values, or a symmetry point can be defined for the $Bl(x)$ characteristic [29]. However, we believed such features controversial for our scope as calculated by fixing some thresholds and this may be a source of uncertainty if to be related to subjective judgements.

Olive's work [50] demonstrated overall perceived Spectral Balance rating to be strongly correlated with loudspeaker's measured listening window, giving credence to the importance of the direct sound in timbre perception. As the same attribute was involved in our tests performing feature extraction on direct sound measurements is highly motivated. He also provided an interesting rationale behind the choice of some numbers quantifying amplitude deviation from flatness. To accomplish this task, we chose selected features for both training sets among the ones typically employed in MIR contextualized to our on-axis response and harmonic distortion measures. These are the curves we want to reduce to a set of very significant numbers. Basic ones are derived from the MPEG-7 low-level descriptors (LLDs) to

observe the desired characteristics from different viewpoints. Descriptors chosen included three spectral moments (Centroid, Spread, Kurtosis), other spectral indicators (Roll off, Flatness), along with Spectral entropy, Spectral Crest, Spectral Slope, Spectral Decrease [81] for the on-axis response. Three further features originally designed for sound power response has been also considered from Olive's work to be applied to our synthesized data as described later. Along with LLDs, two vectoral features were present such as Spectral Contrast [82] values on six different sub-bands applied to on-axis response and the Spectral envelope on 10 different sub-bands applied to THD curves. A Spectral Flatness value was considered for harmonic plots too. A concise mathematical description is given in Section 3.2.1.

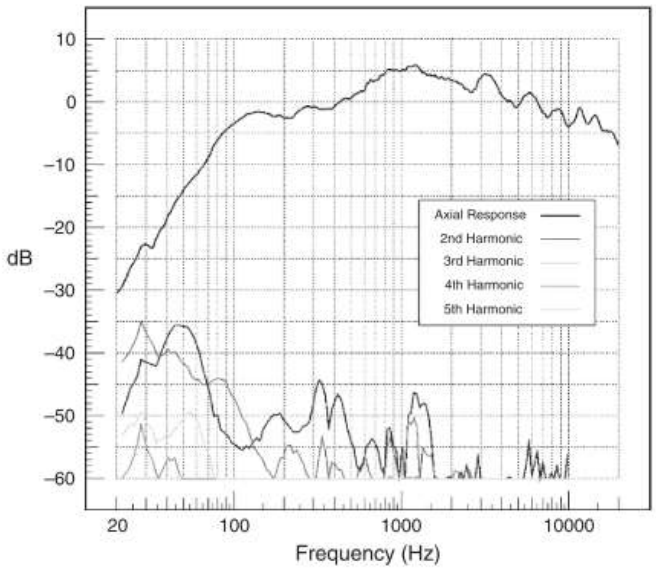


Figure 3.4 – On-axis pressure amplitude and harmonic distortion. Picture from [83]

THD is a frequency-dependent metric by definition and frequency-dependent curves can be plotted. Even though not depicting nonlinearity in its totality this lower-resolution assessment can still be useful, especially if combined with the 15 coefficients mentioned before. Speakers with highest 2nd and 3rd order distortions tend to present the worst frequency responses but also other factors are accountable for listeners' preferences. There is no cause-effect relationship between a lower THD percentage and a better performance of devices [50].

Historically, a distortion evaluation through well-conducted subjective tests resulted in many perceptually significant metrics. We did not include such numbers in the dataset for reasons explained in a while. A correlation study between static nonlinearities and subjective perception [84] by Geddes and Lee evaluated psychoacoustical relevance of their metric [85] in comparison to traditional THD and IMD. Despite adequately characterizing what is audible, calculation of this number is based on the $F[\,\cdot\,]$ curvature for a *general* non-linear system. The metric is applicable only for nonlinearities with well-known mathematical description and is optimal for predicting ratings at low-intermediate levels. If specialized to loudspeakers case, a multiple-band extension is needed to get a frequency-dependent characterization of perceived distortion. Such limits and approximations were confirmed also some years later by a study on direct radiation loudspeakers [18]. Again, due to masking frequency dependence, much of loudspeakers distortion is below the threshold in the metric validity range as low-level musical signals are poor maskers, and those effects are not modeled. A more sophisticated coefficient called MTND (Multitone Total Nonlinear Distortion) also exist [13] averaging all harmonic and IM components from a multitone testing in a frequency domain sliding window. This produces a curve whose value at one frequency accounts for distortion level and "density" in its proximity. Even with being a useful visual tool for nonlinear distortion assessment, no search for correlation with subjective audibility was shown. In the same vein of multitone excitation-based metrics Tan et al. [54] developed a high-correlated metrics with sound quality ratings of even in real cases of transducers' dynamic nonlinearities with music testing signal. The correlation is non-linear, and such curve fitting would be strongly dependent on different stimuli and systems, so that they cannot be considered a complete alternative. A complete assessment distortion per sè is however beyond the scope of our work.

3.2.1 Spectral descriptors analysis

The idea behind applying spectral features to speakers' responses is that, as for statistical moments, these numbers are computed considering spectrum as a distribution of its amplitude

values interpretable as probabilities to observe frequencies. In general, for a discrete set of points, a nth order moment about a point c can be defined as $M_n = \sum_i (x_i - c)^n f(x_i)$. Moments about the center, for which c = 0, are called "raw moments being the ratio $\frac{M_1}{M_0}$ the mean value with $M_0 = 1$ if $f(x)$ is a pdf. Moments of the mean, i.e. "central moments" can be defined as:

$$\mu_n = \sum_{i=1}^{k} \left(x_i - \frac{M_1}{M_0} \right)^n f(x_i) \tag{3.3}$$

They are invariant to translation, and associated with the shape rather than function location in the domain [93]. Commonly, higher-order ones can be used to characterize the asymmetry and peakyness of our frequency curves. For instance, Spectral centroid indicates the barycenter of the spectrum as a weighted mean:

$$SC = \mu_1 = \frac{\sum_{k=b_1}^{b_2} f_k S_k}{\sum_{k=b_1}^{b_2} S_k} \tag{3.4}$$

where S_k is the spectral value at frequency bin k, and f_k is the frequency in Hz corresponding to bin k and b_1 and b_2 are the band limits, in bins over to calculate the feature. They are set to the full audio band edges. Same notation holds for all the following definitions. Spectral spread comes from a special case of the order-p-order spectral bandwidth with $p = 2$:

$$SS = \mu_2 = \sqrt{\frac{\sum_{k=b_1}^{b_2} (f_k - SC)^2 S_k}{\sum_{k=b_1}^{b_2} S_k}} \tag{3.5}$$

This represents the average spectrum deviation around its centroid, commonly associated with signal bandwidths and used as an indicator of the dominance of a tone over noise-like signals. Spectral Kurtosis is computed instead from a normalized 4th order central moment:

$$SK = \mu_4 = \frac{\sum_{k=b_1}^{b_2} (f_k - SC)^4 S_k}{(SS)^4 \sum_{k=b_1}^{b_2} S_k} \tag{3.6}$$

It measures the non-gaussianity of a spectrum around its centroid or, conversely, is a peakyness indicator with lower values appreciable for us. Another useful measure of magnitude noisiness is Spectral Flatness, that is, the ratio between spectrum geometric and arithmetic means:

$$SF = \frac{\left(\prod_{k=b_1}^{b_2} S_k\right)^{\frac{1}{b_2-b_1}}}{\frac{1}{b_2-b_1}\sum_{k=b_1}^{b_2} S_k} \tag{3.7}$$

The last fourth low-level basic MPEG-7 descriptor is Spectrum Envelope. As originally defined it represents is the short-term power spectrum of a signal with a logarithmic frequency scale in several bands [94]. The sum of coefficients in the bth band gives the coefficient for that frequency range. A variant from MPEG7 standard with FFT coefficient magnitude instead of power spectrum has been considered here [95]:

$$SE(b) = \sum_{k=K_b^{low}}^{K_b^{hi}} FFT(k), \qquad \left(1 \le b \le B_{in} = \frac{8}{r}\right) \tag{3.8}$$

with $FFT(k)$ the magnitude spectrum coefficients and the bth band lower and upper edges integer frequencies corresponding to $K_b^{low} = 62.5 \cdot 2^{(b-1)r}$ and $K_b^{hi} = 62.5 \cdot 2^{br}$ respectively. These depend in turn on the resolution index of the frequency bands, taking some discrete values corresponding to a fraction of octave $r = 2^j$, with $-4 \le j \le +3$. The lower r, the greater the resolution in each sub-band. Our choice was $r = 1$, so that 8 scalars correspond to the 62.5Hz-16kHz frequency range. Calculations were repeated then for the two additional 20Hz-62.5Hz and 16kHz- 20kHz bands for a total of 10-dimensional SE feature in the full audio-band. The Spectral Roll-off is defined as the frequency below which a certain percentage of the accumulated spectrum magnitude is concentrated (85% in our case):

$$\sum_{k=b_1}^{K_{roll}} |S_k| = p \sum_{k=b_1}^{b_2} |S_k| \tag{3.9}$$

where K_{roll} is the frequency corresponding to roll-off frequency. We resorted to another two indicators of spectrum peakyness too. The first is Spectral Entropy which follows classical Shannon's definition applied to our spectra:

$$SEP = \frac{-\sum_{k=b_1}^{b_2} \log S_k \, S_k}{\log(b_2 - b_1)} \tag{3.10}$$

This offers a description generally corresponding, for an extremely flat curve, to a situation of maximum uncertainty concerning the output of the random variable of probability mass

function, thus maximizing entropy [96]. Spectral crest measures instead the ratio of the maximum of the spectrum to its arithmetic mean:

$$SCR = \frac{\max(S_{k\varepsilon[b_1,b_2]})}{\frac{1}{b_2 - b_1}\sum_{k=b_1}^{b_2} S_k} \tag{3.11}$$

with a lower values indicating a more noise-like behavior. The Spectral Slope represents the amount of decreasing of the spectral amplitude and is basically computed using a linear regression over the spectral amplitude values:

$$SSL = \frac{1}{\sum_{k=b_1}^{b_2} S_k} \frac{K\sum_{k=b_1}^{b_2} f_k S_k - \sum_{k=b_1}^{b_2} f_k \sum_{k=b_1}^{b_2} S_k}{K\sum_{k=b_1}^{b_2} f_k^2 - \left(\sum_{k=b_1}^{b_2} f_k\right)^2} \tag{3.12}$$

This is linearly dependent on the spectral centroid. As most of the descriptors presented here, Spectral Decrease is calculated as described in [81]:

$$SD = \frac{\sum_{k=b_1+1}^{b_2} \frac{s_k - s_{b_1}}{k-1}}{\sum_{k=b_1+1}^{b_2} s_k} \tag{3.13}$$

It represents the amount of decreasing of the spectral amplitude while emphasizing lower frequencies slopes. Jiang defined another interesting vectoral feature [82] estimating strength of spectral peaks, valleys and their sub-band difference providing a Spectral Contrast feature:

$$SCTR_k = Peak_k - Valley_k \tag{3.14}$$

Contrast is estimated from each octave k[th] sub-band, $k \in [1,6]$ from 180Hz to 11kHz with $Peak_k = \log\{\frac{1}{\alpha N}\sum_{i=1}^{\alpha N} x'_{k,i}\}$ and $Valley_k = \log\{\frac{1}{\alpha N}\sum_{i=1}^{\alpha N} x'_{k,N-i+1}\}$ after sorting FFT vector of the kth sub-band, in a descending order (e.g. $x'_{k,1} > x'_{k,2} > \cdots > x'_{k,N}$). Neighborhood factor α is set to 0.02 from experimental considerations.

From the merged frequency responses described in next Section 'AAD_ON', 'LFX', 'LFQ', 'SM' features from Olive's model [51] were calculated. AAD feature calculation applied also to our measured on-axis curve. These are the useful statistics quantifying loudspeaker

frequency response flatness departures. The Absolute average deviation (AAD)is defined as follows:

$$AAD = \frac{\sum_{Band=100Hz}^{Band=16kHz} |y_{REF} - y_{band,n}|}{N} \ [dB] \tag{3.15}$$

The value in the nth band is calculated from a reference level y_{REF}being the mean amplitude between 200-400 Hz because of an observation made on his test results. Listeners exhibited anchored spectral balance judgments in this band. Calculation of Spectrum Smoothness (SM) is mathematically related instead to the problem of finding an ordinary least-square (OLS) regression fitting, in our case, to frequency data. This generally means finding the best line $\hat{Y}_i = bX_i + a$ such that a sum of squared errors loss-function $\sum_i \left(Y_i - \hat{Y}_i\right)^2$ is minimized on a set of points (X_i, Y_i) with $\hat{Y}$ being the predicted value Such a problem has a well-known closed form solution for both the intercept a and slope b parameters. Besides these two, the Pearson's coefficient of determination r^2 plays an important role and can be interpreted as the ratio between the $\hat{Y}$ variance to the Y- total variance describing the goodness of fit:

$$r^2 = \frac{Var(\hat{Y})}{Var(Y)} = b^2 \frac{Var(X)}{Var(Y)} = \left(\frac{S_{xy}}{\sqrt{SS_x SS_y}}\right)^2 \tag{3.16}$$

S_x, SS_y and S_{xy} are common notational shorthand for the X,Y sum of squares and cross-products which are well-known statistical formulas [97]. SM coincides with the r^2 coefficient of the least square error regression line that best fits our frequency curve in the 100Hz-16 kHz range. By applying $\bar{X}$ and $\bar{Y}$ definitions in Eq. 3.26 and collecting out a $\frac{1}{n^2}$ term:

$$SM = \left[\frac{N\sum_{i=1}^{N}(X_iY_i) \ - \sum_{i=1}^{N} X_i \sum_{i=1}^{N} Y_i}{\sqrt{\left(N\sum_{i=1}^{N}(X_i)^2 - \left(\sum_{i=1}^{N} X_i\right)^2\right)\left(N\sum_{i=1}^{N}(Y_i)^2 - \left(\sum_{i=1}^{N} Y_i\right)^2\right)}}\right]^2 \tag{3.17}$$

where N is number of data points used in the regression. This is the equation we implemented, with a natural-log transformation applied to the spectrum X_i (Hz) to linearly space points. SM values ranges from 0 to 1, larger values representing smoother curves. Low-frequency extension (LFX) and Low frequency quality (LFQ) are related. The former is:

$$LFX = log_{10}\left(x_{SP_{-6dB}}\ re{:}\ \bar{y}_{LW}(300Hz - 10kHz)\right) \tag{3.18}$$

This is the $\log_{10}$ of the first frequency below 300Hz in sound power curve -6dB relative to a mean level measured in listening window between 300Hz-10kHz. LFQ is then defined by:

$$LFQ(dB) = \frac{\left(\sum_{Band_{SP}=LFQ}^{Band_{SP}=300Hz} |y_{LW} - y_n|\right)}{N} \tag{3.19}$$

with y_n the level within each n band of sound power curve, from the lowest frequency defined by LFX up to 300 Hz. We assume resolution of such bands to be 1/20 octave as well. LFQ is intended to quantify deviations over the bass region between the cut-off and 300 Hz.

3.3 Merging Far field and Near field data

This section is dedicated to a strategy for gaining an enlarged speaker dataset with respect to Section 3.2 to add an interesting perspective on our results through a comparison with Olive's model results. Sound power measurements were replaced with the same set of measurements described here, with legit approximations in the assessment. A *quasi-anechoic* technique is employed for synthetizing a full-audio band frequency response for 35 different woofers including the 4 used in the tests. Dataset serves also for counter-validating linear parameter correlation analysis. At first, the context in which our operations fit is provided concisely describing most important aspects of the technique, then a strategy for merging is defined.

3.3.1 Near-field and Far-field measurements

Far-field on-axis anechoic measurements are somehow intrinsically limited; at low frequencies room dominates what we hear due to resonance modes, while at higher ones, absorption phenomena are non-negligible. Nowadays, accurate data can be calculated through proper software to obtain a complete response combining far and near field transfer functions. Quasi-anechoic techniques work by windowing speaker impulse response to remove early reflections effects, implicitly taking advantage of the equivalence of a linear system's frequency and

impulse responses. Obtained far-field response is convoluted with the Fourier transform of that window. This fails however at low frequencies where window is too short for providing enough frequency resolution. The finite time-range of observation creates a limit below which data is non-valid, and plotted transfer function is interpretable only as an FFT artefact. The lowest extractable frequency is a sine wave whose period coincides with the reflection-free analysis window time limit itself. To overcome such inaccuracy, near-field measures comes into play. A universally accepted method to get low-frequency data without anechoic chambers is the Keele approach [9]. Results need amplitude scaling and postprocessing techniques account for this offset. Assuming a speaker as a flat circular piston of radius a in an infinite baffle, the on-axis peak pressure magnitude p_R can be derived analytically:

$$p_R = \frac{2\rho_0 c U_0}{\pi a^2} \cdot \sin\left[\pi \frac{a}{\lambda}\left(\sqrt{\left(\frac{d}{a}\right)^2 + 1} - d\right)\right] \quad \begin{matrix} \overrightarrow{ka < 1} \\ d \gg a \end{matrix} \quad p_F = \frac{\rho_0 c k U_0}{2\pi d} \tag{3.20}$$

with $k = \frac{2\pi f}{c}$ being the wavenumber and c is the speed of sound (346.1 m/s @25°C). Previous equation depends on the measuring distance d and the volume velocity U_0 caused by the piston: in far-field, distant from the piston and for low frequencies, it converges into a useful expression for peak axial pressure p_F. In near field, pressure p_N at low frequencies is:

$$p_N = p_R(d \approx 0) = \frac{\rho_0 c U_0}{\pi a^2} \cdot \sin\left(\frac{ka}{2}\right) \quad \overrightarrow{ka \leq 1} \quad \frac{\rho_0 c k U_0}{\pi a} \tag{3.21}$$

Combining Eq. 3.20and Eq. 3.21 we can notice that:

$$p_N = \frac{2d}{a} p_F \leftrightarrow H_{FF}(f) = H_{NF}(f) - 20\log_{10}\frac{2d}{a} \ [dB] \tag{3.22}$$

Besides being a great advantage for practical measurement issues, Eq. 3.3 states that under above assumptions valid inferences can be made about the low-frequency far field anechoic operation of drivers from non-anechoic nearfield measurements. The relationship is a simple proportionality law independently on frequency and the environment into which the driver radiates[9]. Contrarily, in break up region, pressure waves from different diaphragm areas arrive at the microphone out of phase causing near-field cancellations. This naturally explains

an upper limit in terms of diaphragm diameter [114]. This magnitude offset simply translates to transfer functions where subscripts FF and NF denotes far-field and near-field, respectively. For drivers usually installed in large enclosures at higher frequencies, when wavelength approaches driver dimensions, woofers no longer radiate spherically (4π solid angle) in near field because baffle no more ideally acoustically infinite. This is what explains somehow a transition to a hemispherical radiation (2π solid angle) for an overall mean sound pressure twice the level (or 6 dB higher). In his work, Keele explicitly considered this scenario for *all* its derivation possibly leading to an overestimated result for the low part of the response.

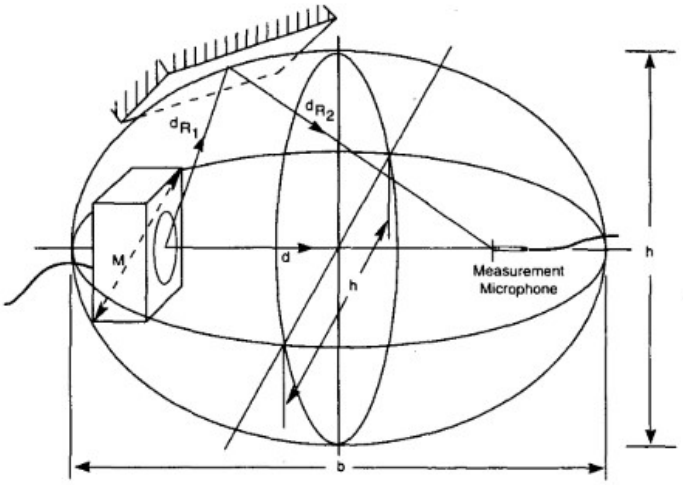


Figure 3.5 - Loudspeaker, measurement microphone, and nearest reflecting surface define an ellipsoid within which free field measurements can be performed using time-selective techniques. Picture from [74].

Independently on this, theoretical frequency upper bound of near-field approach validity being regulated by the most significant dimension M of loudspeaker enclosure. Struck and Temme work [74] describe a full-range response derived with quasi-anechoic measurements. These usually involve time-selective techniques, in which, as a rule of thumb, microphone must be placed at $d > 3M$. Constant speed of sound allows to separate direct sound wave part from reflections arriving later at the microphone due to a difference in the path $d - (d_{R_1} + d_{R_2})$ experienced. Such simple modelling explains the valid time range T to exclude reflections. Assuming $d_{R_1} = d_{R_2}$far-field measurements provide useful data at frequencies greater than:

$$f_L = \frac{1}{T} = \frac{c}{2d_R - d} \tag{3.23}$$

Time T is nothing but the window length applied to impulse response also compensating for the intrinsic delay of sound to reach microphone, i.e. time needed for the first spike to appear. In fact, this is not an actual part of the speaker response. These considerations hold independently on the window shape and for any analysis where the difference in path length appears on the time available for a free-field measurement.

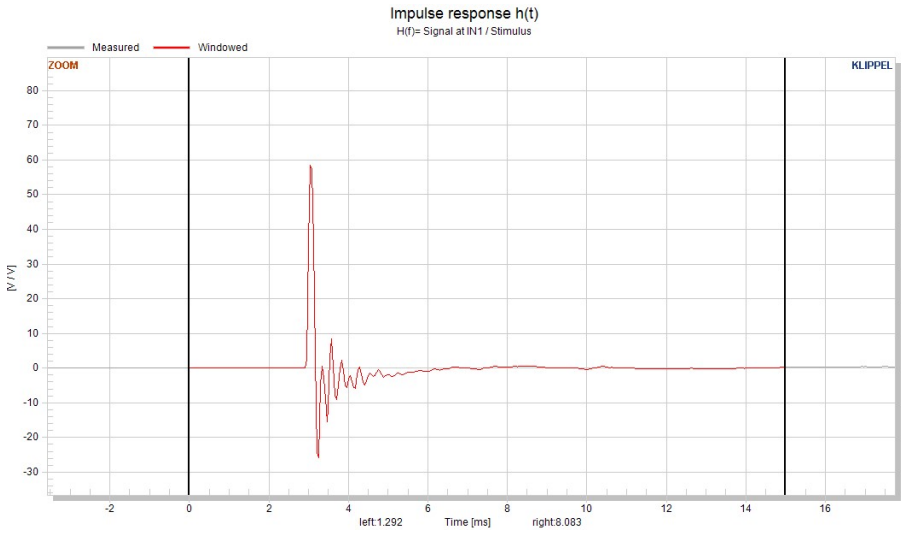


Figure 3.6 - Example of a windowed impulse response of a woofer. Picture from Klippel dB-Lab 210.610.

Near-field measurements for single drivers assume the positioning of microphone as near to the center of the diaphragm as possible. However, they suffer from a theoretical frequency limit simply given by the threshold for pistonic range behavior of the cone:

$$f_{ka=1} \leq \frac{c}{2\pi a} = \frac{c}{\pi M} \cong \frac{11016.7516}{\mathrm{D}\ [cm]} \tag{3.24}$$

The upper bound can be expressed in terms of driver diameter $D = 2a \approx M$ for convenience. Klippel dB-Lab RnD Analyzer Software [75] allowed for the truncation of the impulse response as a merely post-processing operation. Theoretical bounds can be calculated from drivers' specifications in advance. The former needs a simple inspection of the impulse response window, while, for the latter D is retained from Linear Parameter Measurement (LPM) properties.

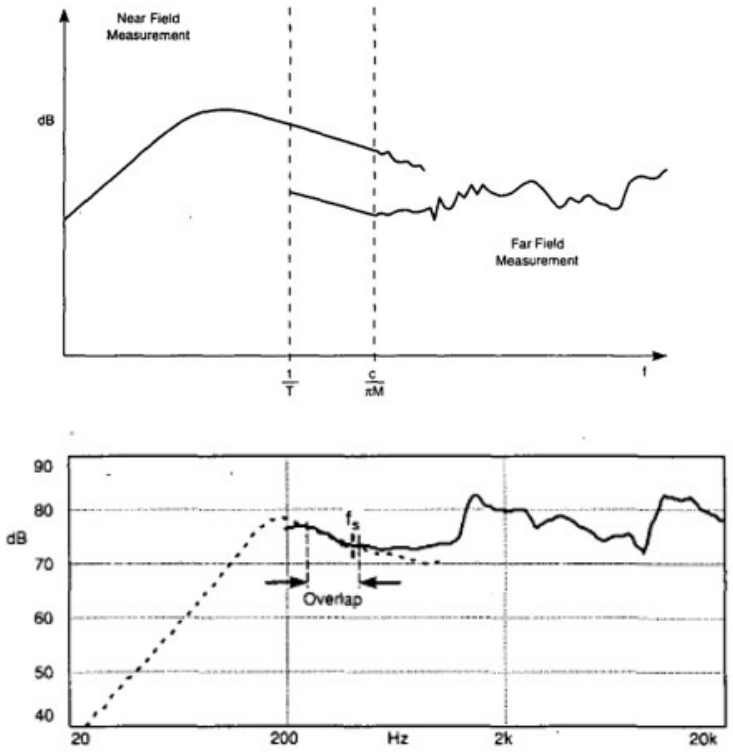


Figure 3.7 - Graphic illustration near field and far field measurements combining. Top; Near field and far field measurement before magnitude compensation. Bottom; overlap region after rescaling. Pictures from [74]

Usually left cursor limit for impulse response windowing is not aligned with the main peak but left to almost a zero value on the left to avoid small microphone positioning errors having a dramatic impact on the frequency response. Right cursor limit of the window approximately coincided with the first reflection instead. Software internally derives the far-field response from that portion where T is the difference between such edges.

3.3.2 The merging procedure

From previous discussion it seems clear that curves are combined by shifting the near-field response down in level to far-field one, possibly without altering it, assuming it was calibrated. Splicing operation assumes that, for a sufficiently large room, an overlap frequency range where both near and far-field measurements are valid exists [76]. A transition frequency f_{sp} must be selected and main concern is choosing a suitable one. Frequency data above and below

is then eliminated it so that responses no longer overlap and can be complex added to get a single dataset. Typically, setting f_{sp} as high as possible for preserving low-frequency resolution seems a good point. Still, it must lie within the region:

$$f_L \leq f_{sp} \leq f_{ka=1} \leq \frac{c}{\pi M} \tag{3.25}$$

A systematic merging procedure could either use Eq. 3.22 with $f_{sp} = \frac{1}{T}$ or, with the same equation, check for an intersection of measures at some frequency: if this is compliant with the constraint it can be used as a transition frequency at all effects. First choice let near-field measurement appear "more detailed" around f_{sp} because of assuming a more demanding time windowing with a smoothing effect on response. Process can however be automatized through Klippel software. To synthetize reasonable responses, we performed a simplified operation whose methodology is limited to a graphical splicing and the choice of the proper parameters. Software comes along with a step-by-step application note [77] offering a detailed discussion on that; if precise LPM is available, it should be included only as a "second opinion" in the process to adjust the true measured near field response according to its predicted level. Alternatively, we considered the latter as an approximation of the former. Exploiting precise LPM both amplitude and phase information for the low-frequency pressure complex spectra were computed through predicted mechanical transfer function:

$$H_x(j\omega) = \frac{X(j\omega)}{U(j\omega)} = \frac{X(j\omega) \cdot j\omega Bl}{U(j\omega)} \cdot \frac{1}{j\omega Bl} = \frac{Z_R}{Z_R + Z_P + j\omega L_E + R_E} \cdot \frac{1}{j\omega Bl} \tag{3.26}$$

given that following equalities holds:

$$Z_R(j\omega) = \frac{j\omega L_{ces}}{(j\omega)^2 L_{ces} C_{mes} + j\omega \left(\frac{L_{ces}}{R_{es}}\right) + 1} \tag{3.27}$$

$$Z_P(j\omega) = \frac{j\omega L_2 R_2}{j\omega L_2 + R_2} \tag{3.28}$$

This operation was needed because original available measurements lacked phase information. Numerator and denominator of Eq. 3.26 represents the displacement signal spectrum read by a laser and the input voltage, respectively. In Eq. 3.27 and Eq. 3.28 values of $L_{ces} = \frac{C_{ms}}{Bl^2}$,

$C_{mes} = \frac{M_{ms}}{Bl^2}$ and $R_{es} = \frac{Bl^2}{R_{ms}}$ form, in addition to R_e and L_e, the motional impedance offered to the electrical circuit side [79] with the analogy explained in Chapter 2. Measures were performed with a multitone input and usage of Eq. 2.4 in time domain then FFT-transforming $x(t)$ would not be a good calculation choice, as such line spectrum manifests as holes in $|P(j\omega)|$ and in a messy phase information due to discontinuities. Therefore, $P(j\omega)$ can be derived according to Eq. 2.5 assuming a convenient reference flat voltage spectrum $|U(j\omega)| = 1$ in the full audio-band. The $H_{FF}(f)$ to be merged with this predicted spectrum was measured with a reference voltage input stimulus of 1Vrms at a standardized distance of $d = 1$m. LPM fitted curves available in Klippel's analyzer were correlated with a different peak-voltage value instead. These must be interpreted as the rms values of each single representable sinusoid by the FFT and not the one of the whole multitone line spectra signal, the two coinciding only in case of a single sweep stimulus. For both checking the validity of our results and comparing curves before merging a pair must be referred to the same input signal avoiding misleading vertical shifts. Because of $|U(j\omega)| = 1$ this translates in both cases into multiplying Eq. 2.5 by a simple scalar factor accounting for the classical sinusoidal rms-peak value conversion. Phase plots are unaffected as they are a frequency-wise ratio between imaginary and real part.

Several parameters are required to run the splicing module. Because of replacing true near-field measurements with predicted LPM curves only a suitable $(f_L, f_{ka=1})$ range. A heuristic is tweaking the two frequencies to reach a satisfying result when software's algorithm adjusts f_{sp} by itself. Imposing minimum and maximum values by Eq. 3.25 and then decreasing the upper bound with lowest edge coinciding with f_L or slightly above allows to confine as much as possible the merging in a region where true far-field measurements are still valid. Excluding merging results with upper limit left to the edge $f_{ka=1}$ is also a safe choice which intrinsically assumes a narrower limit for the threshold of pistonic range behavior. Then, exploiting the result viewer a good f_{sp} can be selected for the attempts whose RMS error between near and far field curve around the splicing frequency was <1dB. This metric alone is obviously a coarse indicator; a too restricted frequency range may lead to an artificially low value, because of

averaging too few points. This depends in turn on the specific resolution of measures. The lowest RMS criterion is not necessarily reliable; however, it is still useful for an a-priori discarding of trials >1dB and to hierarchically order attempts with similar local minima. Final choice can be derived from intersecting previous considerations and graphical inspection of results. A reasonable behavior, apart from small peaks or notches relative to a speaker specific characteristic, is a response with an increasing slope before passband behavior and some ripples appearing at higher frequencies due to resonance modes. The process concerned all the 35 woofers in our context. Below f_{sp}, far field response is replaced by near field one.

3.4 Setting up experiments

A pure tone continuously swept up across the audible range can be sent to devices for a full audio band magnitude response and harmonic curves measurements. State of art is a logarithmic sweep with variable speed profile with which testing time for harmonics is reduced even to a fraction of a second [7]. Filters spaced integer octaves above the input main tone retain higher-order harmonics whose level is given in relative percentages or decibels level below it. Output can be recorded by a calibrated microphone connected to PC-based acoustic measurement software for high-resolution that can be selected as the highest in the same software. This allows to be close to true anechoic measurements excluding early reflections. A full description of the measurements setup and signal routing is postponed to Section 4.1.2.

Measuring listening room for such curves was the same of the one for listening sessions. In this scenario, before the beginning of the test an inter-system and inter-stimulus level calibration can be applied for a neutral reproduction. A peak normalization against clipping of test signals is an important safeguard too. Loudness balancing of multiple loudspeakers reproducing the same material during listening tests is important (See also section 8.1, 8.2 of [55]) as differences in reproduction levels lead to uncontrolled bias. This correction can be performed in several ways. Typical approaches employ metrics and models based on RMS measurements of test

signals to adjust level instrumentally. Subjective perception of stimuli changes according to the frequency-level dependence of loudness characteristics of the ear, therefore, to obtain a measure of perceived loudness, we must frequency-weight SPL. A simple way are weighting curves (A, B, C, D) commonly adopted and defined in standards as shown in Figure 3.8. These basically work as frequency-dependent inverse filtering roughly compensating for human hear mechanism, but still enough for calibration purposes. Their performance was compared with different loudness metrics through listening tests with reference anchors by 1992 Ronald M. Aarts [86]. By analyzing intra-subject ratings variation, B-weighting appeared to be the most consistent method with subjective ratings, but, if loudspeakers under test are similar, there is no serious objection to a simpler weighting for calibration purposes. ITU-R BS.1116- 1 [58] directions are limited to one basic approach defining a reference SPL listening level per single channel in terms of dB(A) as "a preferred listening level produced with a given measuring signal at the reference listening point for the same excerpt". Without resorting to more complicate metrics this seems a pragmatic way to proceed with levels calculated with a sound level meter offered in the free-source software Room Eq Wizard (REW) [87].

Besides A-weighting, we opted for another inter-system calibration procedure preparatory to the listening sessions. For the woofers case, this was meant to adjust all systems to the cut-off behavior of the most penalized one, i.e. whose slope after cut-off frequency was higher. The goal is adapting such slopes to be coincident without altering artificially speakers' passband behavior and denaturalizing their intrinsic brand differences. In the case of tweeter measurements, cut-off frequency is chosen as the highest among the raw measurements after sweep, to put ourselves analogously in a worst-case situation. We decided to perform such slope adjustments designing digital filters and equalizers through a free-source plugin inside Reaper [88] Digital Audio Workstation (DAW) thanks to how we handled signal routing by essentially sending to the final output only to the interested band in both cases. Task was reduced to a customizable DSP processing inside the DAW.

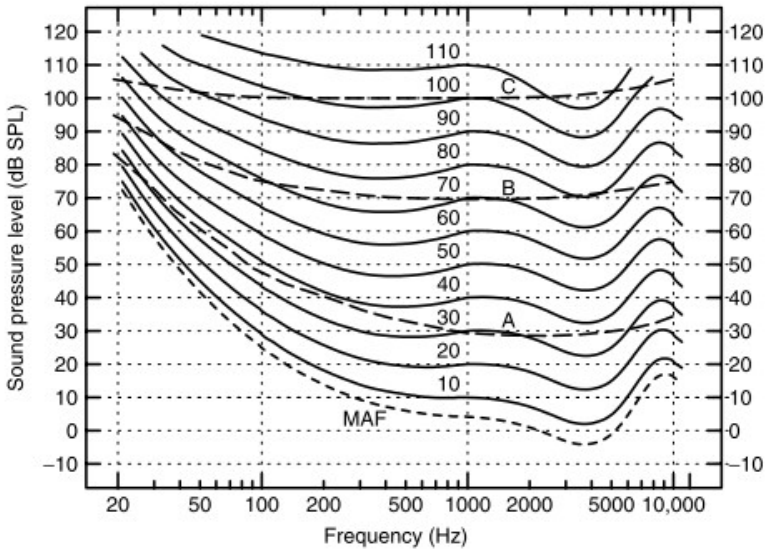


Figure 3.8 - Equal-loudness contours of the ISO 226 standard (solid curve) and A-, B-, C- weighting curves superimposed (dotted lines). Picture from [55].

An inter-stimuli Loudness calibration can be carried out by changing the volume by a fixed amount considering the *overall* loudness of a file. Performing this automatically in the multitrack DAW project as well is the best approach. Reaper offers again a free-source Subjective Loudness Measurement levelling thanks to the SWS extensions [89] embracing the standard of Loudness Normalization with LUFS[4] unities. Process consists in basically analyzing each audio item to determine its individual LUFS loudness, then, fixing a target, Loudness level of others is aligned to it. Tracks' levels were decreased to match the loudness of the quietest one. With this solution audio is played back with original dynamics with still enough headroom before clipping, that is, not exceeding digital 0dBFS upper bound for peaks.

Last interesting things to discuss are related to the custom GUI we coded in MATLAB® from scratch to respond to all our needs for a complete setting of the listening test. We needed a

[4] LUFS is an absolute loudness scale unit that reads like a standard Level Meter (0 LUFS = 0 dBFs) defined in EBU R 128 [115] European standard. This is exactly the same unit of the "LFSK" defined by ITU's [116] for which letter K is only an unlucky choice that must not be confounded with the B-Katz "K-Weighing Filter" [117] to measure Perceived Loudness.

specific handling of attributes and scales, DoE control within trials and assessors and consequent collection of data for successive export, analysis and understanding. Moreover, such interface also dealt all the technological aspects like signal routing and communication with speakers for a robust playback of the audio stimuli in the listening environment.

First at all, GUI design and assessment scale have impact on the experimental data. Response format based on line scales with magnitude intervals was the choice. The general idea is that attaching meaningful words or ticks to intervals or endpoints helps in assuring a reliable interpretation, but listeners might use labels differently, causing data scattering. Decision was of maintaining the most simple and user-friendly interface possible involving not too many graphical objects. Labels must correctly represent the intensity of scale position avoiding non-equidistant distribution of quality terms on internal perceptual listener's scale. A first recognizable cause is a language issue, when translated words do not correspond perceptually to the same position of the scale. For this reason, we translated the interface to Italian, that is, participants' native language. Then, listeners are known to use the scale points that are associated with labels, numbers, or ticks more frequently than the remaining parts as shown in Figure 3.9. Consequently, scores are clustered nearby these points [56]. To limit such quantization effect, scale must preserve as less breaks or divisions as possible, especially with a small panel size where averaging this effect out is more difficult. On the other hand, the lack of a calibrated and anchored preference scale most likely results in contextual effects. To avoid mapping process non-linearity problems, 10 labels were retained only for scaling Preference ratings in accordance with the classical 9-point hedonic scale for which however, as for other attributes, the GUI did not display any number. In agreement with Note 1 in ITU-R BS.1116 [58] we left the usage of the only end scales labels only for Loudness and Timbral Balance. Unsuitable antonyms choices are also a risk, as subjects become more reluctant in using endpoints saving extreme ratings resulting in a similar clustering of one mentioned before [71].

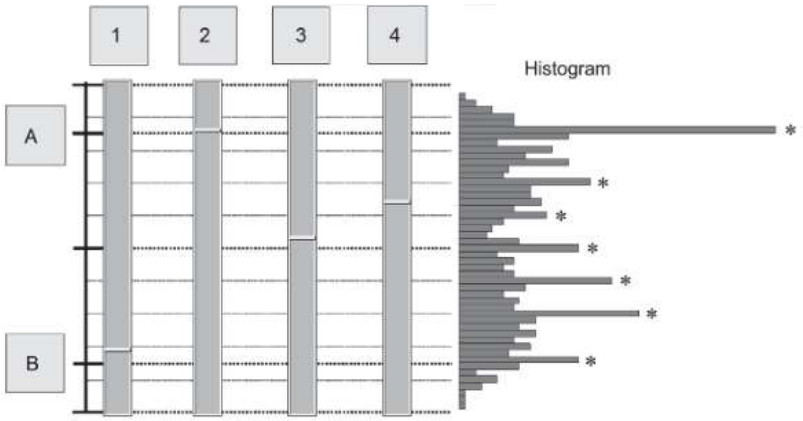


Figure 3.9 - Quantization effect in distribution of data due to horizontal lines crossing sliders. Pictures from [56]

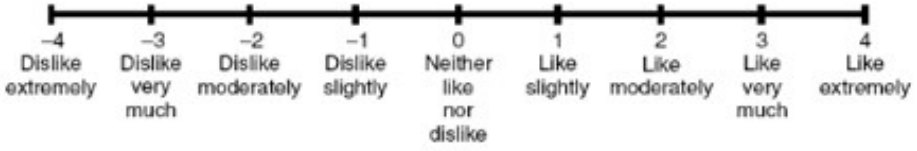


Figure 3.10 - 9-point hedonic scale for preference rating. Picture from [55].

For this task we chose the same baseline scale that, in the former case was labelled with no other words but the opposite anchors “too soft” and “to loud”. A rating of 1 would imply the sound perceived is excessively loud whereas a 0 rating would imply the signal is merely perceived audible. For Timbral Balance, scale came with bipolar descriptors too, with “dark” and “bright” endpoints plus a semantic differential only in the middle given by a “neutral” label in the center. In both these cases, the best-case positive evaluation coincides with the center of such scale, but listeners must not be graphically aware of it.

The GUI handled all logistic behind the test with the most essential graphics possible. Depending on the current state, proper attribute name, description and end scales had to be displayed every “next” button pressing according to the DoE described in Section 3.1.3. Reaper was configured to listen to multiple Open Sound Control (OSC) [90] messages on a UDP local port, sent from the application with a third-party developed MATLAB® script [91].

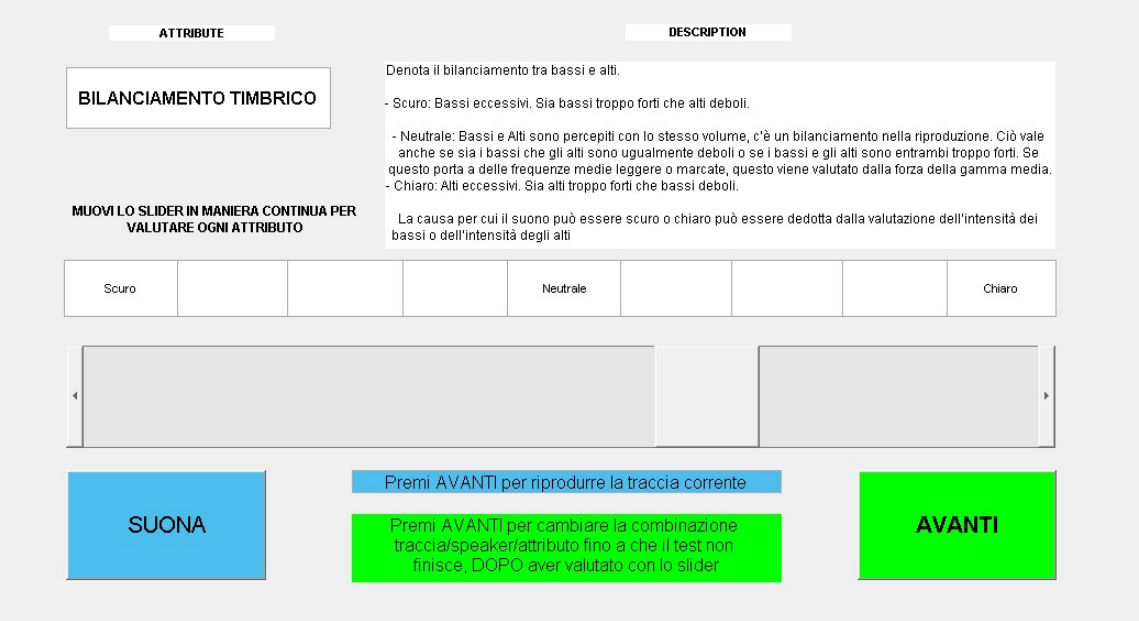


Figure 3.11 - Example of a rating (timbral balance case) with the GUI used by listeners;

DAW allows for a mapping of such messages and shortcuts to mute/unmute functions for its multi-tracks project. Therefore, we dealt with computer automation of speakers and audio signal switching easily once listeners pressed “play”. This activated the combination of stimuli to sound. We set 4 four channels in Reaper to which we sent the right excerpt to be reproduced, then channel 1 (loudspeaker A), 2 (B), 3 (C) or 4 (D) were activated alternatively. Solution for real-time stream audio processing and writing on digital audio board was the MATLAB® Audio Toolbox™ [92] I/O system object optimized for the task. Because of the employment of very short sound samples of ~20 each, we allowed the subjects to control individually when play a sound. Once stimuli finished, they could also repeat the listening every time they desired and take their time for judgment. If rating is completed, “next” button automatically stopped reproduction moving the test one step forward. Slider moved in between labels corresponding to a normalized rating whose value was hidden to listeners during the process; each evaluation was stored automatically. At the end of the test all functionalities of the interface automatically disabled in order not to corrupt saved data.

3.5 Criteria for relating sensory and objective data

3.5.1 Statistical validation method of listening test data

Listeners' performance can be quantitatively inferred by statistically analyzing responses to understand if variability stems from the true stimuli or is just accidental. Simple descriptive statistics serves to understand how much independent variable(s) contribute to attribute variation(s). To prove experimental outcomes and find results incoherencies, univariate techniques such as 95% confidence intervals mean, and significance tests are a valid policy. The principle is determining the likelihood that results are found by chance with a p-value compared to a significance level α, conventionally set to 0.05 for tests. If $p < \alpha$, there is less than 5% probability of random results. For continuous interval scales responses, parametric procedures are a suitable choice, provided that statistical assumptions are not violated severely. Among these, an ANOVA implies independent and normally distributed observations with equal variance. The rationale behind is decomposing the total data variation into deviation between the treatments mean values (here loudspeaker ratings) and random deviation within samples due to repeated ratings. They are termed between-groups (α_t) and within-groups ($\varepsilon_{t,i}$) variances, respectively. Then, a check if this ratio (F-statistic score) is higher than a critical value is done; the null hypothesis H_0 of all equal means against the alternative one H_a that they are not, is tested giving a % probability again in terms of p-value. The two scores are related: under H_0, deviations are comparable ($F \cong 1$) condition which is undesired. We performed such analysis concentrating on one attribute ratings at a time, which is our predictive model and correlation analysis objective, with six different standard one-way univariate ANOVAs. The underlying linear additive model for the ith rating on the tth stimuli of a subject for a single factor experiment is formally:

$$Y_{t,i} = (\mu + \alpha_t) + \varepsilon_{t,i}, \quad \varepsilon_{t,i} \sim N(0, \sigma^2) \tag{3.29}$$

where μ is the overall mean, α_t is the effect of the specific t^{th} treatment and $\varepsilon_{t,i}$ is the assumed normally distributed experimental error (non-acoustical factors, bias effects etc.). If one studies the effect of one factor (group) made of 15 listeners (levels) affecting the ratings (dependent variables) the question becomes how much an assessor, the only factor assumed to cause different ratings' mean, is influential for each attribute, independently on speaker-track combination. Setting the analysis in such way was intentional but conclusions are limited to these specific levels of speaker-track combination, fixed in statistical sense, i.e. non-generalizable. Listeners were instead *random* factor representative of a greater population whose results are generalizable still in the context of a similar methodology to ours. Listener data exhibiting an overall guessing can be rejected from further analysis. Aim here is not to test specific effects but validate properly results for being related to our physical data. Further considerations go beyond, and other experimental results are inferred through model building and correlation analysis.

Besides knowing from ANOVA that there is a difference between *at least two of* the mean scores, a Tukey's Honestly Significant Difference (HSD) post-hoc testing is usually a good choice. This is essentially a two-tailed null hypothesis multi-comparison standard t-test with the usual specified significance level here determining a cut-off value for the t-statistic. Such procedures are simply meant to upper bound the probability that *any* μ comparison is significant. HSD procedure is well-suitable for larger groups of paired comparison and optimal for balanced one-way ANOVA and similar procedures with equal sample sizes [55].

3.5.2 Regression and Correlation analysis

Once ratings are statistically validated, it may be advantageous to improve sensory profiling continuous data with a pre-treatment based on panel averages before performing multivariate analyses In the general context of relating three-way data structures to external data or replacing sensory data with fast measurements the two sample (vertical) dimensions are generally required to be the same. Therefore, three-way subjective matrix (Figure 3.12) is

averaged across assessors such to find the wanted relations. A consensus sensory profile across assessors, is a single sensory profile where all results are merged in an exemplificative matrix containing the main trends of magnitude ratings. A strategy for doing that is an averaging that tries to give less weight to more unreliable assessors [98]. The general formula for the weighted average profile for sample j can then be written as:

$$\bar{y}_{jk} = \frac{\sum_{i=1}^{I} \hat{y}_{ijk} w_{ik}}{\sum_{i=1}^{I} w_{ik}} \tag{3.30}$$

Here $\hat{y}$ represents raw data, eventually averaged over replicates that in our case were considered ratings given on different tracks. From the point of fitting a model, taking means over replicates is usually the most sensible choice. If the only reason for differences between the replicates is noise, then it makes little sense to model this [99] independently on the statistical method to obtain a representative consensus profile. Weights w_{ik} are computed according to the criterion discussed below. A possible one is the classical Pearson's Linear Correlation Coefficient between an average assessor and a hypothetical average profile:

$$\overline{\mathbf{X}} = \frac{1}{I} \sum_{i} \mathbf{X}_i \tag{3.31}$$

possibly with the actual assessor eliminated. The matrices $\mathbf{X}_i$ are the $I = 15$ matrices of ratings for each assessor. The absolute value of the resultant 15x3 matrix elements is plugged into w_{ik} expression of Eq. 3.30 considering in this way the differences between attribute variances for a given assessor. The resultant $\mathbf{Y}$ matrix is then used as the 12x3 response variable for prediction. Predictors matrix ($\mathbf{X}$) is given by all the features described in Section 3.2. When relating $\mathbf{Y}$-data to continuous $\mathbf{X}$-data standard regression methods including linear models are the most suitable choice. Once again, the interest is finding valid results for each single attribute separately in a multivariate scenario not dealing with multiple $\mathbf{y}$-variables. Typical assumptions to be met are a constant variance of the error terms (residuals), their independence and finally normality of the error term distribution. Independent variables chosen must also be as less collinear (correlated) as possible.

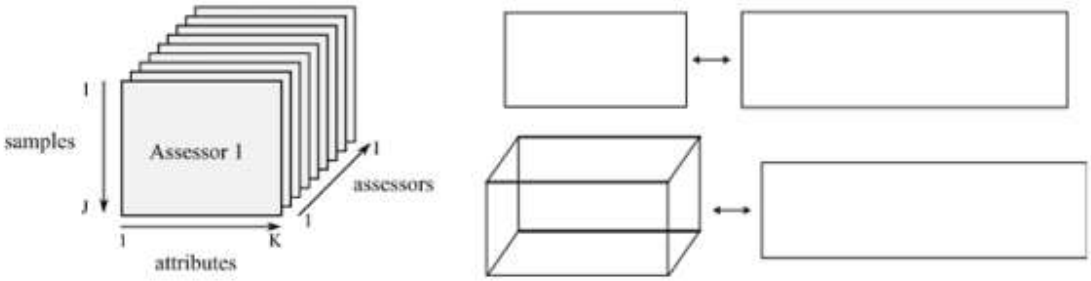


Figure 3.12 - Three-way structure of data (left), three-way sensory data to be related to a regular two-way objective data matrix (right). The vertical dimension must be the same. Picture from [98]

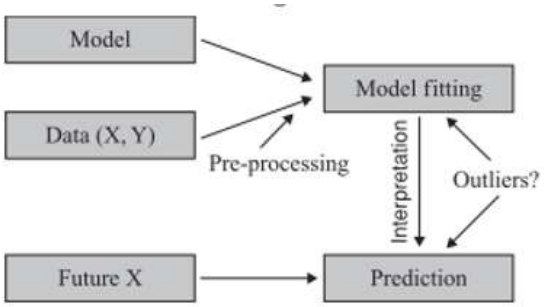


Figure 3.13 - Relating X to Y data for the purpose of prediction and improved understanding. Picture from [98]

The multivariate linear regression model for a single observation (n) can be written in the following way:

$$y_n = \beta_0 + \sum_{\mathrm{p}=1}^{P} x_{np}\beta_p + \varepsilon_n \tag{3.32}$$

Or, in compact matrix form:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \tag{3.33}$$

In our case prediction is based on of the single attribute (ratings) and in each of the models $\boldsymbol{\beta}$ is a vector of weights (coefficients). Each independent variable weight represents the relative contribution of that predictor on the outcome. Common approaches for estimating the right set of (weighted) variables include progressive searches such as stepwise models or forward- and backward feature elimination procedures where variables are added or removed to the model up to satisfying a certain criterion.

An exploration of alternative regression methods is mention worthy too. In case of classical OLS regression problem being ill-posed for many predictors, n highly collinear exploratory variables in $\mathbf{X}$ and comparatively little response data at disposal, Principal Component Regression (PCR) and Partial Least Square (PLS) come into hands. These are alternatives to regularization techniques and variable selection methods meant to solve collinearity and dimensionality problems smartly while providing good interpretation tools. The general idea is extracting a few major $K \ll P$ components capturing the most useful information in $\mathbf{X}$ for predicting $\mathbf{Y}$. Dimensionality of regression problem can be reduced selecting fewer terms on the basis of Principal Component Analysis (PCA) on $\mathbf{X}$:

$$\mathbf{X} = \sum_{k=1}^{K} \mathbf{t_k} \mathbf{p}_k^T + \sum_{K+1}^{K'} \mathbf{t_k} \mathbf{p}_k^T = \sum_{k=1}^{K} \mathbf{t_k} \mathbf{p}_k^T + \mathbf{E} = \mathbf{T}\mathbf{P}^\mathbf{T} + \mathbf{E} = \widehat{\mathbf{X}} + \mathbf{E} \tag{3.34}$$

with $\mathbf{P} = (\mathbf{p}_1, \dots, \mathbf{p}_k)^T$ being the loadings matrix, i.e. the eigenvectors of $\mathbf{X}^\mathbf{T}\mathbf{X}$. First addend accounts for most of the variation in $\mathbf{X}$ while the rest is treated as noise. Original predictors are transformed into a new components space where analysis is translated, and information compressed down to a linear combinations of them to regressed. In PCR case, components are derived solely from $\mathbf{X}$ according to a standard PCA. Differently, in PLS of our interest here, both $\mathbf{X}$,$\mathbf{Y}$ data are projected influencing the regression model construction:

$$\mathbf{X} = \mathbf{T}\mathbf{P}^\mathbf{T} + \mathbf{E} \tag{3.35}$$

$$\mathbf{Y} = \mathbf{U}\mathbf{Q}^\mathbf{T} + \mathbf{F} \tag{3.36}$$

From a prediction standpoint the difference between the two is usually small [98]. However, PLS exploits two sets of $\mathbf{X}$-loadings for the same model, the canonical PCA matrix $\mathbf{P}$, but also PLS-weights $\mathbf{W}$, somehow merging a multivariate regression, a PCA and covariance analysis. The key is finding combinations of predictors $\mathbf{x}_p$ with large covariance with the response, then, a standard regression used for fitting [100]. Thus, components (latent variables) are found including both $\mathbf{X}_{(\mathrm{nxp})}$, and $\mathbf{Y}_{(\mathrm{nxq})}$ in the optimization criterion:

$$\arg\max_{\mathbf{w}_k,\mathbf{q}_k}\{\mathrm{cov}(\mathbf{X}_{k-1}\mathbf{w}_k, \mathbf{Y}_{k-1}\mathbf{q}_k)\} \tag{3.37}$$

subject to:

$$\|\mathbf{w}_k\|_2 = \|\mathbf{q}_k\|_2 = 1, \qquad \mathrm{cov}(\mathbf{X}\mathbf{w}_k, \mathbf{X}\mathbf{w}_{k'}) = 0 \text{ for } k \neq k' \tag{3.38}$$

where $q = 1$ in our case for a univariate attribute-wise regression. The $\mathbf{t}_\mathrm{k} = \mathbf{X}_{k-1}\mathbf{w}_k$ and $\mathbf{u}_\mathrm{k} = \mathbf{y}_{k-1}\mathbf{q}_k$ are the $\mathbf{X}$- and $\mathbf{y}$-projections; a matrix $\mathbf{T}_{(\mathrm{nxk})} = (\mathbf{t}_1, \ldots, \mathbf{t}_\mathrm{k})$ collects such PLS scores from directions that maximize covariance between *latent* scores. Vectors $\mathbf{w}_k$ and $\mathbf{q}_k = \mathbf{y}^T\mathbf{t}_k$ are the $\mathbf{X}$-and $\mathbf{y}$-loadings weights, respectively. These allow to perform the bilinear decomposition of Eq. 3.35 and Eq. 3.36. Maximization is carried out sequentially $\forall k$ and $\mathbf{w}_k$ are the first right eigenvalues from the cross-covariance $\mathbf{X^T y}$, i.e. the true loadings linked to regression relationship building. The kth PLS-component direction in $\mathbf{X}$- space characterization is generally different to the one found by PCA on the basis of $\mathbf{X}$-data alone ($\mathbf{w_k} \neq \mathbf{p_k}$) because of PLS's simultaneous ($\mathbf{t}_k, \mathbf{u}_k$) maximization and interdependence on which algorithm relies. It is *this* difference that indicates how much $\mathbf{y}$ controls the $\mathbf{X}$ decomposition. Both $\mathbf{P}_{(\mathrm{p\times k})}$ and $\mathbf{W}_{(\mathrm{p\times k})}$ matrices are used to estimate scores. Essentially, the regression coefficients from the $\mathbf{y}$-variable onto scores are derived iteratively: a $\hat{\mathbf{q}}_k = (q_1, \ldots, q_k)$ is temporarily set and this is where the usual LS approach comes in. This vector, transformed back in terms of the original variables, *implies* the true px1 regression coefficient:

$$\hat{\boldsymbol{\beta}}_k^{PLS} = \mathbf{W}_\mathrm{k}(\mathbf{P}_\mathrm{k}^T\mathbf{W}_\mathrm{k})^{-1}\hat{\mathbf{q}}_k \tag{3.39}$$

Size of $\hat{\boldsymbol{\beta}}^{PLS}$ is independent on latent components number can be used directly on the $\mathbf{X}$ matrix. The complexity of PLS regression model is controlled by the choice of hyperparameter k. Once it is set, usually with cross-validation methods, the regression step is:

$$\hat{\mathbf{y}}_k = \mathbf{T}_\mathrm{k}\hat{\mathbf{q}}_k = \mathbf{X}\mathbf{W_k}(\mathbf{P}_\mathrm{k}^T\mathbf{W}_\mathrm{k})^{-1}\hat{\mathbf{q}}_k^T = \mathbf{X}\hat{\boldsymbol{\beta}}_k^{PLS} \tag{3.40}$$

Principal components corresponding to the dominating eigenvalues are used directly in the regression model and univariate PLS regression is competitive to similar regression techniques.

Often, univariate PLS regression of each column of Y is preferred to multivariate PLS [101]. Most common way of executing univariate PLS is SIMPLS algorithm which came as an already software-implemented solution to be applied attribute-wise.

In regression analysis, the degree of conformity of the searched relation can be measured in terms of RMSE, r^2 (Eq. 3.16) and adjusted r^2, higher values indicating, in our case, that listening tests results can be described well. However, predictors coefficient must be interpreted and statistically validated along with this. For PLS, as components are defined sequentially, to determine optimum k three metrics on training data were considered; the estimated mean-squared errors (MSE) for PLS models with $0 : \mathrm{K}$ components, a non-inflationary measure $\mathrm{Q}^2\mathrm{Y}_k = 1 - \frac{MSE_k}{RS_{k-1}}$ of predictive ability and the cumulative explained variance on $\mathbf{y}$. The minimum k is retained as optimum according to a comparison of such criteria; later ones are usually dominated by noise in data and may decrease predictive ability. Moreover, $\mathbf{w}_\mathrm{k}$-vectors are mutually orthogonal so score and loadings bi-plots can be inspected as usual.

Due to the impossibility of gathering further data, we tried to exploit the enlarged dataset collected as described in Section 3.3. This considers also the four speakers involved in the test to understand the extent to which features choice has been meaningful for our task. Correlation analysis is made of two essential parts. At first, we explored the possibility of exploiting an already existing predictive modelling to counter validate its applicability to our specific context where experimental conditions and independent variables are completely different to the one used for building. We Look Forward To extrapolating general conclusions from such analysis. For doing that we can operate a Preference ratings mapping process in a proper range to understand limits and differences with our approach. Furthermore, light can be put on some linear electro-mechanical parameters which composed the objective training dataset too as regards simple correlation with subjective data. This is a relatively unexplored and unclear research direction and test whether some new general inferences can be done in a simple way with data at our hand. The investigation may concern especially finding a

connection with some experimental choices in the listening test such as equalization procedure and driver box enclosure design.

3.6 Conclusions and remarks

In this chapter we presented important criteria to achieve our objectives. As regards subjective assessment of loudspeaker quality, our test method is a QDA-like approach with no training and attribute elicitation phases, where proper stimuli are presented sequentially. Assessors rated a predefined list of attributes and a blocking presentation order, ideally different for each of them, has been addressed. Size of the experimental paradigm grows rapidly in terms of both independent and dependent variables and this limited test choices not to exceed a reasonable duration and tire listeners. The logic behind calibration preliminary procedures and all technological and implementation aspects behind setting up experiments has been presented too. The possibility of finding the most representative set of numbers to relate with the different attributes was explored when must be applied to single driver case. Previous studies stressed the importance of frequency response in justifying most of listeners' quality ratings variance. Still, distortion should not be ignored. We tried to model compactly these aspects. A rationale behind merging strategy for a further feature-based assessment on a different dataset was provided as well. After clarifying a validation strategy for subjective data, two different ways of addressing the regression problem were described with the necessary mathematical background. Finally, a general idea about carrying out a correlation analysis between subjective and objective results was presented.

4. Evaluation

In this chapter we want to present our experimental activity results. At first, we detail the settings adopted for both tests and measurements and last experimental variables to be considered. Acoustic characteristics of a listening room are known to play an important role, likewise, a correct handling of all secondary aspects must be considered. Alongside with experimental conditions, the impact of design variables in relation to the chosen methodology discussed in previous chapter can be tested with proper analysis techniques for inspecting results reliability with all numerical details. Model results and correlation analysis with extracted objective data are critically evaluated also based on a comparison to other related work along all the implications.

Chapter is structured as follows. Section 4.1 discusses all experimental conditions. Section 4.2 and Section 4.3 summarize objective and subjective datasets respectively. Lastly, Section 4.4 highlights our attempt of predictive modelling and a comparison between results coming from precedent works. Moreover, a correlation analysis with electro-mechanical linear lumped parameters is inspected. At the end, in Section 4.5 we want to highlight the scope and limitation of our study, mainly imposed by the number of variables and their levels.

4.1 Experimental conditions

To ensure reproducibility of the results, test conditions are set to certain rules. Also in case of a quality control of the test environment and correct test handling, requirements are outlined in several standards; as an example, ITU-R BS.1116-3 [102] delivers rigorous guidelines.

Section 7.1 of Bech and Zacharov [55] systematically expose various experimental errors arising from inadequate environments. Main concerns are about room aspect ratio, reverberation time, attenuation of early reflections and low background noise levels. Even non-acoustical specifications such as lighting, temperature, furnishing and subject comfortability can be influent, and room must also fulfil bias-free listening and viewing conditions

4.1.1 Test environment and logistics

All tests and measurements were conducted in Elettromedia s.r.l. headquarter located in Potenza Picena (MC). The place made available for experiments is configured as a non-rectangular shape listening room whose greater dimension is about ~6m sizes and smaller one of ~4m. The space is acoustically treated to minimize reverberation phenomena and permits a close free-field propagation condition. Reverberation time is $T_{60} = 0.15s \div 0.2s$ averaged on all frequencies and through preliminary measurements it has been tested that position inside the room does not influence listening. A low background noise (<27dB) ensured a neutral listening experience and let the devices be assessed without any disturb. To obtain meaningful results, it is also important to carefully instruct listeners before taking part in the session about the planned test, its purpose and expected duration. However, instructions must be as neutral as possible and not reveal too many elements that may bias their judgments. Paper sheets reported in Appendix A were printed and handed out to the subjects along with precise oral explanations before the beginning. Subjects could ask further details regarding formalities but without going in details about any technical aspect of devices. In the briefing, interviewees were ensured that no wrong answers exist as true subjective perception is what matters and were encouraged not to skip ratings. We also gathered additional remarks and a-posteriori comments after completing the individual session. A touch screen in the center was used for projecting the test interface to assessors so that they could interact with the GUI either with it or remotely by a touchpad; as part of oral instructions, subjects were made familiar with the process before starting, but not notified about the progress of the test. Automatically

collected data in matrix form for each corresponding subject avoided any transcription errors. To preserve assessors' performance independence, it was made sure that subjects did not interfere each other by exchanging comments in-between sessions.

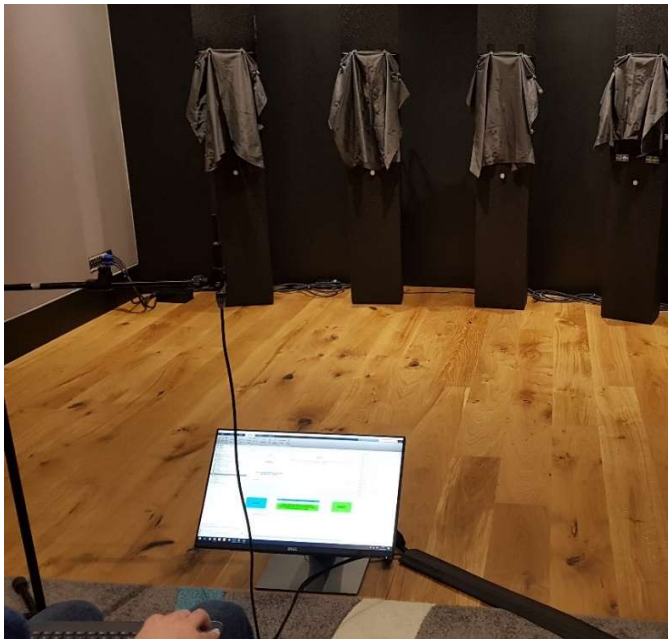


Figure 4.1 - Picture from listening room: a subject's performing the test with blinded drivers

On average, it took 25-30 minutes for the completion of one, but no time limit was imposed. A test operator was always physically present in the listening room in case of further clarification needed and to handle any sudden issue potentially happening. The two sessions were completed all at once and independently to minimize risks of set-up errors.

4.1.2 Hardware setup and playback settings

To provide that signal level and equalization is the same for all recordings and completely automatize tests, the audio chain in the room was composed as follows:

- A laptop to control the reproduction system and the acquisition device
- A Beyerdynamic MM1 condenser microphone for speaker response measurements

- A Hypex FusionAmp FA501 power amplifier, which outputs the signal coming from the audio board and drives the speaker(s)
- An audio interface that allows the communication between the PC and the speakers. This also included A/D D/A converters and microphone pre-amps.

All listening test-related tasks are performed by the MATLAB® custom-made GUI interfaced with the TotalMix FX [103] software of the digital audio board. As far as reproduction of stimuli is concerned, the complete signal path from the player to speakers is described henceforth along with playback settings. A macroscopic view of the audio flux is given in Figure 4.2. Signal is first sent to a virtual "AN1/2" input completely left-panned and this corresponds to a unique physical input called "AES_L" on the audio board interface.

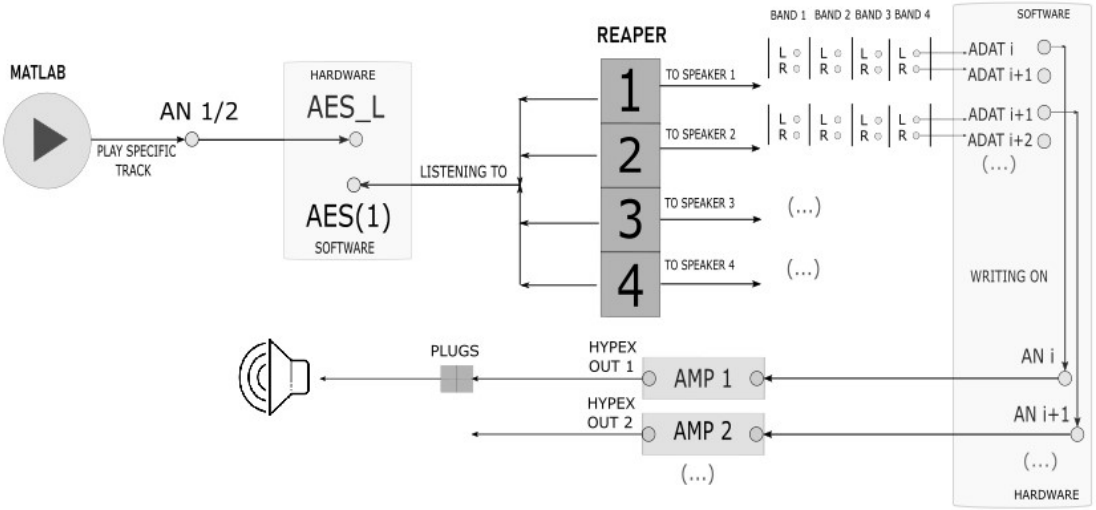


Figure 4.2 - Block diagram of the stimulus playback architecture for listening tests. Exemplificative signal flow in case of a single speaker. Picture made with free-source software INKSCAPE 0.92 [119]

Communication is set with an "AES (1)" Reaper Channel on which the audio board loopback software was activated. All 4 DAW tracks, corresponding to a different speaker with a stack of plugin-implemented effects on four sub-bands, listened to that input. This was according to the equalization tasks described in Section 3.2.1. Each one of them corresponded to a software

stereo channel called "ADAT i" completely left-panned related to a "AN i" physical channel fed to the correct power amplifier whose output "HYPEX OUT" is sent through proper cabling to plugs placed in the opposite side of listening room to which speakers were connected. As a result, each properly equalized speaker could be completely manipulated from the control laptop (Figure 4.3), core of the routing system for both measurements and listening test tasks.

Besides test room, location of reproduction loudspeakers significantly modifies listening experience relatively to of the assessors' one. For monophonic reproduction, recommendations [58] do not specify a guidance on non-reference listening location whose ideality is in relation to specific test to be conducted. Slightly differently from standard approach, speakers were placed in front of the listeners' frontal axis at a 2.5m distance, in a semicircle within an angle of $\pm$ 10°. Position was symmetrical with sitting position to guarantee on-axis listening condition for all. This is the same setup in which speakers have been measured and equalized. All the 15 subjects performed the test alone, in the middle-seat of a chair row at the center.

Although not formally scientifically documented, general recommendation is that tests must be conducted "blind" as both trained and untrained listeners are likely to prefer devices which meet their preconception the most. To not allow test participants to see brand names or any kind of labelling, speakers were hidden by acoustically transparent black curtains. Moreover, they have been assembled in their respective box before sessions and positioning with electrical cabling has not been carried out under assessors' eyes. This helped the presentation order to provide a further obscuration not perceiving which speaker was playing in a precise moment. All enclosures were sealed boxes of $V_{box} = 17l$ volume to minimize diffraction and reflection effects. Speakers were Elettromedia s.r.l. specified in Table B.0.1 and Table B.0.2.

As far as lumped parameters are concerned, mechanical measurements were obtained with a calibrated laser with speaker placed on a low vibration stand. Acoustical measurements have been conducted on a semi-anechoic room instead. In both measurement cases the Klippel KA3

Analyzer software [104] has been exploited with a Heartworks M30 microphone located at 1m from the speaker and a Crown Macro-Tech MA-5002VZ amplifier.

4.2 Objective Data results

Measurements were performed according to the above-mentioned environmental variables considering the scenario in which a single speaker at time was reproducing sound and without any kind of room compensation. Referring to previous geometry and signal chain of Sections 4.1.1 and 4.1.2, we placed the condenser microphone at the listening position in the room mounted on rotating rig as shown in Figure 4.3 (left) with equal distance to each driver acoustic center.



Figure 4.3 - Beyerdynamic MM1 microphone mounted at one end of the rotor (left), Drivers mounted in the sealed boxes without black curtains (center), Control location to manage experiments and hardware rack (left)

The excitation signal adopted was a Log Swept pure tone with a starting frequency of 20.187378 Hz and a frequency step of 0.3364563Hz at a user defined level of -31,0dBFS. Two complete sweeps in the audio band with a total duration of 5sec were used. Measures were recorded in REW with no dB-weighting compensation settings.

Recorded raw data with 1/48 smoothing was selected for further processing. Following graphs demonstrate either on-axis amplitude (dB) or THD (dB) of each frequency bin in logarithmic scale and axes are scaled according to the frequency range under investigation. Usually woofer crossover frequency is set at around 2.5kHz depending on the component itself if it is thought to sound in a two-way system. Because of testing single drivers, by inspecting measures a reasonable cut-off frequency adjust was chosen as 4kHz for equalization purposes described Section 3.2.1. For tweeter measurements, worst-case is chosen as the one for which the THD measurement start to increase later; the start of the distortion curve climb is than selected as a the most restrictive cut-off frequency to adjust other drivers.

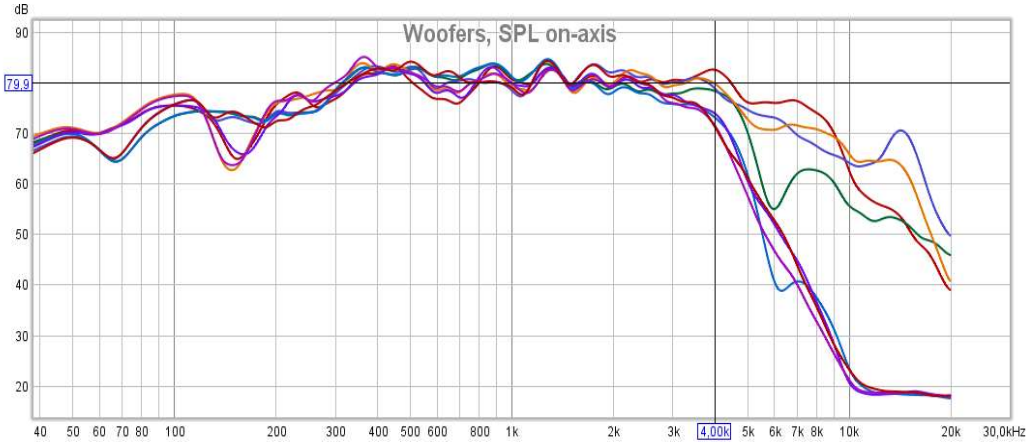


Figure 4.4 – Woofers All SPL on-axis. Picture exported from REW.

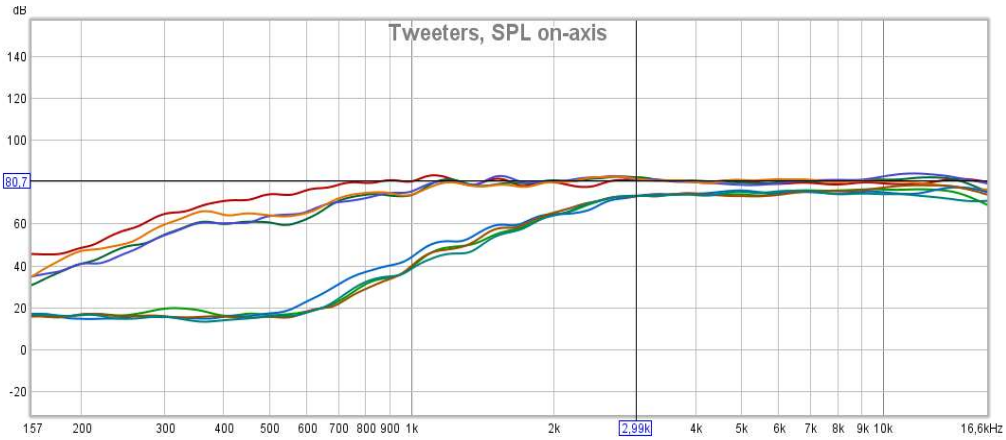


Figure 4.5 – Tweeters All SPL on-axis. Picture exported from REW.

Both conditions of pre- and post-equalization are plotted on the same graph, with each speaker appearing with different color and a *graphical* 1/6 octave smoothing for visualization purposes. Distortion legenda explains equivalent percentage values of various harmonics. Although tweeter measurements seem corrupted from the software export data used for calculation were complete and is only a matter of graphic visualization.

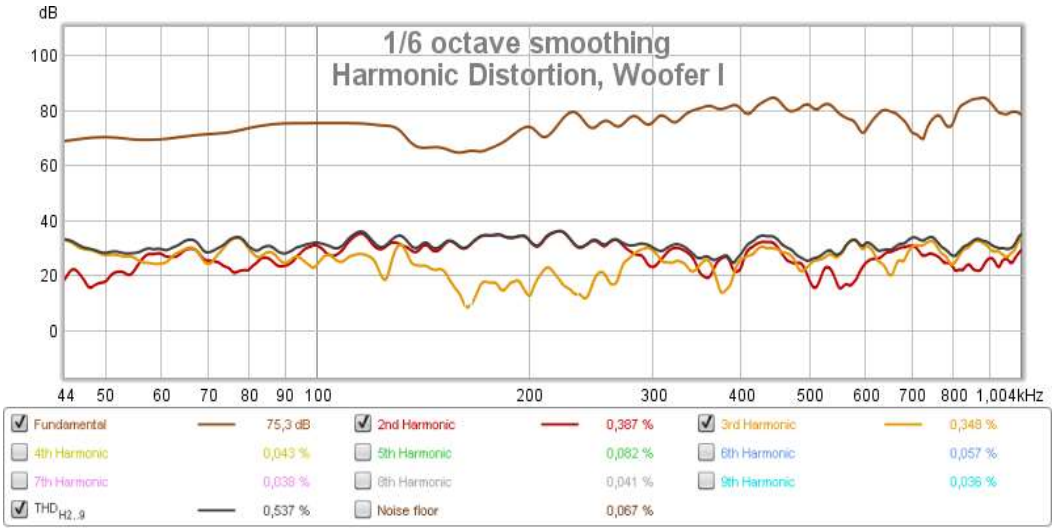


Figure 4.6 – Hertz C165 Cento Harmonic distortion plot. Picture exported from REW.

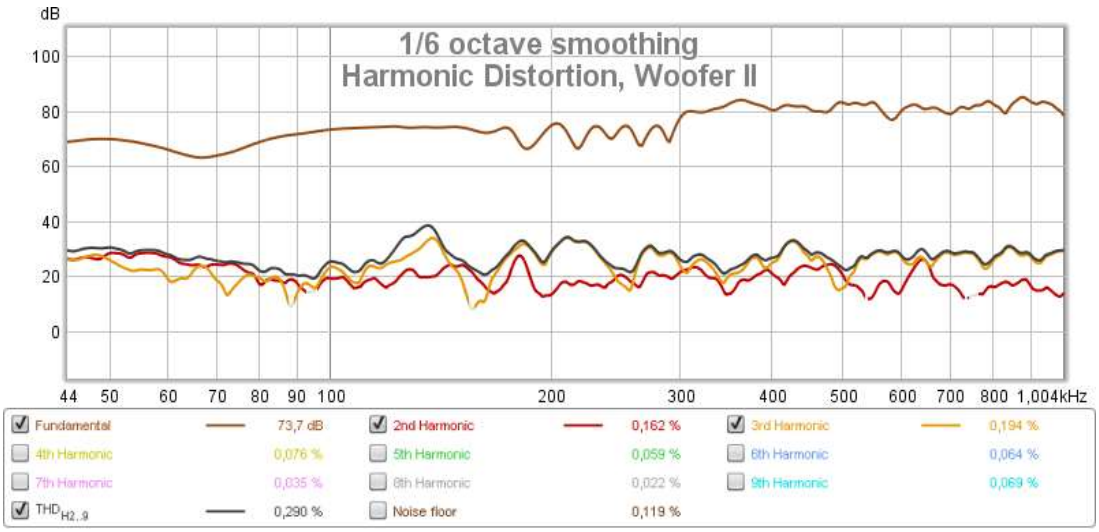


Figure 4.7 – Audison TH 6.5 II Sax Harmonic distortion plot. Picture exported from REW.

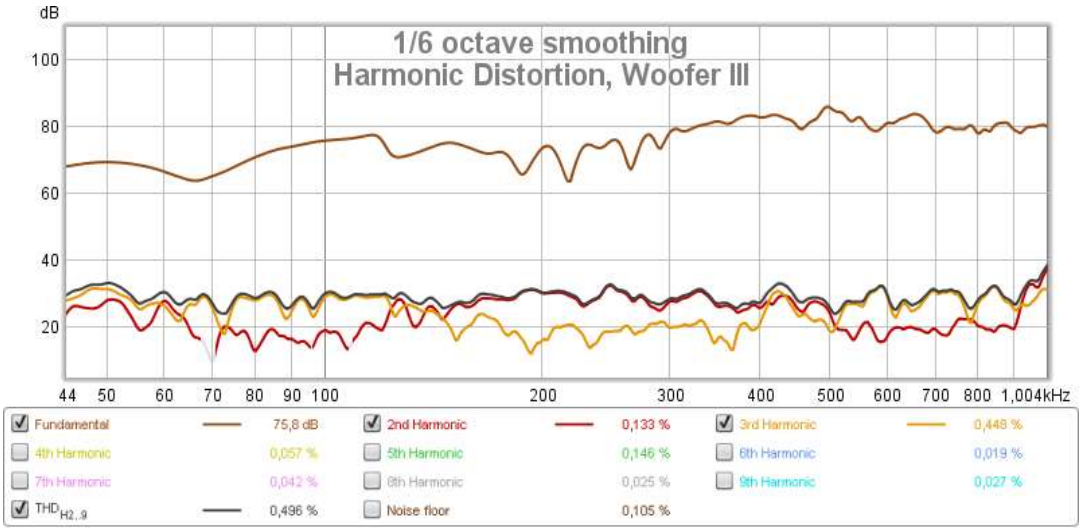


Figure 4.8 – Hertz ESK EV 165.5 Harmonic distortion plot. Picture exported from REW.

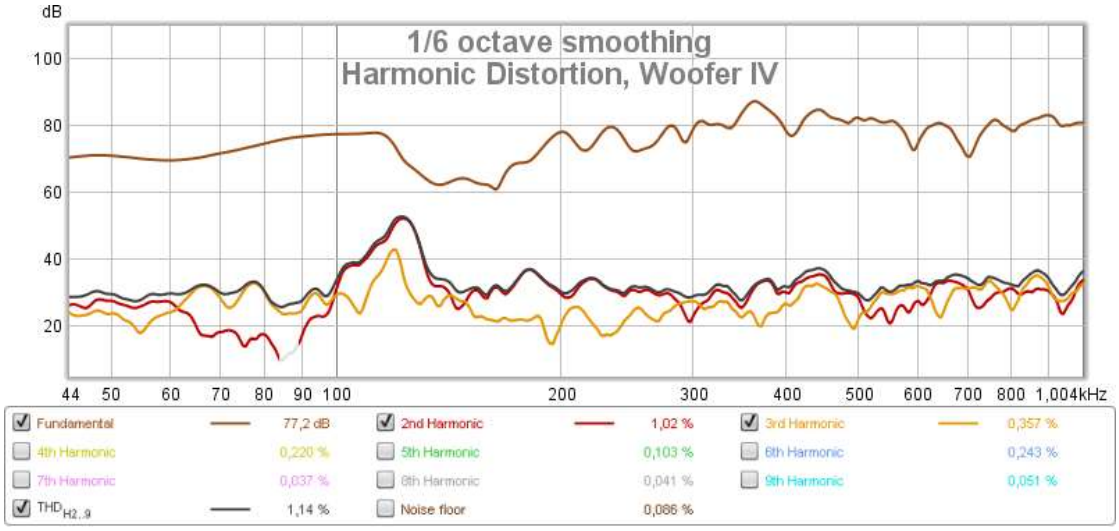


Figure 4.9 – Hertz MP 165.3 Mille Pro Harmonic distortion plot. Picture exported from REW.

As regards inter-system level-matching, all channel master faders were put to 0dB, then the channel corresponding to the quietest SPL sounding speaker was tweaked to the level 75dB(A) weighted SPL (-31dBFs in both woofer and tweeter cases) recorded in listening position with a 5s pink noise signal fed to each. All other faders were then adjusted to match that level and mixing configuration was saved for the test.

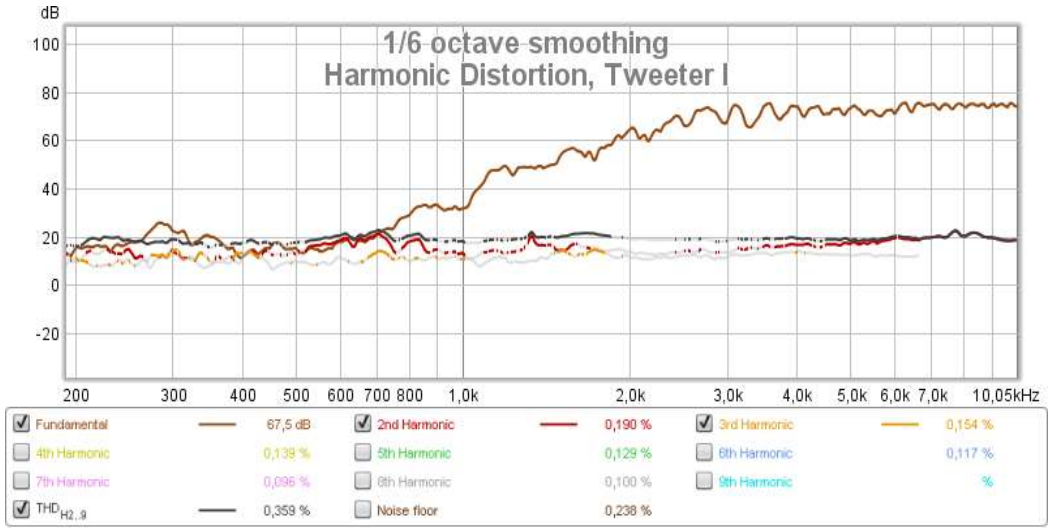


Figure 4.10 – Hertz C26 Harmonic distortion plot. Picture exported from REW.

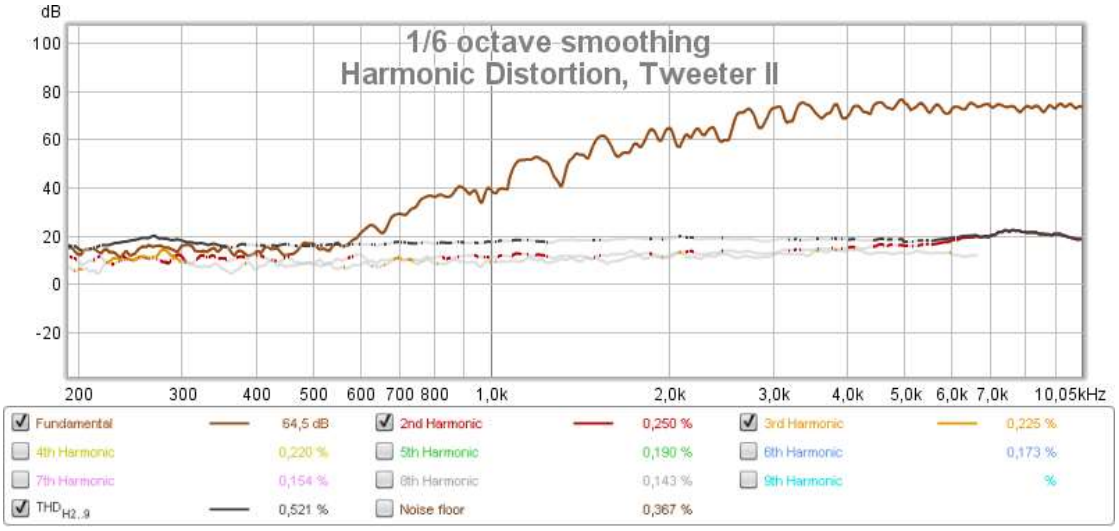


Figure 4.11 – TH 1.5 II Violino Harmonic distortion plot. Picture exported from REW.

Intra-stimuli normalization was done instead considering an average excerpt duration chosen was 22 seconds (shorter track lasts 15s while longer one 32s). Quietest track level (DP) reference was pretty low (-21.5LUFS). After normalization procedure ended a Reaper mixing configuration for both sessions was saved.

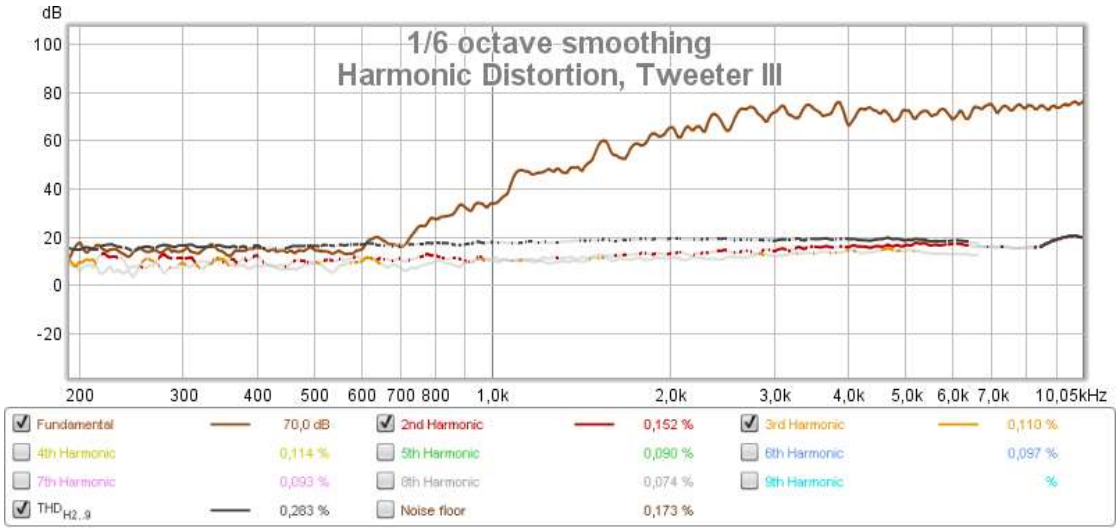


Figure 4.12 – ET 26.5 Harmonic distortion plot. Picture exported from REW.

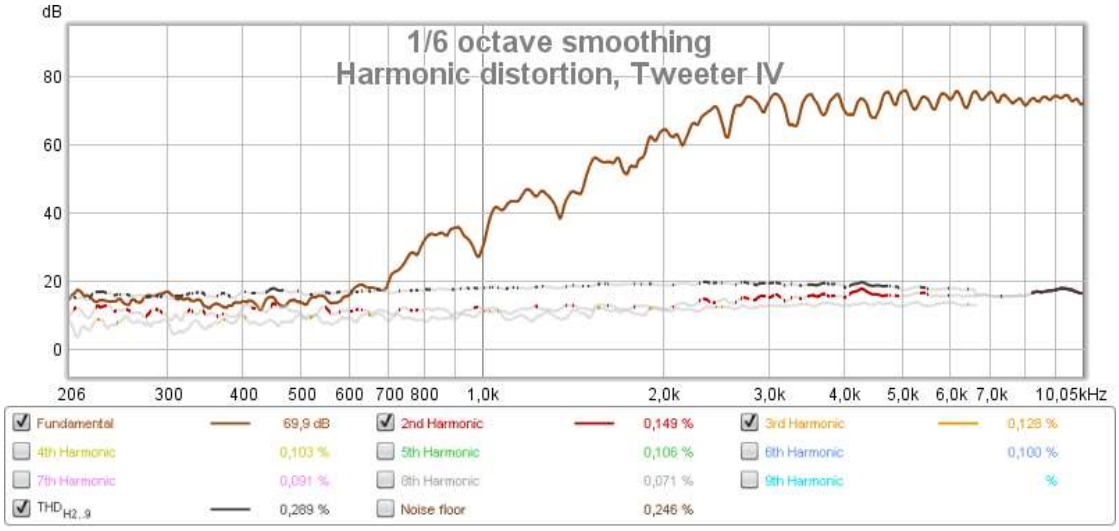


Figure 4.13 – Hertz MP 25.3 Mille Pro Harmonic distortion plot. Picture exported from REW.

Speakers' merged responses were computed on a per-frequency basis, with a specific resolution for each woofer, denoted as B001, B002, …, B035. In particular, B034, B035, B014, B017 correspond to Hertz C165 (I) Audison TH 6.5 II Sax (II), Hertz EV165.5 (III), Hertz MP165.3 (IV), from left to right as disposed in the listening session (Figure 4.3, center). They will be denoted in this way for the rest of the discussion. Selected transducers were thoroughly analyzed with the Klippel R&D System. For comparison of our MATLAB® generated data with amplitude response already available in Klippel analyzer a normalization factor was

applied. A 9dB increment was necessary for both because of rms-peak conversion (+3dB) and Klippel's $H_{NF}(f)$ referenced to a voltage stimulus of 2Vrms (+6dB). For a quantitative assessment of merging operation validity, errors were calculated between already available amplitude response and our generated one:

$$\Delta x_{i,j} = \left| x_{KL,i,j} - x_{MAT,i,j} \right| \ [dB] \tag{4.1}$$

where the subscripts denote the $i = 1,2,\dots,I$-th value of magnitude response for the $j = 1,2,\dots,35$-th woofer. For each woofer, we computed the mean of these differences:

$$\overline{\Delta x}_{j,dB} = \frac{1}{I} \sum_{i=1}^{I} \Delta x_{i,j} \ [dB] \tag{4.2}$$

Finally, we derived an error with respect to the "true" analyzer measure:

$$\varepsilon_{\%} = \left(\overline{\Delta x}_{j,lin} - 1 \right) * 100 \tag{4.3}$$

where $\overline{\Delta x}_{j,lin}$ is the result of Eq. 4.2 converted in linear domain. Apart from tiny uncertainty propagation errors results were good: on average, error was <0.7%. Phase information were reasonable as well. Once established that, data were imported in the software; both far-field and fitted SPL responses were matched in their peak voltage form with the splicing algorithm. After running it, a window shows both curves for comparison. Results for one of the selected models are shown henceforth. Table 4.1 shows instead merging data. Features were instead computed from on-axis frequency, THD measurements and synthetized curves exploiting MATLAB® Audio Toolbox [105] except for *SE*, *SD*, *SC*, *SCR*, *SCTR*, *AAD*, *SM*, *LFX*, *LFQ* that were coded by hand. A third-party software was exploited [106] to deal with fractional octave bands splitting. All excluding *SM*, *LFX*, *LFQ* were stacked in a matrix as potential candidates for model. Objective database resulted in a NxP matrix with N = 4 is speakers and P the number of features (P = 59, P = 31 for woofers and tweeters respectively). Values are reported in Appendix C for clearness.

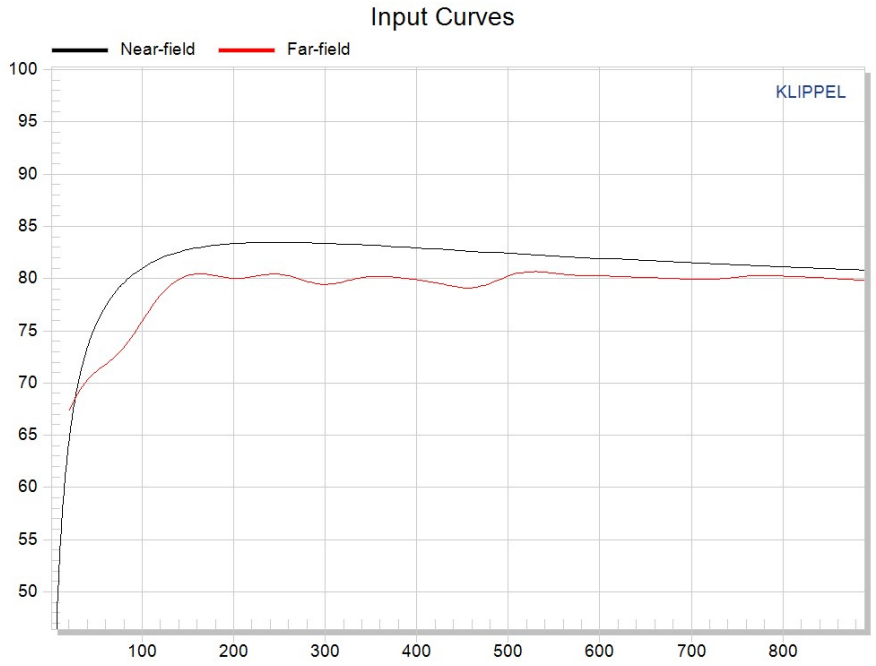


Figure 4.14 – Example (B035) of input curves before merging. Picture exported from Klippel dB-Lab Analyzer

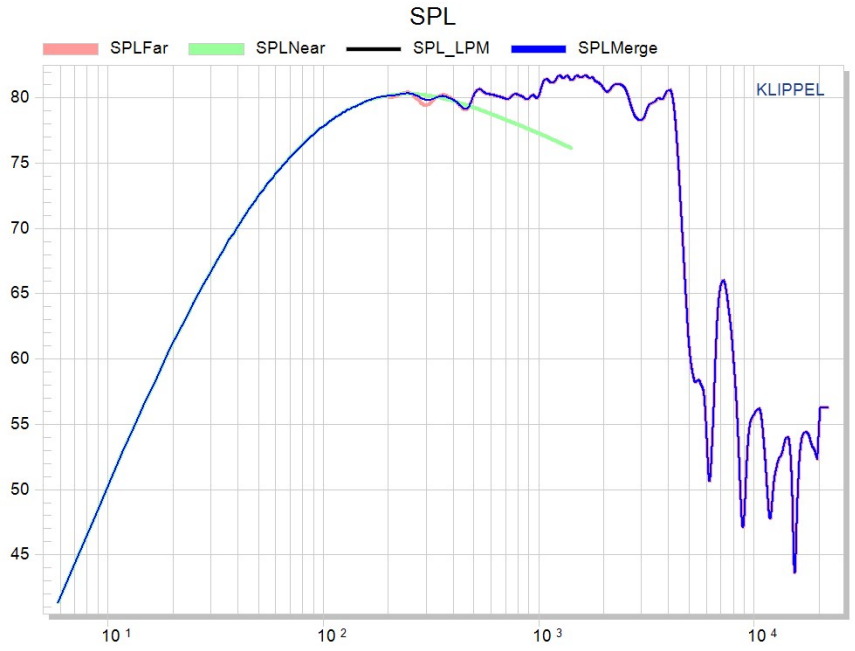


Figure 4.15 – Example (B035) of input curves after merging. Picture exported from Klippel dB-Lab Analyzer.

Table 4.1 – Merging Results

WOOFER	D [cm]	T [s]	RMS [dB]	f_L [Hz]	$f_{ka=1}$ [Hz]	f_{sp} [Hz]	Resolution [Hz]	RMS #points
B001	8.5	0.0078	0.737	128.35	925	527	2.9297	271
B002	8.5	0.0078	0.561	128.35	660	398	2.9297	181
B003	10.8	0.0078	0.462	135.59	600	369	1.4684	316
B004	12.5	0.0078	0.403	128.35	700	416	1.4684	389
B005	13.4	0.0074	0.536	135.59	600	369	2.9297	158
B006	16	0.0074	0.761	135.59	300	221	1.4684	111
B007	16.5	0.0074	0.661	134.82	600	369	1.4684	316
B008	12.9	0.0080	0.454	125	405	265	1.4684	190
B009	13	0.0080	0.649	125	500	316	0.7324	512
B010	13	0.0074	0.813	135.59	700	422	1.4684	384
B011	13.2	0.0074	0.432	135.59	600	369	1.4684	316
B012	13	0.0075	0.964	200	700	451	1.4684	340
B013	13.6	0.0075	0.493	133.33	700	422	1.4684	385
B014	13.5	0.0075	0.425	200	600	404	1.4684	272
B015	13.2	0.0080	0.577	200	600	404	1.4684	272
B016	13.5	0.0080	0.303	250	700	480	1.4684	306
B017	13.2	0.0080	0.452	250	700	480	1.4684	306
B018	12.8	0.0080	0.498	125	800	463	1.4684	459
B019	13	0.0080	0.424	125	700	416	0.7324	785
B020	12.3	0.0068	0.597	147.25	600	375	2.9297	154
B021	12.8	0.0078	0.523	128.35	700	416	1.4684	389
B022	12.8	0.0080	0.59	125	700	413	1.4684	391
B023	12	0.0080	0.505	125	850	488	1.4684	493
B024	13	0.0080	0.45	200	700	451	1.4684	340
B025	13	0.0080	0.494	200	700	451	1.4684	340
B026	21.5	0.0130	0.563	76.92	200	141	0.7324	168
B027	13.7	0.0100	0.611	150	600	375	0.7324	614
B028	11	0.0072	0.499	138.71	500	322	2.9297	123
B029	11.8	0.0074	0.541	200	700	451	2.9297	170
B030	8.4	0.0083	0.52	120.00	1000	563	0.7324	1201
B031	13.4	0.0078	0.565	200	600	404	2.9297	136
B032	13.5	0.0080	0.518	125	700	416	2.9297	196
B033	13.5	0.0080	0.519	200	700	451	2.9297	170
B034	13.15	0.0080	0.279	125	600	363	1.4684	323
B035	12.92	0.0068	0.344	147.25	500	325	1.4684	240

4.3 Subjective Data Results

The dataset results collected during listening test and statistical testing described in Section 3.5.1 are offered in this section. Attributes' ratings are presented with vertically conclusive mean scores plots for different tracks on each system. Uncertainties are displayed in the form of 95% confidence error intervals, i.e. an estimate of the range within which the true means lie, calculated with the CIFcn function in MATLAB®. This graphical impression helps with the decision whether our normalized judgments (0<1) of different subjects can be averaged: if inter-individual variation would be too large, such operation would be imprudent [71].

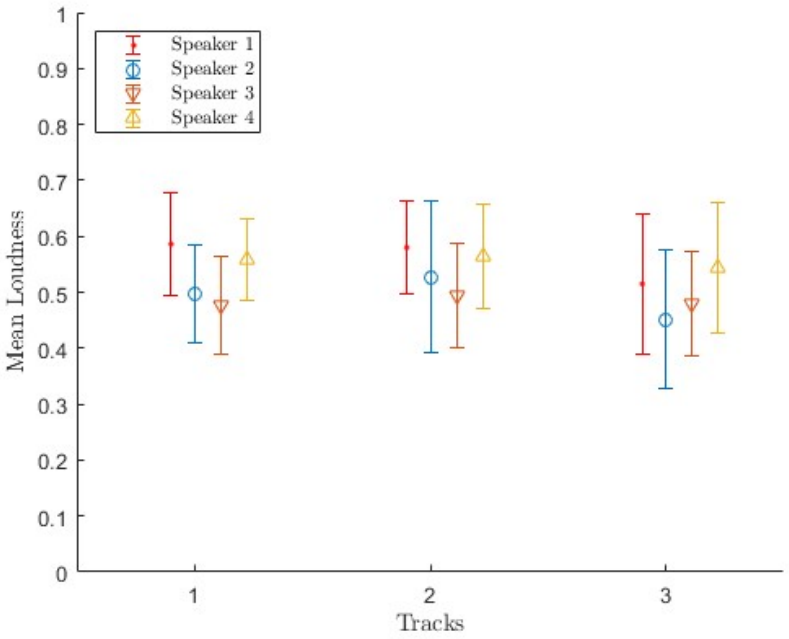


Figure 4.16 – Mean loudness ratings (woofers), 95% confidence interval

Some basic assumptions for statistical analysis were also verified with simple descriptive statistics. We collected ratings from all listeners (denoted as L1 – L15 from now on) into two 15x36 matrices corresponding to the two sessions, circularly shifted back according to DoE logic and finally separated in six 15x12 matrices, containing all attributes' ratings.

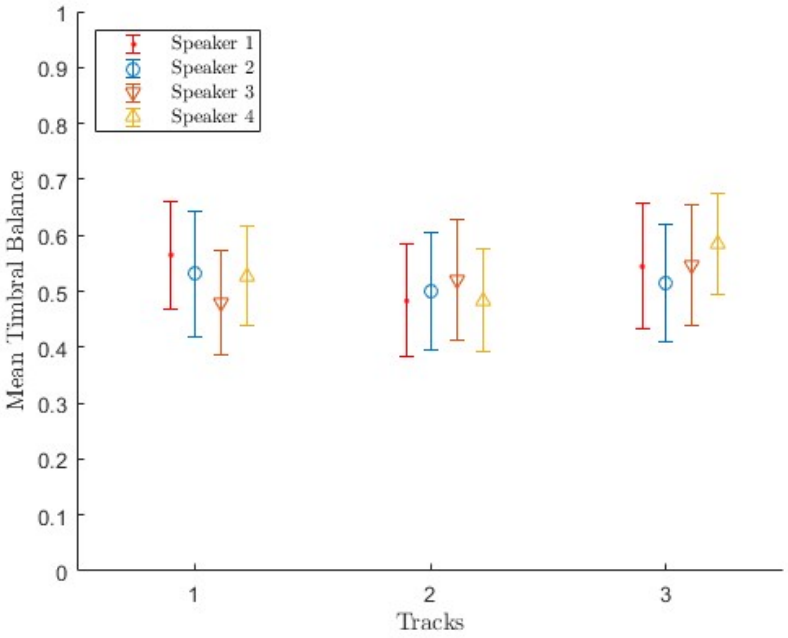


Figure 4.17 – Mean Timbral Balance ratings (woofers), 95% confidence interval

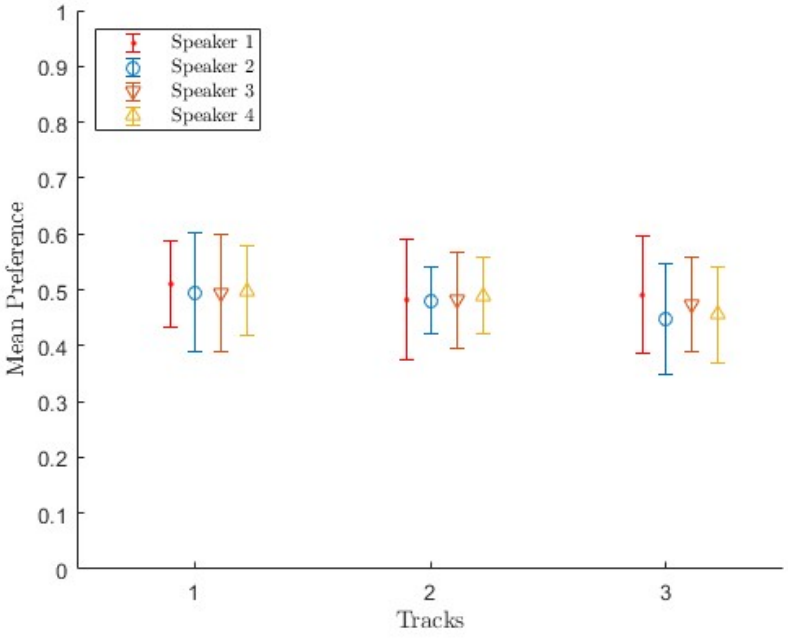


Figure 4.18 – Mean Preference ratings (woofers), 95% confidence interval

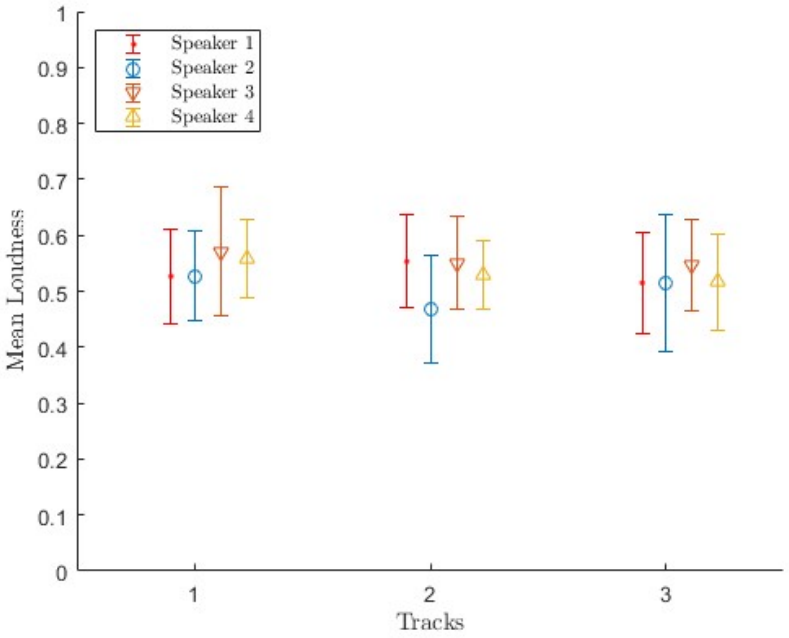


Figure 4.19 – Mean Loudness ratings (tweeters), 95% confidence interval

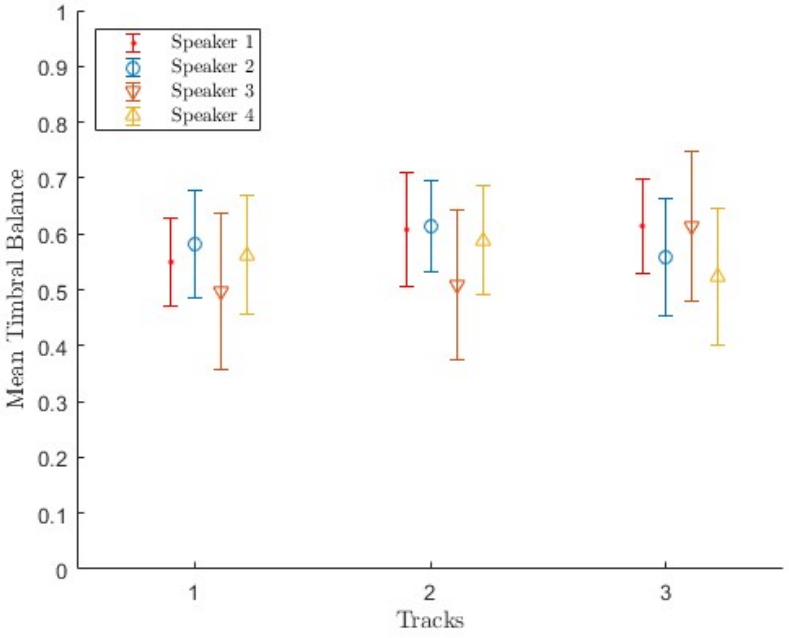


Figure 4.20 – Mean Timbral Balance ratings (tweeters), 95% confidence interval

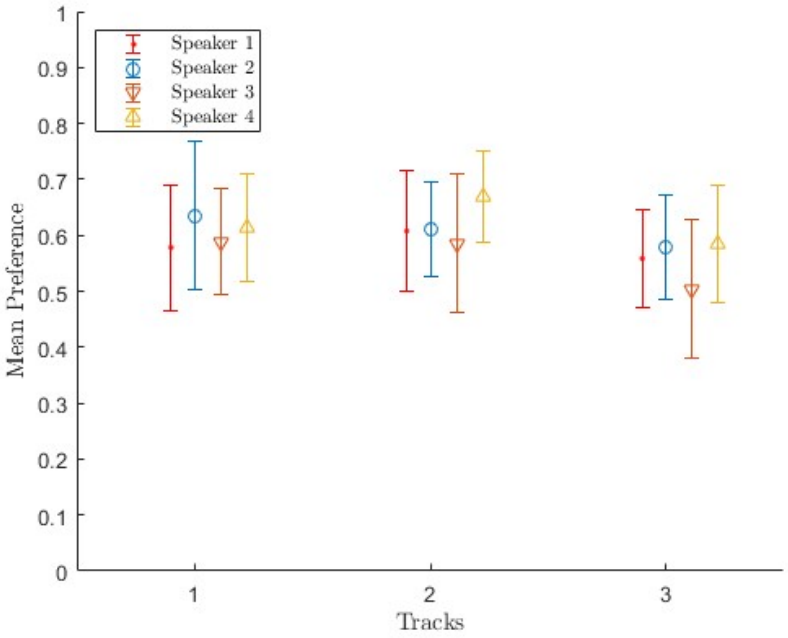


Figure 4.21 – Mean Preference ratings (tweeters), 95% confidence interval

An explorative analysis for each matrix was taken to check the common ANOVA statistical assumptions which is informative on data anyway. We assume the observations independent thanks to statistical experimental design described in Section 3.1.3 and a careful control over all issues mentioned in Section 4.1.1. Homogeneity of (groups) variances (HOV) was checked with a Bartlett and Levene test in MATLAB® [107] indicating a p-value greater than the significance level for all, in favor of the null hypothesis that all are equal. Repeated listeners' ratings, i.e. the dependent variable in each group being compared, should be normally distributed in one-way ANOVA. This was controlled exploiting a MATLAB® normality test package [108] including a Jarque-Bera (JB) test, exploiting the usual p-value principle in the opposite way like HOV test. Test were statistically positive and significant for each group for each attribute in woofers. Same exact considerations apply for tweeters, except for Timbral Balance ratings from L12. In fact, as we see in the correspondent boxplot data distribution this was an extreme case of scoring the exact same 0.5 value for every speaker-track combination. He has been considered an outlier this reason.

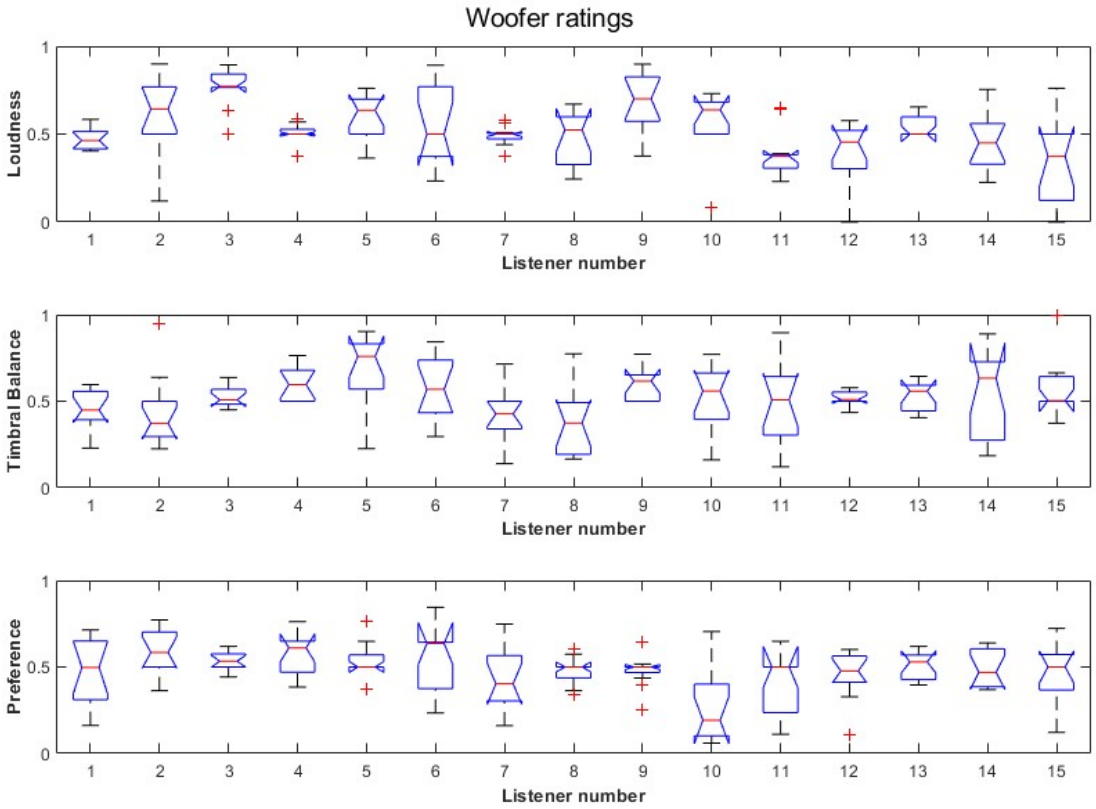


Figure 4.22 – Boxplots for Woofers ratings

By checking interquartile ranges and median values, besides such extreme case for the main box width, we can consider data to be significant in all other cases. Other considerations are about balanced or unbalanced distribution of ratings for each listener. We can see that, for loudness ratings, on woofers dataset, L4,L7, L11, L13 have a median value very close to one of the upper or lower quartiles (25% quantile is even coincident with median for L4, L13) meaning biased ratings towards a particular value. This phenomenon happens also for listeners 12 and 15 of woofer ratings on Timbral Balance and for L6, L11 on Preference. For what concerns tweeter boxplots on Loudness, L1, L3, L10 and 13 behave like that, on Timbral Balance L5, L9 while for Preference on L2, L5, L10, L12, L13.

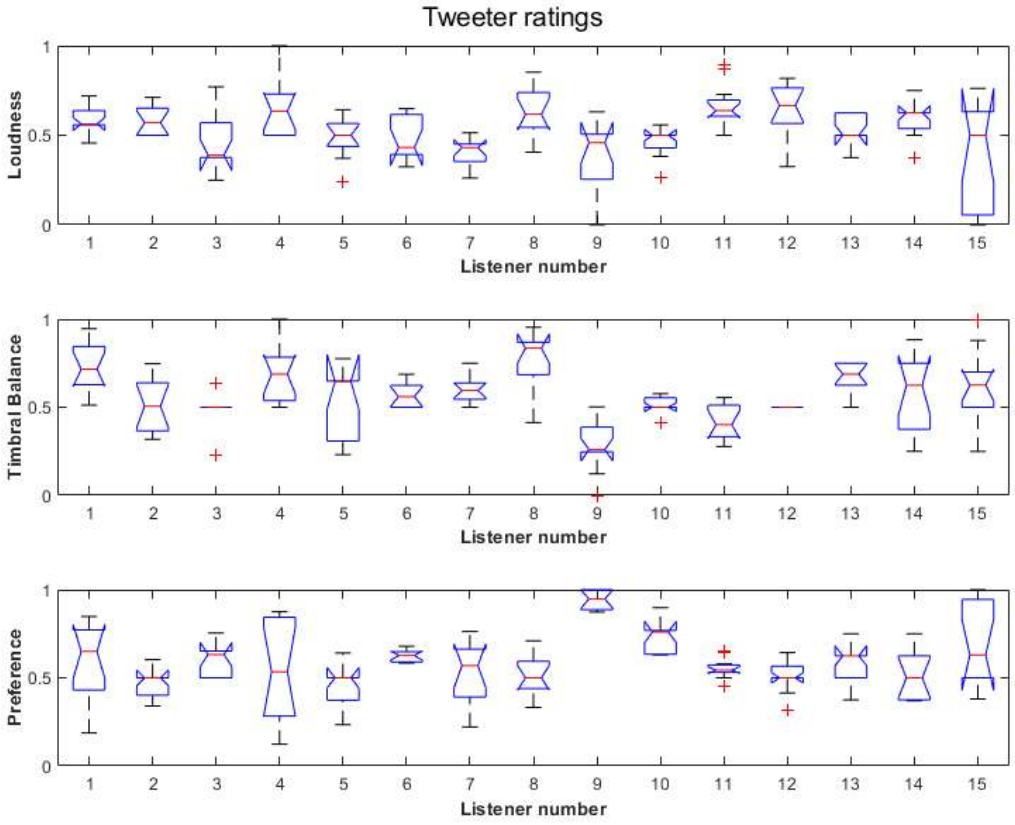


Figure 4.23 – Boxplots for Tweeter ratings

A univariate one-way ANOVA as described in Section 3.5.1 was implemented in MATLAB®. We are comparing deviation related to differences between listeners' ratings mean values and random deviation due to repeated ratings; each data matrix was framed with listeners on column and speaker-track combinations as rows. Because of how ANOVA test is set, a small p-value indicates statistically significant differences between means, i.e. loudspeaker ratings for each subject for each attribute, were not a merely guessing. The compliance of these results is also confirmed by the associated F-values.

Source	SS	df	MS	F	Prob>F
Columns	2.0819	14	0.1487	6.0806	1.2528e-09
Error	4.0352	165	0.0245		
Total	6.1170	179			

Source	SS	df	MS	F	Prob>F
Columns	1.1022	14	0.0787	2.8843	6.4133e-04
Error	4.5039	165	0.0273		
Total	5.6061	179			

Source	SS	df	MS	F	Prob>F
Columns	0.9827	14	0.0702	3.1573	2.0942e-04
Error	3.6682	165	0.0222		
Total	4.6509	179			

Figure 4.24 – Woofer ratings one-way ANOVA; Loudness (top), Timbral Balance (center), Preference (bottom)

Source	SS	df	MS	F	Prob>F
Columns	1.6113	14	0.1151	5.6659	6.6263e-09
Error	3.3516	165	0.0203		
Total	4.9629	179			

Source	SS	df	MS	F	Prob>F
Columns	2.5271	14	0.1805	9.4657	3.6226e-15
Error	3.1465	165	0.0191		
Total	5.6735	179			

Source	SS	df	MS	F	Prob>F
Columns	2.5333	14	0.1810	8.5559	9.5928e-14
Error	3.4896	165	0.0211		
Total	6.0229	179			

Figure 4.25 – Woofer ratings one-way ANOVA; Loudness (top), Timbral Balance (center), Preference (bottom)

Provided that HOV assumption holds, a MATLAB® function *multcompare* [110] allows an interactive HSD testing to quantify *statistically* which groups are more different in means. Results are more-easily presentable here in descriptive way and can also be confirmed by visual inspection of Figure 4.22 and Figure 4.23. As regards woofers, L3 performance resulted different from almost every other in scoring Loudness. On average, L9 results are meaningfully different from L11, L12, L14, L15 while listener 15 resulted far from L2, L3, L5, L9 ratings which appear visually farther. For Timbral Balance, L5 performed very differently especially from L1, L2, L7, L8. In case of Preference scores, we can see that almost 7 listeners behaved

differently from L10. Examining tweeters results, for Loudness ratings, listeners 7,9,15 have significantly different means with respect to L4, and L4, L8, L11, L12 performed differently from L7, L9, L15 in a non-trascurable manner. For Timbral Balance L1 resulted significantly different with respect to 7 other listeners, while L9 has a mean evidently far away from all other thirteen mean scores. Five listeners also have means different from L11. For Preference is evident that L9 has a significantly different mean with respect to all others.

Table 4.2 – Correlation of single listeners with group mean scoring

Woofers, Correlation with Group Mean				**Tweeters, Correlation with Group Mean**			
Listener	Attr. 1	Attr. 2	Attr. 3	Listener	Attr. 1	Attr. 2	Attr. 3
L1	-0.47743	-0.18167	0.053527	L1	0.580155	-0.25912	0.550235
L2	0.4228	0.357196	0.684667	L2	0.352542	0.64095	0.178111
L3	-0.14371	0.265916	-0.68692	L3	0.685827	0.43213	0.434238
L4	-0.05614	0.660338	0.082913	L4	0.254281	0.28479	0.381671
L5	-0.04916	-0.629	-0.61459	L5	0.176552	0.32137	-0.19354
L6	0.248168	0.480616	0.408605	L6	0.281902	-0.14047	0.064056
L7	0.37237	0.354042	-0.04499	L7	-0.1569	-0.21856	0.407831
L8	0.045411	0.086461	-0.57582	L8	-0.11404	-0.13187	0.587321
L9	0.55683	-0.33365	0.485165	L9	-0.45308	0.59657	0.089558
L10	0.472787	0.683758	-0.03866	L10	0.090762	0.594285	0.416059
L11	0.309672	0.366045	0.598015	L11	0.062264	0.63532	-0.04636
L12	0.4525	-0.15077	-0.06447	L12	0.487469	NaN	-0.49471
L13	-0.13496	0.648642	0.611607	L13	0.182562	0.162884	0.676585
L14	0.497814	0.352756	0.208606	L14	-0.21939	0.716344	-0.15596
L15	0.618916	0.104686	0.049947	L15	0.588393	0.329799	0.630589
Mean	0.209058	0.204358	0.077173	**Mean**	0.18662	0.283173	0.235047

A correlation coefficient was calculated for agreement between the mean group ratings and the individual listener ones. High disagreement ($0.07 < r < 0.28$) among assessors showed up. Rating process was quite heterogeneous even though sessions were performed in the same day.

Listeners agreed on the fact that test length was well-calibrated and not exhausting although being the task a bit challenging on their opinion requiring a very expert listening.

4.4 Predictive model and correlation analysis Results

To find connections with listening test results, in the context of predictive modelling, a correlation between each feature and average score for all the 4 speakers was calculated at first. An a priori discarding of features with $|r| \geq 0.66$ for each attribute separately was a first pre-processing for reducing $\mathbf{X}$-matrix K-high-dimensionality. Regression analysis was then carried out in MATLAB® testing a classical linear model, a Support Vector (Machine) Regression (SVR) with a polynomial Kernel and a Stepwise regression with the *fitlm*, *fitrsvm* and *stepwiselm* functions respectively from Statistics and Machine Learning Toolbox™ [111]. The former two relied on a ranking feature process with *Relief* algorithm with total number determinable in advance. The latter englobes automatically a comparison of all possible models up to a certain size according to a specified criterion during construction. A wide range of different solutions were tried tweaking different algorithms' hyperparameters. Typically, one looks at solutions with up to 5-6 regressors. Training and test sets were defined as a 70% and 30% of subjective data respectively randomly permutating elements on the listener dimension of the original three-way structure. After splitting, Eq. 3.30 weighting is applied on the two sets, resulting in two different 12x1 $\mathbf{y}$ vectors for each attribute independently on speaker-track combination. To equal vertical dimensions, we created replicates for $\mathbf{X}$-matrix appending a column with a normalized (0<1) track information number. Feature matrix was the same for training and testing purposes. Three statistical metrics were tested, but, though no implementation errors and statistically significant data, fitting results are not reliable with so coarse speaker sampling, i.e. $\text{rank}(\mathbf{X}) = 4$. Even assuming no linear relationships of different orders with eventually interaction terms good fitting results cannot be found. Negative values for metrics are simply numerical artefacts of too sparse observations to fit data.

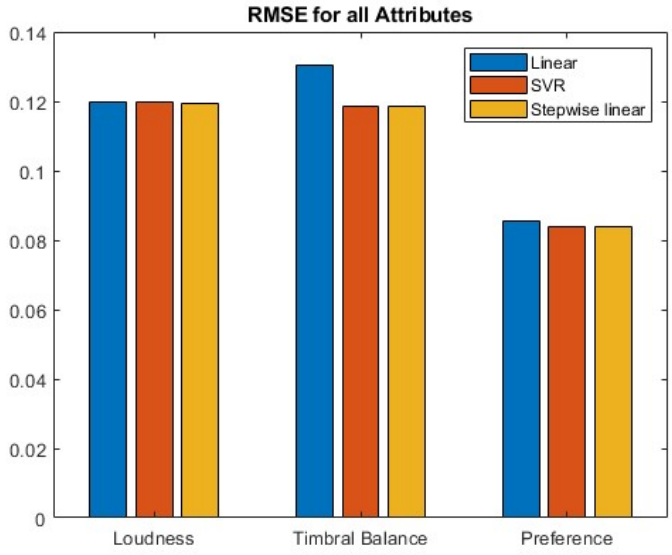


Figure 4.26 – RMSE metric on testing

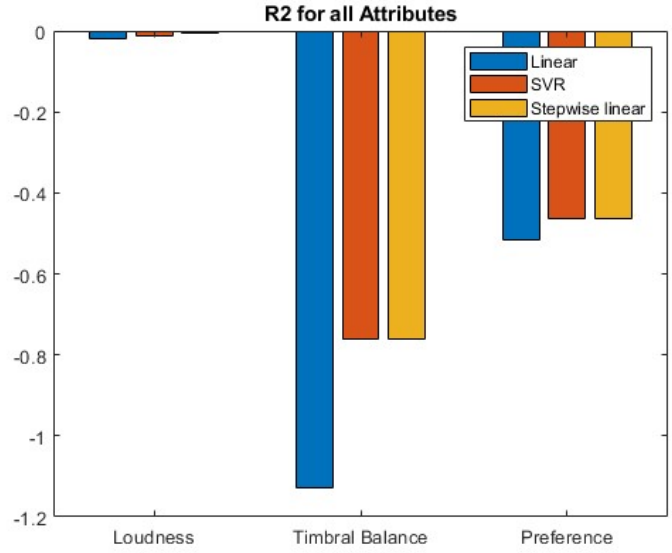


Figure 4.27 – R2 metric on testing

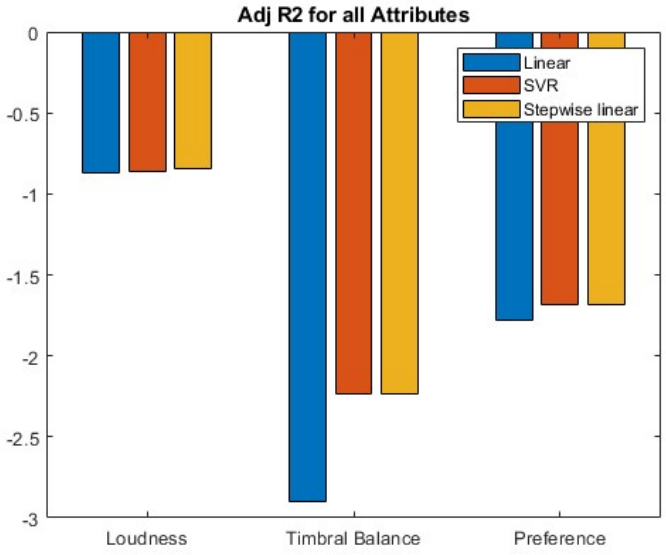


Figure 4.28 – Adjusted R2 metric on testing

The adjR2, weighting the sum of squares of residuals (RSS) and total sum of squares (TSS) ratio with a factor $\frac{N-1}{N-p}$ becomes negative too. In the case of $p = 5$ regressors this means, according to its definition, a fitting approximately twice worse than the simplest mean data model in terms of a multivariate linear model scenario. If one checks the p-value associated with the coefficients of correlation predictors this is confirmed. Synthetic uncorrelated data was generated for $\mathbf{X}$ to simulate a setup with 12 different speakers filling missing rows without replicates. Fitting metrics noticeably improve with the same subjective data especially if polynomial terms up to the 2nd order are included with 4 predictors. Model fitting is theoretically possible with $N = 12$ hypothetical independent points with good results at least from the mere *numerical* standpoint of the metrics. However, a countercheck of regression coefficients showed again not statistically acceptable results but the intercept term.

From such data it is impossible to understand the most representative predictor for explaining ratings and if results are generalizable. Because of how it works, for exploring the underlying structure of the feature space a PLS regression analysis attempt was built in MATLAB® with the *plsregress* function [112]. Values for MSE, Q^2Y_k and cumulative explained var($\mathbf{y}$) by PLS

dimensions were evaluated trying different k for each attribute to choose optimal one. MSE was calculated with a Leave One Out (LOO) validation approach based on 10^3 Monte Carlo repetitions. Due to $\mathbf{X}$ rank deficiency, cumulative sum of var($\mathbf{y}$) exceeded 100% for certain k values. This is another artefact as additional components with respect to the right k explaining no more variance cannot be properly normalized and artificially manifest as deflated values. It was not possible to find an explained var($\mathbf{y}$) greater than 11% or 37% for Timbral Balance and Preference ratings respectively, with even $k \leq 2$, i.e. cumulative variance curve flattens completely in adding more. For an optimal $k = 4$, Loudness resulted in ~53% of explained var($\mathbf{y}$) by 1st component and a ~1% more in the second, which is a negligible contribution. Nonetheless, we tried to have a look for clusters and patterns in scores and loadings plots for such a 1D latent problem, as a summary of data from *both* blocks reading $\mathbf{T}$ set. In this interpretation 99% of var($\mathbf{X}$) mirroring original multidimensional data configuration seems to be explained from the first dimension only. Farthest predictors from center, i.e. with greater loadings, are the one supposed to be more important in the model. Normalized (-1<1) loadings show all features clustered on the first dimension with maximum correlation among them.

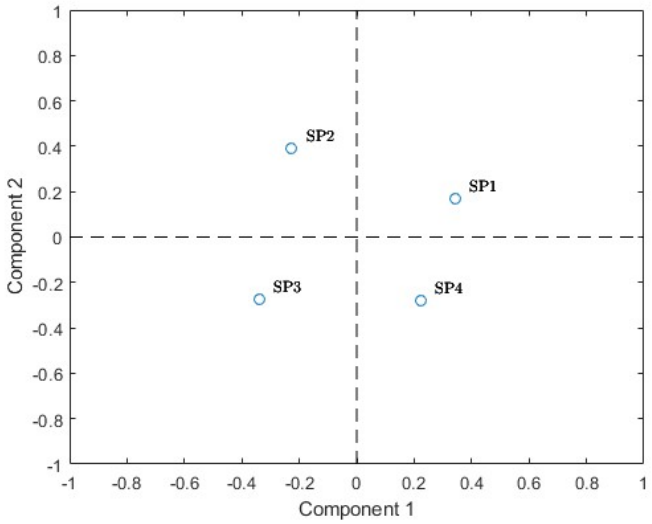


Figure 4.29 – Woofers, Loudness Score plot

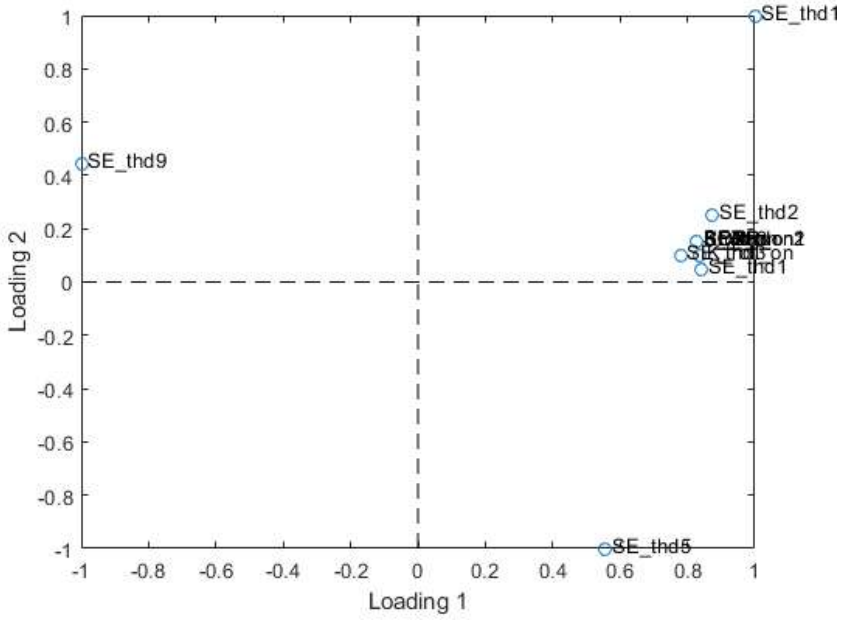


Figure 4.30 – Woofers, Loudness Loadings plot

Of course, due to repetitions in **X**-matrix we can visualize only the 4 observation (rows) in PLS scores. Mapping of speakers I-IV into same factorial space, their relative similarity and location; they seem to be weakly correlated with underlying dimensions and completely four different entities. We got a $R^2 \cong 0.63$ on training data but once again negative on test set showing very poor predictive ability. Prediction residuals exhibited an oscillatory behavior around zero confirming inaccuracy of the model. A power score for each predictor $VIP_p = \sqrt{\left(\frac{P}{\sum_k^K \boldsymbol{\beta}^{\mathbf{T}}\boldsymbol{\beta}\mathbf{t}_{\mathbf{k}}^{\mathbf{T}}\mathbf{t}_{\mathbf{k}}}\right)\sum_k^K \boldsymbol{\beta}^{\mathbf{T}}\boldsymbol{\beta}\mathbf{t}_{\mathbf{k}}^{\mathbf{T}}\mathbf{t}_{\mathbf{k}}\,(w_{pk})^2}$ generally discriminates relevant ones for values >1. Such values and corresponding magnitude of $\boldsymbol{\beta}$ coefficients resulted quite low ($10^{-2} \div 10^{-3}$) describing definitely no influential features in the predictive context.

To not overload the discussion, only woofer dataset results were discussed here as same exact considerations and erroneous trend hold for tweeters under the same methodology. It would be meaningless to show further graphs of this kind or numerical artefacts. It was no possible to collect more data and explore further in this direction. We focused then our attention in

comparing our work with Olive's model based on a set of anechoic measurements [51] to accurately predict speaker Preference, where the proposed estimation is given by:

$$\text{Pref} \;=\; 6.04 - 0.67\text{AAD} - 1.28\text{LFX} - 0.66\text{LFQ} + 7.6\text{SM} \tag{4.4}$$

with *AAD*, *SM*, *LFX*, *LFQ* defined in Section 3.2.1 and calculated for the whole dataset of merged responses. By applying original definitions, the above linear combination happened to be negative for B022, B026, B028, B031 and B035. These number were in fact designed for a full-band frequency response extension in case of 2-way reproduction. When to be applied to our single driver case, to put more emphasis on bass and midrange frequencies more meaningful upper bounds in the summations which define them were inspected.

Table 4.3 - AAD, LFX, LFQ, SM features applied to woofers involved in the listening sessions

	AAD	**LFX**	**LFQ**	**SM**	**Olive**
B034	0.859529846	1.722094	1.581177	0.504908	6.05355847
B035	0.653365465	1.778575	1.853852	0.600614	6.666795987
B014	1.400957738	1.79926	2.252471	0.331704	3.832624494
B017	1.266360817	1.722094	1.482933	0.444913	5.389858633
MAX (all 35)					6.05355847
MIN (all 35)					0.551849232

shows results by modifying all upper bounds to 3500Hz. In Olive's work, listeners were able to discriminate preferences of 7 speaker groups whose main value ranged from ~0.75 for the least preferred to ~6.5 for the most preferred one in a 11-points scale ranging from 0 to 10. After observing a reasonable rating range for prediction with our devices (Table 4.3), we mapped data into Preference range determined by our test results with following formula:

$$\mathbf{x}_{\text{new}} = a + \frac{\big((\mathbf{x} - \min(\mathbf{x})\,(b-a)\big)}{\max(\mathbf{x}) - \min(\mathbf{x})} \tag{4.5}$$

where $\mathbf{x}$ is the vector of synthetized ratings obtained with Eq. 4.4 and (a, b) the Preference ratings min-max interval. We distinguished two analysis cases:

- Case (1): $a = 0.464231, b$ =0.494654 being values of overall *average* preference ratings
- Case (2): $a = 0.059581, b = 0.84541$ being min and max values of all preference ratings without averaging on speaker and tracks

As a reverse-engineering approach we searched for upper bounds producing closest results to our ratings. Precise analytical optimization process is a quite a tough task because of minimizing with respect to a summation. limit Heuristic approach was tweaking values to find a local minimum of the problem with the lowest Euclidean distance between our 4 ratings and the predicted ones. This is done in a certain a-priori range assumption (3.5kHz -5kHz).

Table 4.4 – Heuristic algorithm results

	(1) AAD	LFX	NBD	$\|\cdot\|_2$	**(2)** AAD	LFX	NBD	$\|\cdot\|_2$
TRIAL 1	3500	3500	3500	0.033633	3500	3500	3500	0.511947494
TRIAL 2	4000	3500	4000	0.033618	4000	3500	4000	0.504387354
TRIAL 3	3650	3500	3650	0.033617	3650	3500	3650	0.509891089
TRIAL 4	5000	5000	5000	0.034624	5000	5000	5000	0.511254211
TRIAL 5	4500	4000	4500	0.034004	4500	4000	4500	0.495446823
TRIAL 6	3548	3548	3548	0.033632	3548	3548	3548	0.509094575
TRIAL 7	3514	3514	3514	0.033615	3514	3514	3514	0.510569195

Appreciable differences in the metric are of the magnitude order 10^{-3} and 10^{-2} for case (1) and case (2) respectively, basically providing the same results, confirming the right range. However, case (2) provides a more reasonable variation. It must be noted that in our tests, drivers were all equalized to have the same cut-off slope at 4kHz. All these features except for LFX, having however the greater impact in the weighted combination, basically quantify frequency response flatness deviations .We are estimating a $\|\cdot\|_2$ distance with ratings obtained

for woofers sounding as shown in Figure 4.31. Green curve represents the average of the 4 responses, which in turn has an average value of ~81dB; if we look for the -6dB frequency it coincides approximately with 3.5kHz (3514Hz) making our previous assumption reasonable.

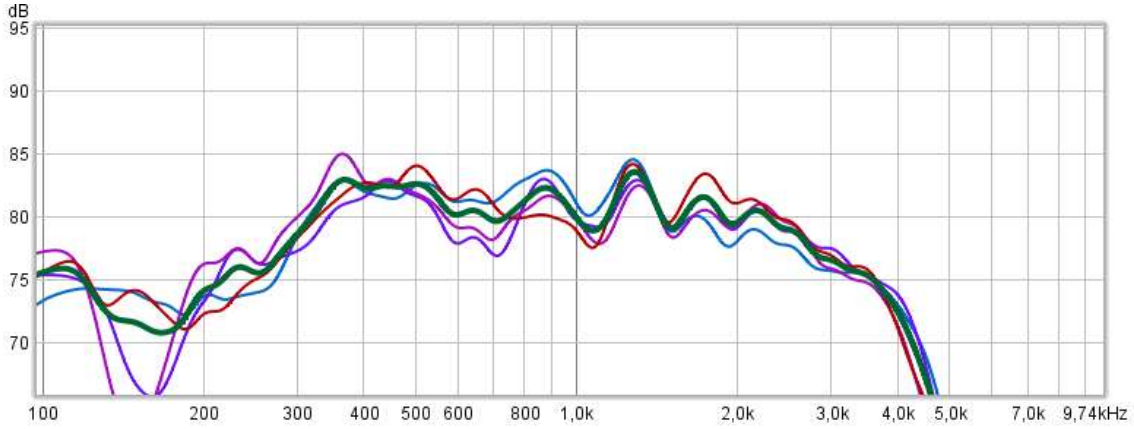


Figure 4.31 – Woofers' measured SPL on-axis and average response (green)

The modified *LFX* quantifies exactly a -6dB frequency w.r.t mean amplitude level in the 300Hz-3.5kHz band and this means should confirm the effectiveness of our equalization for conducting single driver tests. Speaker B035 seems to be the predicted as the most preferred one, by Olive's model, as it must be because it is a far-superior product. This did not happen according to our test results. About this, after test sessions listeners asked for letting them see the devices under test. Once curtains were removed, nearly half of the panel commented that the Audison TH 6.5 II Sax (B035) should be without doubt the most preferred, but, also from experience of previous listening tests, is still a mystery why its superiority never manifests when is compared with other much lower budget options. The other three products ranking effectively reflected their expectations. AAD, LFX, LFQ, SM features (@3.5kHz upper bounds) were applied to our real on-axis equalized measurements too. Smoothing was adjusted to 1/24 octave resolution for comparison. If comparing with the same applied to merged curves we can notice a discrepancy on the absolute value of features. However, the hierarchical rating of devices is once again confirmed according to the quality of the devices involved.

Table 4.5 – Loudspeaker predicted Preference ranking Case 1 and Case 2

	Ranking Test one	**Ranking (Average our test)**	**CASE 1 @3514Hz, 3514Hz, 3514Hz**	**CASE 2 @4500Hz, 4000Hz, 4500Hz**	**CASE 2 @3514Hz, 3514Hz, 3514Hz**
Lowest	B014 (III)	B035 (II)	0.480405	0.493705	0.477365
	B017 (IV)	B017 (IV)	0.488154	0.691352	0.680114
	B034 (I)	B014 (III)	0.491537	0.724590	0.764915
Highest	B035 (II)	B034 (I)	0.494654	0.845410	0.845410

Table 4.6 – Loudspeaker preference ranking; predicted versus listening test results

Speakers	Features								Preference	
	AAD@3.5kHz		**LFX@3.5kHz**		**LFQ@3.5kHz**		**SM**		**Olive**	**Test (x10)**
	On-ax	Merge	On-ax	Merge	On-ax	Merge	On-ax	Merge		
B034 (SP1)	3.6707	0.8595	2.3351	1.7221	2.5668	1.5929	0.7563	0.5049	4.6459	4.9465
B035 (SP2)	3.6431	0.6534	2.4624	1.7786	2.3960	1.8522	0.7756	0.6006	4.7625	4.6423
B014 (SP3)	3.4822	1.4010	2.4316	1.7993	3.3581	2.2663	0.7705	0.3317	4.2354	4.7922
B017 (SP4)	4.1606	1.2664	2.4095	1.7221	3.0552	1.4821	0.7775	0.4449	4.0608	4.7435

Yet being the order of magnitude of ratings very comparable, such synthetic data seem to be even "more correct" in this sense with respect to real results coming from our tests. The SM value for our on-axis measurements is really the same for all the woofer, and this may be due once again to the specific equalization chosen as this value is calculated on the full-audio band. Table 4.7 shows the absolute range of predicted ratings for all woofers along with correspondent features values. In the case minimum range goes outside the supposed range and 3.5Hz still is the more meaningful upper frequency limit. The influence of electromechanical characteristics of electrodynamic transducers on sound quality perception was also investigated. Correlation analysis is carried out for both the 35 predicted Preference scores and average (on speaker) subjective scores for the different sound quality attributes involved in the test. As regards most mentionable values from Table 4.9, mechanical mass exhibits a negative r with all attributes scores especially for preference ratings.

Table 4.7 – Min and Max Preference Range and feature values for Case 2

Case 2					
	AAD@4.5kHz	**LFX@4kHz**	**LFQ@4.5kHz**	**SM**	**Olive Pref.Rating**
B034	0.859529	1.722093	1.592922	0.504907	6.045806437
B035	0.653365	1.778575	1.852182	0.600614	6.667897969
B014	1.400958	1.799259	2.266290	0.331704	3.82350379
B017	1.266361	1.722094	1.482068	0.444913	5.390429383
MAX (all 35)					6.65663332
MIN (all 35)					0.1084991
	AAD@3.5kHz	**LFX@3.5kHz**	**LFQ@3.5kHz**	**SM**	**Olive Pref.Rating**
B034	0.859529	1.722093	1.581176	0.504907	6.05355847
B035	0.653365	1.778575	1.853851	0.600614	6.666795987
B014	1.400957	1.799259	2.252471	0.331703	3.832624494
B017	1.266360	1.722093	1.482933	0.444912	5.389858633
MAX (all 35)					6.667897969
MIN (all 35)					0.59470148

The phenomenon translates even more strongly on Q_{ms}.This may due to the fact that the heavier is the moving part, the slower oscillation reaches the steady-state amplitude, with damping factor proportional to $M_{ms}^{-0.5}$ and Q_{ms}^{-1} (Eq. 2.21). Lighter woofer cones are expected to reproduce higher frequencies better and this could explain a decrease of perceived Timbral Balance with a M_{ms} increase. If for a short pulse signal $Q_{ts} \approx Q_{es}$ [31] with a long signal the moving part has time to oscillate up to the steady-state amplitude. This is what should matter for our stimuli, making a significant contribution for Q_{ms}, i.e. being smaller because the two Q sums in parallel. In this sense, a strong negative correlation with Q_{ms} seems compliant with a positive one with Q_{ts} for two of the three attributes although being weaker in the latter case. Nearby f_s the force-factor has generally a greater impact on the steady-state process and a negative correlation with all attributes, especially preference, seems contradictory in this context. A high Bl commonly translate into better transients sounds as motor faster reacts to the cone, but this may not necessary imply a higher perceived loudness. Time required for reaching a steady-state amplitude depends instead on $\frac{Bl}{M_{ms}}$ that was quite comparable for our 4 devices. In low-frequency modelling that ratio goes squared in increasing the passband

efficiency, that is, the fraction of electrical input power converted to sound waves. This could potentially explain a strong positive correlation with all attributes.

Table 4.8 – Min and Max values for lumped parameters in the experiment for both 35 and 4 woofer dataset

	f_s [Hz]	M_{ms} [g]	Q_{ms}	Q_{ts}	Bl $[\frac{\mathbf{N}}{\mathbf{A}}]$	$\frac{Bl}{M_{ms}}\left[\frac{\mathbf{N}}{\mathbf{A}\cdot\mathbf{g}}\right]$
Max(35)	153.7	61.187	13.63	1.205	19.44	1.0256
Min(35)	27.7	3.7050	1.542	0.23	3.218	0.2022
CK165	63.9	13.446	3.215	0.571	4.906	0.3649
TH6.5II	57.9	23.767	4.022	0.443	8.163	0.3434
ESK 165.5	74.7	12.964	3.396	0.719	4.642	0.3581
MP165	72.6	15.267	4.469	0.687	5.317	0.3483

Table 4.9 – Electro-mechanical parameters and attribute ratings correlation

Linear Parameters				
	Loudness	**Timbral Balance**	**Preference**	**Olive's predicted Preference (35 woofers)**
f_s [Hz]	-0.4181	0.3504	0.1480	-0.6719
M_{ms} [g]	-0.3590	-0.4072	-0.7934	0.4406
Q_{ms}	-0.5851	-0.3486	-0.9070	0.2061
Q_{ts}	-0.2129	0.2992	0.4230	-0.1962
Bl $[\frac{\mathbf{N}}{\mathbf{A}}]$	-0.3304	-0.4089	-0.7775	0.1766
$\frac{Bl}{M_{ms}}\left[\frac{\mathbf{N}}{\mathbf{A}\cdot\mathbf{g}}\right]$	0.6517	0.2808	0.9447	-0.5966

Correlation was carried out on the enlarged dataset too. This should provide a quite wider scenario of devices possibly existing in the market in terms of parameters ranges (Table 4.8) still exploitable for the sole synthetized preference ratings comparison. Scores exhibited quite divergent results with the only notable negative correlation with f_s and a relatively weak

negative $\frac{Bl}{M_{ms}}$ correlation. The signs trend is completely reversed. This may depend either on the fact that no frequency equalization was implied in Olive's model construction or on the fact that such ratings are obtained from merged responses, i.e. not completely true measurements. However, in a woofer frequency response, frequency response below f_s starts to roll off with a mathematically predictable 2nd order high-pass behavior with linear parameters. The lower f_s, the deeper the low-frequency extension, which should be high desirable. In case of a sealed box enclosure for drivers, an equivalent shifted up f_s can be calculated, scaling by a factor $\sqrt{1+\frac{V_{as}}{V_{box}}}$ the free-air scenario. Depending on cabinet size, air acts like a spring contributing to speaker dampening and the equivalent Q factor is incremented by the same scalar factor as behaves a third parallel lossy effect [79]. If one accounts for this for our test ratings, Timbral Balance correlation results unaltered for both f_s and Q_{ts} (r = 0.3585, r = 0.2881) while basically nullifies with Loudness. Sealed box correction increase instead correlation with preference (r = 0.4402, r = 0.5890 for f_s and Q_{ts} respectively). Correcting synthetized ratings, as if such hypothetical test were conducted with drivers in our $V_{box} = 17l$ boxes, the exact same trend is maintained with Preference ratings.

Results partially disagree with the only previous work present in literature related to other mid-bass transducers [31]. In that case analysis was carried out with many differences. Firstly, frequency response was equalized to a common target with customized inverse filter design to remove steady-state linear distortions: any potentially residual audible effect is so due to transient distortions, which was the aim of the work. Besides such equalization, drivers were tested with a playback of percussive-like stimuli with a very short attack and decay time and wide dynamic range. This exhibited a strong *positive* correlation of f_s ($r = 0.74$) with Preference comparable in magnitude with Olive's predicted results. Finally, all these devices explicitly operated in a convenient linear parameter membrane excursion so that one can neglect influence nonlinear distortion upon the subjective Preference in favor to useful signal.

4.5 Conclusions and remarks

In this chapter we presented the experimental and simulative outcomes of our methodology. As regards subjective results, conclusions are stressed out along with collected listener comments after test with which we can speculate some hypothesis. Confidence intervals on mean ratings seemed acceptable even though data were quite clustered around the middle of the scale. Almost all assessors mentioned that they tended to create an internal reference in scoring attributes according to how they were used to perform listening tests because absolute evaluation of single drivers sounded "unnatural" to them. This is because devices were perceived differently with respect to usual multiple-way reproduction scenarios where a crossover network is realized, and responses cut accordingly. Some assessors biased slider positioning when started to recognize all one-way devices. This potentially explains most of the ratings variance concentrated in that range. Maybe our evaluation procedure requires a proper-designed training phase, even with an expert panel along with a more careful GUI planning. Testing single drivers was strictly related to the idea of a feature-based approach and we still do not believe it an unrealistic choice to characterize perceived sound. Correlation of single listeners with group mean scoring showed low results which is quite strange for listeners belonging to the same identical background and working context. Perhaps this happened because test was addressed to be quite difficult: the general idea for subjects was that only later true differences among devices showed up. Nonetheless, statistical tests performed good on raw data indicating meaningful differences between their performances.

Objective results were provided following from measurement processing and software implementation. We deem chosen features to be representative and reasonable for the methodology in absolute sense, but the whole dataset suffered from dimensionality limitations. Two different regression attempts in an attribute-wise fashion were presented. All numerical modelling artefacts probably stem from the small data size involved and the extent to which presented plots are trustworthy is questionable. Along with a representative number of scores,

the choice of speakers seems the most critical factor. The question is however a little subtler., as regards how many devices we need to implement a reliable specific regression model and an optimal sampling of this population. For a multivariate regression task, 4 speakers are too few, especially if compared to the high-multidimensional parametrization of the feature space. Statistics books report that a classical regression with p predictors approximately need a sample size greater than $50 + 80 \cdot p$ for overall model testing and then $104 + p$ for testing the influence of a specific regressor. Augmenting sampling population of linearly independent speakers' data must be done accordingly to their controlled parametrization. A possibility of building a training model with a comparable number of speakers and subjective ratings was tried with artificially uncorrelated data from the mere numerical point of view. This is however an intuitive assessment as results always must be considered along with subjective data because of the relations we are searching for. We learned that a more distributed collocation of devices in the features' hyperspace is in order. To grasp the idea, from a simple inspection of Table 4.8 and Table C.0.1 and things discussed in Section 4.4 is evident, even from the only linear parameters that features values are densely collocated in a small range inside the true possible variation defined by a greater representative variety of devices possibly found in the market. A single driver evaluation scenario surely has the advantage of reducing the entire span of the possibilities in this sense, but still more systems are required. Maybe, a smaller feature-dimension from the beginning of the analysis should be considered in a single modelling attempt to make pre-screening operation more significant not contaminating truly correlated variables with the attribute in question. The strategy of splitting training and test data along with three-way structure pre-treatment for computing averages was inherited from general consumer science statistic. Different possibilities for doing this operation also exists and does worth to be tested with bigger datasets. In-depth studies of all issues that can be encountered with common audio data structures enforces validity of our choices for processing operations and statistical analysis presented.

The speaker quantity of devices must be considered also in relation to the other independent variables in the listening test planning. One can estimate an optimal number n of observations stimuli according to the smallest subjective difference that is to be resolved and the accepted levels of Type I (α-value) and Type II statistical errors. Assuming to detect an effect size of 0.05 on normalized magnitude scale units and ratings with a $\mu = 0.5$ and a $\sigma^2 = 0.25$ with a 95% confidence interval, simple calculations with t-distribution show ~860 necessary ratings with the problem tested with μ being average value on speaker and on tracks for single attribute. Fixing n, distribution of other parameters is given by other experimental design selections. Assuming the same panel size, adding one or two more track, 14 or 11 devices are needed, respectively. Maybe number of expert listeners can be increased a bit to retain less devices but having the highest number possible is more desirable provided that all things discussed so far hold. These exemplificative numbers of the problem must be handled accordingly with a DoE or suggest a change of perspective in favor of a more suitable methodology for monophonic single drivers' evaluation. Exceeding reasonable test duration is truly an easy pitfall in this scenario. Finally, especially with descriptive-based experiments, it would be recommended for speakers to cover a more perceptually wide range. The problems are not uncorrelated, especially if a specific test methodologies implied. Moulin's recent study[60] demonstrated several differences between the selected speakers in terms of both timbral and temporal aspects with expert listeners in a QDA context. Our test size was a scientifically reasonable comparable one. However, a predicting modelling attempt was not even mentioned in the work. For these purposes, the current state-of-art of loudspeakers studies involve a huge number of independent variables. As a general idea, Klippel's study (1990) took 45 systems, 53 excerpts, with 94 people taking take part in the session, not to mention Olive's work (2004) requiring 70 speakers and 268 subjects to test generalizability of his research hypothesis. Model construction relies on 13 systems and thousands of Preference ratings were collected in a nearly two-year period. A shorter strategy to obtain comparable

reliable results would be noteworthy. The partially alternative approach proposed here also aimed to disregard evaluation from such demanding size.

In a feature-based modelling scenario, direct sound has demonstrated to be the most important aspect despite a prediction of speakers' sound quality regardless a first reflection and sound power frequency response curve information is inaccurate. It was not possible to effectively test the extent to which the last two components are negligible. Olive's features were applied to our 1/24 smoothed real on-axis equalized measurements in the anechoic listening room, and still the most valuable driver in the test was distinguished with his model. The supposed true ranking confirms also in the case of merged response. All calculation has been carried out without a sound power information with features like LFX even purposely specified for quantifying bass fullness in such curve. Set of merged curves replaced it with legit approximations in the assessment. It can be interesting to check performance of pure anechoic measurements of a greater sample of drivers with important inherent spectral differences among them. These aspects need however a specific dedicated work. The model seemed to hold also with our specific equalization which seemed a valuable choice.

Correlation of subjective data results with electro-mechanical parameters is still an open issue. The problem of assessment of sound quality with them becomes reasonably multidimensional if one wants to also consider the impact of different reproduction settings. Such parameters continue to show only a small part of the big picture. Our conclusions are not necessarily wrong or mutually exclude with literature results which are however limited and related to completely different experimental conditions. We tried to provide an insight to interpret our results despite not being completely trustworthy. Maybe a deeper analysis with different kind of stimuli and a more neutral equalization can be more inferential as well as a more distributed speaker sampling population. A valuable improvement of the current state-of-art scenario is still a long path. As regards the areas of study involved in this work, we hopefully provided a solid base to advance with a still relatively uncovered question in its entirety.

5. Conclusions and Future works

The thesis addressed the most perceptually relevant sound quality characteristics with the awareness of finding relationships with plausible physical description methods of speakers. Full-objective characterization alone does not account human perception or very partially does, and our choices made have been inspired by relevant characteristics to be predicted. A feature-based approach for sound quality assessment has been explored identifying most important topics in literature to motivate paths followed here. An attempt in finding a discriminative sets of such numbers to magnitude estimations of sound perception for a single driver monophonic reproduction never appeared in literature especially about predictive modelling.

Single driver absolute evaluation may unbalance ratings and contaminate data reliability due to an assessor inner expectation of how they should sound in advance. This could be independent on how-well other experimental factors are set. In our work, all the rationale behind subjective test design were deeply covered and meticulously addressed along possible drawback and bias-related issues in every single step of the process. Moreover, it is widely confirmed from previous large loudspeaker studies that nearly all sound quality ratings' variance is explicated by frequency response. Still, nonlinear distortion should not be ignored. The strategy adopted here relies on common direct sound measurements and simple modelling of non-linear distortion speakers' phenomena. Specific classes of representative features were chosen accordingly. These were related to quantifying departures of on-axis response from flatness, the envelope shape of THD curves and a straightforward set of linear and non-linear parameters. The results were not coherent with the judgments collected from listeners in the formal listening tests due to wrong dataset dimensionality issues. Generalizability of results

to system not included in the experiments is an important task for which conclusions were really limited here. Aspects related to reverberant sound may still impose specific constraints on subjective perception of sound quality that cannot be completely discarded. Enlarged dataset provide a small portrait about this aspect and equalization choices. With respect also to all the premises of our work, it was no possible to test the effectiveness of different listening tests methodologies for subjective evaluation of single drivers and gather a greater amount of data. This was another motivation for the merging procedure to gain further insights on the potential errors introduced by our devised methodology. It would be also interesting to understand the extent to which the PLS approach can be exploited. Intention was gaining a compact low-dimensional insight on features' hyperspace elegantly accounting for subjective response variability in the model construction itself. Because of how is built, decomposition attempt was meant to achieve more meaningful results with the same datasets but still exhibited the same problems. This is a powerful approach to be exploited in the audio field for concentrating analysis on one attribute at time sticking with a smart multivariate linear regression model approach. The philosophy behind such a model may fit the task very well.

Linear parameters at the present state-of-art scenario are yet very coarse indicators of sound quality and correlations may not be representative of the entire speaker population. Past literature findings are not generalizable as well. All results given here are only a small screen of the big picture and very context-related to the possibilities of this research. Building a new predictive reliable model is still far from our achievements. We believe that the proposed evaluation scenario, methodology and rationale behind the model setting and correlation analysis is mention-worthy. Caveats have been highlighted in previous chapters; our strategy can be still useful to assess important drivers' characteristics if supplemented by more data and it does worth exploring deeper in future tests. The hope is that this thesis constituted a solid background for infer important relations between speakers' low-level characterization and quality attributes and important experimental design choices as well.

Future work

This thesis aimed at proposing a new methodology to evaluate the perception of sound quality for single drivers in the most time- and cost-effective fashion possible. About that, interesting novel Rapid sensory profiling approaches also exist as efficient ways to obtain reliable data. Due to the less formal structure of these associative methods, the independent variables requirements are not yet fully known when they must fit in the context of finding quantitative relationships between subjective and objective data. It would be of interest to explore this area and more correlated sophisticated statistical techniques as possible alternatives. Still this requires an almost brand-new research path to be opened. A first step is instead a simple extension of the present strategy. A multiple comparison approach providing all stimuli simultaneously to assessors could be evaluated especially for increasing test size not at the expense of duration. Still a QDA-like magnitude rating scales are retained modifying according to the specific attribute with a balanced test design. The choice of x-dependent parameters in the model for nonlinear distortion characterization can be extended to perceptual-relevant metrics for stand-alone tests. Other metrics can be large signal mechanical domain-based numbers relying on time-expensive vibro-acoustic cone scans such as directivity index (DI). The effectiveness of feature-extraction on multidimensional IMD graphs would be an interesting standpoint too. Different alternative is accounting for human auditory perception in a similar way Klippel did almost 30 years ago from measurements by reproducing musical excerpts over speakers and stimuli recorded in the listening position with a HATS. Recordings can be processed with auditory models to be used as input for predicting discriminative sensory descriptors for similar devices. Anyhow, such techniques must be accompanied by an accurate and extensive subjective dataset as well.

Appendix A

Documentation of the programme material

In the following two pages the instructions for conducting the listening test presented to each participant before starting the test are shown. These serves to notify the subjects about of the nature of the task, without giving too much details which could bias the experiment. For example, test applicants could not know anything about technical specification of the speakers or which type of component were involved in the test. As the entire panel was Italian, the second sheet was the one given for the test.

Listening Test general information sheet

Thank you for taking part in one of our listening tests and provide us with your responses. Please make yourself comfortable in the seat provided and read through these instructions before starting the experiment.

In this experiment, you will hear a short excerpt of three different tracks through each speaker in front of you, and you will be asked for rating sound quality attributes on proper scales. The set of stimuli to be rated are presented in sequential manner using a list of predefined or selected attributes with an intensity scale.

The test will comprehend not a complete designed 2-way system, but the evaluation is only about single components such as woofers or tweeters.

Remember that there are normally no right or wrong answers during the listening test, but we are only interested in your responses because of what they tell us about the way sound signals are perceived and/or described. For this reason, there is no "I don't know" option: even if you are aware of which is the right answer, feel free to ask to the interviewer eventually for further explanations or for clarifying any issue related to the current evaluation before going on.

ATTRIBUTE NAME	**DESCRIPTION**
LOUDNESS	That attribute of auditory sensation in terms of which sounds can be ordered on a scale extending from quiet to loud.
TIMBRAL BALANCE	Denotes the balance between bass and treble: - Dark: Excessive bass. Either loud bass or weak treble. - Neutral: Bass and treble are perceived equally loud, there is a balance in the reproduction. This also applies if both bass and treble are equally weak or if the bass and treble are both too loud. If it leads to prominent or soft midrange this is assessed by the Midrange strength. - Bright: Excessive treble. Either loud treble or weak bass The cause for the sound being dark or light can be deduced from the assessments of Bass strength and Treble strength
PREFERENCE	Score the stimuli according to the continuous quality scale (CQS)

Informazioni generali per il test d'ascolto

Grazie per aver partecipato in uno dei nostri test d'ascolto e di fornirci le tue risposte. Per favore mettiti comodo nel posto a sedere che ti viene dato e leggi da cima a fondo queste istruzioni prima di iniziare l'esperimento.

In questo test, ascolterai un piccolo estratto di tre differenti tracce tramite l'utilizzo di ognuno degli speaker che si trovano di fronte a te, e ti verrà chiesto di giudicare attributi sulla qualità del suono con appropriate scale di valutazione. Il set di stimoli da valutare è presentato in maniera sequenziale usando una lista di attributi predefiniti o selezionati con una scala di intensità.

Il test non comprenderà la valutazione di un Sistema a due vie completo ma soltanto di singoli componenti come woofers e tweeters.

Ricorda che normalmente non esistono risposte giuste o sbagliate durante il test d'ascolto, ma siamo unicamente interessati alle tue risposte per via di quello che possono rivelarci riguardo la maniera in cui i segnali sonori sono percepiti e/o descritti. Per questo motivo, non c'è nessuna opzione "non lo so": anche se sei sicuro di quale sia la risposta giusta, sentiti libero di chiedere eventualmente all'esaminatore per ulteriori spiegazioni o per chiarire qualsiasi problema relativo alla valutazione corrente prima di proseguire con il test.

ATTRIBUTE NAME	**DESCRIPTION**
LOUDNESS	Quell'attributo della sensazione uditiva secondo il quale i suoni possono essere ordinati in una scala che si estende da silenzioso a rumoroso.
BILANCIAMENTO TIMBRICO	Denota il bilanciamento tra bassi e alti. - Scuro: Bassi eccessivi. Sia bassi troppo forti che alti deboli. - Neutrale: Bassi e Alti sono percepiti con lo stesso volume, c'è un bilanciamento nella riproduzione. Ciò vale anche se sia i bassi che gli alti sono ugualmente deboli o se i bassi e gli alti sono entrambi troppo forti. Se questo porta a delle frequenze medie leggere o marcate, questo viene valutato dalla forza della gamma media. - Chiaro: Alti eccessivi. Sia alti troppo forti che bassi deboli. La causa per cui il suono può essere scuro o chiaro può essere dedotta dalla valutazione dell'intensità dei bassi o dell'intensità degli alti.
PREFERENZA	Valuta gli stimoli in base alla scala di qualità continua (CQS)

Appendix B

Technical Specifications

Full documentation of the instrumentation used for recording, reproduction and for the experiment setup in general.

B.1 Drivers

Table B.0.1 - Experimental setup: woofers' specifications

Specifications	**Model**	
		Hertz C165 Cento
Power Handling	W peak	210
	W cont.	70
Woofer size mm(in.)		165(6.5)
Voice coil ∅		25(1)
Frequency Response (Hz)		55 ÷ 7k
Impedance		4Ω
Magnet size D x d x h	mm	85 x 32 x 15
	(in.)	(3.35 x 1.26 x 0.59)
Weight of one speaker	Kg(lb.)	0.96 (2.12)

		Audison TH 6.5 II Sax
Power Handling	W peak	300
	W cont.	70
Woofer size mm(in.)		165(6.5)
Voice coil ∅		50(2)
Frequency Response (Hz)		65 ÷ 23k
Impedance		4Ω
Magnet size D x d x h	mm	80 x 54 x 4.5
	(in.)	(3.15 x 2.13 x 0.18)
Weight of one speaker	Kg(lb.)	1.25 (2.76)

		Hertz ESK EV 165.5
Power Handling	W peak	300
	W cont.	90
Woofer size mm(in.)		165(6.5)
Voice coil ∅		25(1)
Frequency Response (Hz)*		50 ÷ 23k
Impedance		4Ω
Magnet size D x d x h	mm	85 x 32 x 17
	(in.)	(3.35 x 1.26 x 0.7)
Weight of one speaker	Kg(lb.)	0.98 (2.16)

		Hertz MP 165.3 Mille Pro
Power Handling	W peak	200
	W cont.	100
Woofer size mm(in.)		165(6.5)
Voice coil ∅		25(1)
Frequency Response (Hz)		45 ÷ 4.5k
Impedance		4Ω
Magnet size D x d x h	mm	95 x 45 x 17
	(in.)	(3.74 x 1.77 x 0.67)
Weight of one speaker	Kg(lb.)	12 (2.65)

* Range refers to common usage as
a two-way system with tweeter ET 26.5

Table B.0.2 - Experimental setup: tweeters' specifications

Specifications		**Model**
		Hertz C26
Power Handling	W peak	120 (Hi-Pass filtered @ 2,5kHz - 12 dB Oct.)
Tweeter size mm(in.)		26(1)
Voice coil ∅		20(0.8)
Frequency Response (Hz)		1.8k ÷ 22.5k
Impedance		4Ω
Magnet size D x d x h	mm	19 x 0 x 3
	(in.)	(0.75 x 0 x 0.12)
Weight of one speaker	Kg(lb.)	0.04 (0.08)
		TH 1.5 II Violino
Power Handling	W peak	200 (Hi-Pass filtered @ 1,8kHz - 12 dB Oct.)
Tweeter size mm(in.)		38(1.5)
Voice coil ∅		34(1.34)
Frequency Response (Hz)		65 ÷ 23k
Impedance		6Ω
Magnet size D x d x h	mm	60 x 36 x 5
	(in.)	(2.36 x 1.42 x 0.2)
Weight of one speaker	Kg(lb.)	0.355 (0.78)
		ET 26.5
Power Handling	W peak	150 (Hi-Pass filtered @ 3,5kHz - 12 dB Oct.)
Tweeter size mm(in.)		165(6.5)
Voice coil ∅		26(1)
Frequency Response (Hz)		2 ÷ 23k
Impedance		4Ω
Magnet size D x d x h	mm	19 x 3
	(in.)	(0.75 x 0.12)
Weight of one speaker	Kg(lb.)	0.03 (0.07)

Hertz MP 25.3 Mille Pro

Power Handling	W peak	120 (Hi-Pass filtered @ 2,5kHz - 12 dB Oct.)
Tweeter size mm(in.)		29(1.14)
Voice coil ∅		25(1)
Frequency Response (Hz)		1.4 ÷ 22.5k
Impedance		4Ω
Magnet size D x d x h	mm	24.5 x 6 x 3.5
	(in.)	(0.96 x 0.24 x 0.14)
Weight of one speaker	**Kg(lb.)**	**0.07 (0.14)**

B.2 Microphones

Table B.0.3 - Experimental setup: measurement room microphone's specifications

Model	**Heartworks M30**
Frequency Range	3Hz -30kHz ±1/-3dB Free-Field
Polar Pattern	Omnidirectional
Sensitivity	34mV/Pa (-29dBV/Pa)
Max Acoustic Input	140dB SPL
Signal-to-Noise ratio:	20dB SPL (A weighted)
Power supply required:	24-48V Phantom Power
Output Connector:	XLR-3 (pin 2+)
Weight:	0.5 lb. (227g)
Dimensions L x D:	229 x 22 mm (9 x .860 in.)

Table B.0.4 - Experimental setup: listening room microphone's specifications

Model	**Beyerdynamic MM1**
Frequency Range	50 - 16,000 Hz ± 1.5 dB
Polar Pattern	Omnidirectional, Diffuse field calibrated
Sensitivity	15 mV/Pa (= -36.5 dBV) ± 1 dB
Max. SPL @ f = 1 kHz, THD= 1%	122dB
Signal-to-Noise ratio rel. to 1Pa:	26dB SPL (A weighted)
Power supply required:	12-48V Phantom Power
Output Connector:	3-pin XLR male
Weight (w/out cable):	88g
Dimensions L x D:	133 x 9 mm

B.3 Power Amplifiers

Table B.0.5 - Experimental setup: measurement room power amplifier

Model	**Crown Macro-Tech MA-5002VZ**
Frequency Response @1W:	20Hz – 20kHz (±0.1dB)
Voltage Gain (8Ω load, rated output @1kHz, maximum level settings, V sensitivity)	0.775V: 132:1 ±6% or 42 dB ±0.5 dB
Input Impedance (Nominally balanced, unbalanced):	20kΩ, 10kΩ (nominally
Load Impedance (Safe with all type of loads)	
Stereo:	2-8 Ω
Bridge Mono:	4-16 Ω
Parallel Mono:	1-4 Ω
Output power @ 1kHz, <0.1% THD+N (1**20VAC, 60 Hz Units)**:	
Stereo (Ω per channel):	2500(2), 200(4), 1300(8)
Bridge Mono (Ω):	5000(4), 4000(8), 2600(16)
Parallel Mono(Ω):	5.000(1), 3.700(2), 2.600(4)
Weight:	77 pounds, 9 ounces (35.2
Dimensions H x D:	13.3 x 40.3 cm
Width:	EIA RS-310-B Standard 19-inch rack

Table B.0.6 - Experimental setup: listening room power amplifier

Model	**Hypex FusionAmp FA501**
Frequency Response @1W:	20Hz – 20kHz (±0.1dB)
Inputs:	1x Balanced 3 poles XLR 1x High level 4 poles input
Outputs:	1x Single-Ended Stereo RCA 1x Balanced 3 poles XLR (Direct output)
Mains voltage Input:	100-120Vac / 200-240Vac ±10%
Input power:	650W max
Output power in 4Ω @ <0.1% THD+N:	1x 500 Wrms
Output power in 8Ω @ <0.1% THD+N:	1x 270 Wrms
THD(D+N) (at -1dBFS) AD/DA:	-100dB
THD(D+N) (at -1dBFS) DA:	-102.5dB
Weight:	925g
Dimensions H x W x D:	280 x 135 x 55 mm

B.4 A/D and D/A Converter

Table B.0.7 - Experimental setup: 30 Inputs/30 Outputs channels, 4 mic/instrument Preamps, 24-Bit/192 kHz resolution Digital Audio USB interface

Model	**RME Fireface UFX II**
	Analog I/O
	12 Inputs/12 Outputs
Type:	
Input (Microphone In 9-12, front):	XLR, balanced
Input (AD, Line In 1-8, rear):	6.3 mm TRS jack, electronically balanced
Input (Instrument In 9-12, front):	6.3 mm TS jack, unbalanced
Output (Line Out 3-8, rear):	TS jack, 6.3 mm TRS jack, servo-balanced
Output (Stereo Monitor XLR 1-2):	XLR, balanced
Output (Stereo Monitor Output Phones 9-12):	2 x 6.3 mm TRS stereo jack, unbalanced
Level (maximum nominal):	
Input (Microphone In 9-12, front):	Gain 0 dB: +18 dBu, Gain 75 dB: -57 dBu
Input (AD, Line In 1-8, rear):	+19 dBu
Input (Instrument In 9-12, front):	Gain 8 dB: +21 dBu, Gain 50 dB: -21 dBu
Output (Line Out 3-8, rear):	@4dBu:+13 dBu, @10dBV:+2 dBV
Output (Stereo Monitor XLR 1-2):	@24dBu:+24 dBu, @-10 dBV:+2 dBV
Output (Stereo Monitor Output Phones 9-12):	High: +19 dBu, Low: +2 dBV
Input Impedance:	
Input (Microphone In 9-12, front):	3.4kΩ
Input (AD, Line In 1-8, rear):	8 kΩ unbalanced, 12 kΩ balanced
Input (Instrument In 9-12, front):	1MΩ
Output Impedance:	
Output (Line Out 3-8, rear):	75Ω unbalanced, 150Ω balanced
Output (Stereo Monitor XLR 1-2):	75Ω unbalanced, 150Ω balanced
Output (Stereo Monitor Output Phones 9-12):	2Ω
A/D and D/A resolution:	24 bit

Frequency Response:	
Input (AD, Line In 1-8, rear):	@ 44.1 kHz, -0.1 dB: 5 Hz – 20.8 kHz
	@ 96 kHz, -0.5 dB: 3 Hz – 45.8 kHz
	@ 192 kHz, -1 dB: 2 Hz – 92 kHz
Input (Instrument In 9-12, front):	same as above
Input (Microphone In 9-12, front):	same as above
Output (Line Out 3-8, rear):	@ 44.1 kHz, -0.5 dB: 5 Hz – 20.8 kHz
	@ 96 kHz, -0.5 dB: 5 Hz – 45 kHz
	@ 192 kHz, -1 dB: 5 Hz - 89 kHz
Output (Stereo Monitor XLR 1-2):	same as above
Output (Stereo Monitor Output Phones 9-12):	same as above
THD+N:	
Input (AD, Line In 1-8, rear):	< -104 dB, < 0.00063 %
Input (Instrument In 9-12, front):	same as above
Input (Microphone In 9-12, front):	same as above
Output (Line Out 3-8, rear):	< -104 dB, < 0.00063 %
Output (Stereo Monitor XLR 1-2):	same as above
Output (Stereo Monitor Output Phones 9-12):	same as above
Dynamic Range (all channels):	115 dB RMS unweighted, 118 dBA
Channel separation (all channels):	> 110 dB

Digital I/O

	Up to total 18 Inputs/18 Outputs
Type:	AES/EBU I/O transformer-balanced
	ADAT I/O 2 x 8 channels 24 bit
	S/PDIF I/O optical, IEC 60958
Sample rate:	up to 192kHz
Channels:	
@44.1/48 kHz	2/2 AES/EBU, 16/16 ADAT
@88.2/96 kHz	2/2 AES/EBU, 8/8 ADAT
@176.4/192 kHz	2/2 AES/EBU, 4/4 ADAT

On-board Digital Mixer

Type:	DSP-based, digital real-time mixer, multiple latency-free monitor
Routing and Mixing:	Independent Routing and Mixing of input and playback channels to all physical outputs, TotalMix FX [103]
Status:	Levels adjusted according to the calibration procedure

Generals

Power supply:		Internal switching PSU, 100 - 240 V AC, 36 Watts
Idle power consumption:		19 Watts
Typical power consumption:		22-25 Watts
External Clock:		75Ω BNC Word Clock Input & Output
MIDI:		2 x MIDI I/O via 5-pin DIN jacks
Dimensions including rack ears (W x H x D):		483 x 44 x 210 mm (19" x 1.73" x 8.5")
Total depth:		240 mm (9.4”)
Weight:	Kg(lb.)	3 (6.6)

***** Output levels @0dBFs

Appendix C

Feature Extraction results

Table C.0.1 – Woofer Feature Extraction results

WOOFER	CK165	TH6.5 II	ESK 165.5	MP 165.3
R_e(x=0) [Ω]	3.09	3.79	3.1	3.15
L_e(x=0) [mH]	0.199	0.349	0.19	0.294
L_2(x=0) [mH]	0.297	0.337	0.268	0.39
R_2(x=0) [Ω]	1.48	2.34	1.47	1.75
Bl(x=0) [N/A]	4.906	8.163	4.642	5.317
K_{ms} (x=0) [N/mm]	2.17	3.15	2.86	3.18
C_{ms} (x=0) [mm/N]	0.461	0.317	0.35	0.314
M_{ms} [g]	13.446	23.767	12.964	15.267
R_{ms} [kg/s]	1.679	2.152	1.546	1.559
S_D [cm^2]	128.74	128.44	147.41	136.85
Q_{ms}	3.215	4.022	3.396	4.469
Q_{es}	0.694	0.492	0.881	0.776
f_s	63.9	57.9	74.7	72.6
Q_{ts}	0.571	0.438	0.72	0.661
V_{as} [l]	10.8163	7.41	10.7706	8.3317
L_m [dB]	88.13	86.7	89.1	88.18
L_{nom} [dB]	89.24	86.94	90.21	89.22
L_{x0} [mH]	0.2503772	0.375055	0.2434451	0.3468253
L_{x1} [mH/mm]	-0.0246764	-0.0016316	-0.0271912	-0.024916
L_{x2} [mH/mm^2]	0.0003206	0.0014973	0.0011335	0.0003108
L_{x3} [mH/mm^3]	0.0002656	0.0000013	0.0003473	0.0001967
L_{x4} [mH/mm^4]	-0.0000062	-0.0000306	-0.0000272	0.0000032
Bl_0 [N/A]	4.940000	8.163	4.62	5.32

Bl_1 [N/Amm]	0.017674	0.008139	-0.0736965	-0.0523871
Bl_2 [N/Amm^2]	-0.0949433	-0.0221246	-0.074991	-0.0238318
Bl_3 [N/Amm^3]	0.0003964	0.0005152	0.0008243	0.0006828
Bl_4 [N/Amm^4]	-0.0001008	-0.0001567	-0.0008613	-0.0007986
K_{ms0} [N/mm]	1.7660277	14.875094	2.1741285	2.0968978
K_{ms1} [N/mm^2]	-0.0042663	-0.0174955	-0.0024996	0.0227445
K_{ms2} [N/mm^3]	0.013397	0.0119012	0.061249	0.0299759
K_{ms3} [N/mm^4]	-0.000665	0.0005094	0.003658	0.0005779
K_{ms4} [N/mm^5]	0.0003717	0.0000889	0.0005367	0.0001925
SD (ON)	4.27336e-05	3.97303e-05	4.72876e-05	3.55763e-05
K_{roll} (ON)	13893.6270	13868.7290	13797.4000	14001.2930
SC(ON)	4.27336399	3.97303858	4.72876744	3.55763627
SS(ON)	13893.6270	13868.7290	13797.4000	14001.2930
SK(ON)	2.66082117	2.59785450	2.64324789	2.59185814
SEP(ON)	0.98252219	0.98287359	0.98312128	0.98279521
SF(ON)	0.82376415	0.82993884	0.82987836	0.82950402
SCR(ON)	2.27960596	2.39873599	2.31465657	2.40653641
SSL(ON)	-0.00374481	-0.00365626	-0.00375675	-0.00363159
SCTRsub1(ON)	0.19124526	0.26215319	0.28009718	0.22351036
SCTRsub2(ON)	0.20005294	0.11873196	0.12562698	0.25278199
SCTRsub3(ON)	0.26533781	0.19613690	0.25792034	0.18950787
SCTRsub4(ON)	0.19007568	0.13829204	0.20731043	0.20847813
SCTRsub5(ON)	0.42233535	0.69091634	0.37655596	0.48620158
SCTRsub6(ON)	1.09583739	0.79904606	1.05428944	1.00309098
AAD3.5k (ON)	4.00708873	3.83332692	3.72737916	4.49122457
SE sub1(THD)	8508.26600	8492.48800	8310.37400	8630.52600
SE sub2(THD)	13861.9580	13237.5130	13544.6340	14004.6350
SE sub3(THD)	26374.5760	26965.8190	26766.4390	26245.6040
SE sub4(THD)	59307.4810	59441.9190	59726.9639	60133.6970
SE sub5(THD)	117239.400	121948.700	119788.723	118304.877
SE sub6(THD)	234744.560	234243.468	236582.071	232568.734
SE sub7(THD)	456267.511	451696.888	455946.892	453106.219
SE sub8(THD)	615656.979	556985.331	611414.882	564247.493
SE sub9(THD)	483608.378	504476.250	504872.817	483798.471
SE sub10(THD)	216670.857	213304.412	214358.262	214898.865
SF(THD)	0.96842087	0.97205348	0.97350727	0.96285014

Table C.0.2 – Tweeters' Feature Extraction results

TWEETER	Hertz C26	TH1.5 II	ET 26.5	MP 25.3
Z_{nom} [Ω]	4	6	4	4
f_{low}[Hz]	1800	800	2000	1400
f_{high}[Hz]	22500	26000	23000	22500
Sensitivity [dB SPL]	92	93	92	91
SC(ON)	3.55763627	3.55763627	3.55763627	3.55763627
SS(ON)	16702.6990	17028.7270	16996.7640	17084.2420
SK(ON)	10283.8769	10481.8316	10501.5343	10444.1199
SEP(ON)	5365.89287	5466.54586	5419.68420	5461.71277
SF(ON)	1.87676231	1.83521511	1.85503583	1.83554765
SCR(ON)	0.99793731	0.99820909	0.99793225	0.99800976
SSL(ON)	0.96819268	0.97119631	0.96655510	0.96788245
SCTRsub1(ON)	1.14134995	1.12355021	1.12764966	1.14345579
SCTRsub2(ON)	0.00056440	0.00100031	0.00104470	0.00089795
SCTRsub3(ON)	0.75123403	0.82552115	0.69492170	0.67004932
SCTRsub4(ON)	0.63644120	0.99014325	0.56905970	0.70907521
SCTRsub5(ON)	0.84247375	0.68736343	1.02952007	0.74441846
SCTRsub6(ON)	0.42010296	0.37399539	0.43405364	0.52261076
AAD3.5k (ON)	0.18253402	0.22239458	0.18261902	0.16671658
SE sub1(THD)	0.10504212	0.09118816	0.12866462	0.12061050
SE sub2(THD)	17.5142139	20.2693378	18.2987142	18.8782096
SE sub3(THD)	2637.56400	3457.80400	2953.48400	2862.97800
SE sub4(THD)	2883.53500	3256.41600	2885.35000	3125.83500
SE sub5(THD)	5812.10400	5431.23299	5687.52400	5748.75800
SE sub6(THD)	12673.5410	11240.9330	11284.6820	10573.8040
SE sub7(THD)	37727.7510	45671.4200	35960.0570	37577.4250
SE sub8(THD)	156754.492	163495.411	157741.538	150919.433
SE sub9(THD)	419665.423	414645.546	423962.379	421851.356
SE sub10(THD)	885156.860	885495.562	875371.236	881802.489
SF(THD)	1791353.04	1796889.75	1835191.12	1729761.67

Appendix D

Electromotive force

In electrodynamic drivers, a mechanical force is generated by a current flowing in a wire the two being related by a direct-proportionality law. The starting physical point is the Lorentz' force law; if we place a single moving charge with velocity $\vec{u}$ in a magnetic field $\vec{B}$, we can measure a force acting on that electron which depends on the magnetic field strength vector:

$$\vec{F} = q\vec{u} \text{ x } \vec{B}$$

Due to vector product properties, when $\vec{u}$ and $\vec{B}$ are perpendicular that force is maximum, and the modulus is proportional to the charge velocity and to the modulus of the magnetic field strength B (measured in $[T]$). If we now imagine inserting in between the magnet a wire of length l in which a current i flows (consequently there will also be a voltage v at its ends), by denoting with $dV = Sdl$ an infinitesimal portion of volume of such wire, the infinitesimal force acting on each electron can be re-written according to the previous law:

$$d\vec{F} = (ndV)q \ \vec{u} \text{ x } \vec{B}$$

The total force acting on the wire is now obtained by summing continuously each microscopic contribution acting on each electron along the whole length l:

$$\vec{F} = \int_0^l d\vec{F} = \int_0^l (ndV)q \ \vec{u} \text{ x } \vec{B} = \int_0^l nq \ \vec{u} \text{ x } \vec{B} \, Sdl$$

By introducing the definition of current density $J = qn|\vec{u}|\left[\frac{A}{cm^2}\right]$:

$$\vec{F} = \int_0^l J\ \vec{u}_T \text{ x } \overrightarrow{B} S dl = (JS)\vec{u}_T \text{ x } \overrightarrow{B} \int_0^l dl = il\ \vec{u}_T \text{ x } \overrightarrow{B}$$

with $\vec{u}_T$ being the unity vector having the same direction of the line. As its magnitude is unitary, we have:

$$\left|\vec{F}\right| = ilB\, sin\, \alpha$$

Of course, placing the coil orthogonal to $\overrightarrow{B}$ vector in any point of the magnetic field ($\alpha = 90°$) is a convenient choice of design [5]. For a comprehensive discussion moving coil loudspeaker topologies refer to [3], [4]. This leads us to the well-known equation involving the Bl factor. It is also clear to see how the assumption of validity of that simple equation implies the constant proportionality of magnetic field strength with respect to the displacement of the voice coil, the more it goes away from the rest position, the lesser it can be considered true.

Bibliography

[1] R. Magalotti, "Seminar 2019 Loudspeaker Drivers Roberto Magalotti – B & C Speakers," 2019.

[2] T. J. Mellow and L. L. Beranek, *Acoustics: Sound fields and transducers*, vol. 2014-Janua, no. 2. 2014.

[3] V. Dickason and D. R. Raichel, *The Loudspeaker Design Cookbook, 5th Edition*. 2006.

[4] B. Lansing and M. R. Gander, "Moving-Coil Loudspeaker Topology as an indicator of linear excursion capability," *Audio Eng. Soc.*, 1979.

[5] G. Bertuccio, "Electromotive force on loudspeaker, Electronic and electroacoustics for sound engineering, Lectures notes A.A.2018/19." Milano, 2019.

[6] W. Klippel, *Sound Quality of Audio Systems*. Dresden, 2017.

[7] S. Irrgang and W. Klippel, "Fast Measurement of Harmonic Distortion," pp. 335–336, 2008.

[8] V. Gunnarsson, "Assessment of Nonlinearities in Loudspeakers - Volume dependent equalization," vol. d, 2010.

[9] J. D. B. Keele, "Low-Frequency Loudspeaker Assessment by Nearfield Sound-Pressure Measurement," *Engineering*, 1974.

[10] U. Seidel and W. Klippel, "Fast and Accurate Measurement of the Linear Transducer Parameters," *Audio Eng. Soc. Conv. 110*, pp. 1–7, 2001.

[11] A. Dobrucki, "Nonlinear distortions in electroacoustic devices," *Arch. Acoust.*, vol. 36, no. 2, pp. 437–460, 2011.

[12] A. Voishvillo, "Assessment of nonlinearity in transducers and sound systems - From THD to perceptual models," *Audio Eng. Soc. - 121st Conv. Pap. 2006*, vol. 2, pp. 557–583, 2006.

[13] A. Voishvillo, A. Terekhov, E. Czerwinski, and S. Alexandrov, "Graphing, interpretation, and comparison of results of loudspeaker nonlinear distortion measurements," *AES J. Audio Eng. Soc.*, vol. 52, no. 4, pp. 332–357, 2004.

[14] S. T. Glad, "Nonlinear system theory," *Automatica*, vol. 23, no. 4, pp. 545–546, 1987.

[15] W. J. Cunningham, "Non-Linear Distortion in Dynamic Loudspeakers due to Magnetic Effects," *J. Acoust. Soc. Am.*, vol. 21, no. 3, pp. 202–207, 1949.

[16] B. Merit, V. Lemarquand, G. Lemarquand, and A. Dobrucki, "Motor nonlinearities in electrodynamic loudspeakers: Modelling and measurement," *Arch. Acoust.*, vol. 34, no. 4, pp. 579–590, 2009.

[17] W. Klippel, "Loudspeaker Nonlinearities – Causes, Parameters, Symptoms." Dresden, 2006.

[18] C. Cruanes; and F. Monllor;, "Application of the Geddes-Lee model for the subjective evaluation of sound emitted by loudspeakers," *19th Int. Congr. Acoust. MADRID, 2-7 Sept.*

2007, no. September, pp. 2–7, 2007.

[19] H. Scott, "Audible Audio Distortion," 1945.

[20] J. K. Hilliard, "Distortion Tests by the Intermodulation Method," *Proc. IRE*, vol. 29, no. 12, pp. 614–620, 1941.

[21] M. O. J. Hawksford, "System measurement and identification using pseudorandom filtered noise and music sequences," *AES J. Audio Eng. Soc.*, vol. 53, no. 4, p. 275, 2005.

[22] A. Farina, "Silence Sweep : a novel method for measuring electro - acoustical devices Angelo Farina Industrial Engineering Dept . A Silence Sweep test signal is basically the " negative," 2009.

[23] A. W. Rix, J. G. Beerends, M. P. Hollier, and A. P. Hekstra, "Perceptual Evaluation Of Speech Quality (PESQ) - A new method for speech quality assessment of telephone networks and codecs," pp. 2–5, 2001.

[24] T. Thiede *et al.*, "PEAQ - the ITU standard for objective measurement of perceived audio quality," *AES J. Audio Eng. Soc.*, vol. 48, no. 1–2, pp. 3–29, 2000.

[25] P. Brunet, "Nonlinear System Modeling and Identification of Loudspeakers," no. April, pp. 1–152, 2014.

[26] W. Klippel, "Linear Lumped Parameter Measurement," *Hands-On Train. 1*, pp. 1–19, 2013.

[27] W. Klippel, "Vibration and Radiation Behavior of Loudspeaker ' s Membrane," pp. 1–19, 2014.

[28] W. Klippel and J. Schlechter, "Dynamical measurement of the effective radiation area SD," *128th Audio Eng. Soc. Conv. 2010*, vol. 1, no. 1, pp. 568–577, 2010.

[29] W. Klippel, "Loudspeaker Non-Linearities," pp. 81–84, 2004.

[30] W. Klippel, "Predicting the Nonlinear Loudspeaker Behavior," pp. 1–21, 2017.

[31] S. Son, J. Jeon, J. Choi, and M. Pakhomov, "Investigation of influence of electromechanical characteristics of electrodynamic transducers on sound quality perception," in *AES 146th International Convention*, 2019.

[32] F. E. Toole, "Loudspeaker Measurements and Their Relationship to Listener Preferences: Part 2," 1986.

[33] S. E. Olive, P. L. Schuck, J. G. Ryan, S. L. Sally, and M. E. Bonneville, "The detection thresholds of resonances at low frequencies," *AES J. Audio Eng. Soc.*, vol. 45, no. 3, pp. 116–127, 1997.

[34] P. Newell and K. Holland, *Loudspeakers for music Recording and Reproduction*. Amsterdam, Boston, Heidelberg, London, New York, Oxford, Paris, San Diego, San Francisco, Singapore, Sydney, Tokyo: Elsevier Ltd, 2007.

[35] K. Gundry and S. E. Olive, "Differences in Performance and Preference of Trained versus Untrained Listeners in Loudspeaker Tests: A Case Study," *AES J. Audio Eng. Soc.*, vol. 52, no. 3, pp. 261–262, 2003.

[36] A. Gabrielsson, "Perceived Sound Quality of High-Fidelity Loudspeakers," *Audio Eng. Soc.*,

vol. 33, 1985.

[37] A. Gabrielsson, B. Lindström, and O. Till, "Loudspeaker frequency response and perceived sound quality," *J. Acoust. Soc. Am.*, vol. 90, no. 2, pp. 707–719, 1991.

[38] L. Lee and Earl R. Geddes, "Audibility of Linear Distortion with Variations in Sound Pressure Level and Group Delay," *Audio Eng. Soc.*, pp. 1–11, 2006.

[39] S. E. Olive and F. E. Toole, "Detection of reflections in typical rooms," *AES J. Audio Eng. Soc.*, vol. 37, no. 7–8, pp. 539–553, 1989.

[40] B. C. J. Moore and C.-T. Tan, "Perceived naturalness of spectrally distorted speech and music," *J. Acoust. Soc. Am.*, vol. 114, no. 1, pp. 408–419, 2003.

[41] E. Hugo Fastl and E. Zwicker, *Psychoacoustics - Facts and models*. New York: Springer, 2006.

[42] L. D.Fielder and M. J.Smithers, "Perceptual Assessment of Distortion In Low-Frequency Loudspeakers," *Audio Eng. Soc. Conv. Pap. 10242*, pp. 1–20, 2019.

[43] Z. Nick, *Sensory Evaluation of Sound*. Taylor & Francis Group, LLC, 2019.

[44] S. Henrik, "Correlation Between Subjective and Objective Data for Quality Loudspeaekrs," *J. Audio Eng. Soc.*, no. January, 1974.

[45] H. Staffeldt, "Measurement and Prediction of the Timbre of Sound Reproduction.," *AES J. Audio Eng. Soc.*, vol. 32, no. 6, pp. 410–414, 1984.

[46] J. M. Grey, "Multidimensional perceptual scaling of musical timbres," *J. Acoust. Soc. Am.*, vol. 61, no. 5, pp. 1270–1277, 1977.

[47] U. Rosenberg, S. East, and N. York, "Loudspeaker Measurement and consumer information," *Audio Eng. Soc. Conv.*, vol. 4, 1973.

[48] F. E. Toole, "Loudspeaker Measurements and Their Relationship To Listener Preferences: Part 1," *AES J. Audio Eng. Soc.*, no. 5, pp. 323–348, 1986.

[49] W. Klippel, "Assessing the Subjectively Perceived Loudspeaker Quality on the Basis of Objective Parameters," *88th AES Conv.*, 1990.

[50] S. E. Olive, "A Multiple Regression Model for Predicting Loudspeaker Preference Using Objective Measurements: Part I - Listening Test Results," *Proc. 116th Conv. Audio Eng. Soc.*, 2004.

[51] S. E. Olive, "A Multiple Regression Model for Predicting Measurements: Part II - Development of the Loudspeaker Preference Using Objective Model," in *Audio Engineering Society 117th Convention*, 2004.

[52] J. G. Beerends, K. Van Nieuwenhuizen, and E. L. Van Den Broek, "Quantifying sound quality in loudspeaker reproduction," *AES J. Audio Eng. Soc.*, vol. 64, no. 10, pp. 784–799, Oct. 2016.

[53] R. A. Belcher, "New Distortion Measurement.," *Wirel. World*, vol. 84, no. 1509, pp. 36–41, 1978.

[54] C. T. Tan, B. C. J. Moore, N. Zacharov, and V. V. Mattila, "Predicting the perceived quality of nonlinearly distorted music and speech signals," *AES J. Audio Eng. Soc.*, vol. 52, no. 7–8, pp. 699–711, 2004.

[55] S. Bech and N. Zacharov, *Perceptual Audio Evaluation-Theory, Method and Application*. 2006.

[56] S. Zielinski, F. Rumsey, and S. Bech, "On some biases encountered in modern listening tests," *J. Audio Eng. Soc.*, vol. 56, no. 6, pp. 427–451, 2008.

[57] I.-R. BS.1534-1, "Method for the subjective assessment of intermediate quality level of coding systems," *International Telecommunication Union*. pp. 1–18, 2003.

[58] I.-R. BS.1116-1, "Methods for the subjective assessment of small impairments in audio systems including multichannel sound systems," *Int. Telecommun. Union, Geneva*, pp. 1–26, 1997.

[59] ITU-R BS.1283, "Subjective assessment of sound quality – A guide to existing Recommendations," pp. 1–6, 2003.

[60] S. Moulin, S. Bech, and T. Stegenborg-Andersen, "Sensory profiling of high-end loudspeakers using rapid methods - Part 1: Baseline experiment using headphone reproduction," in *Proceedings of the AES International Conference*, 2016, vol. 2016-Augus.

[61] S. R. Jaeger *et al.*, "Check-all-that-apply (CATA) questions for sensory product characterization by consumers: Investigations into the number of terms used in CATA questions," *Food Qual. Prefer.*, vol. 42, pp. 154–164, 2015.

[62] L. Hicks, S. Moulin, and S. Bech, "Sensory Profiling of High-End Loudspeakers Using Rapid Methods—Part 3: Check-All-That-Apply with Naıve Assessors," *AES 146th Int. Conv.*, vol. 66, no. 5, pp. 329–342, 2018.

[63] I. A. Sagredo, S. Moulin, and S. Bech, "Sensory profiling of high-end loudspeakers using rapid methods - Part 4: Flash profile with expert assessors," *AES 146th Int. Conv.*, pp. 1–11, 2019.

[64] G. Ares *et al.*, "Evaluation of a rating-based variant of check-all-that-apply questions: Rate-all-that-apply (RATA)," *Food Qual. Prefer.*, vol. 36, pp. 87–95, 2014.

[65] M. Meyners, S. R. Jaeger, and G. Ares, "On the analysis of Rate-All-That-Apply (RATA) data," *Food Qual. Prefer.*, vol. 49, pp. 1–10, 2016.

[66] I.-R. P.800, "Telephony transmission quality, Methods for subjective determination of transmission quality," *Int. Telecommun. Union*, vol. 800, p. 22, 1996.

[67] S. E. Olive, P. L. Schuck, S. L. Sally, and M. E. Bonneville, "Effects of loudspeaker placement on listener preference ratings," *AES J. Audio Eng. Soc.*, vol. 42, no. 9, pp. 651–669, 1994.

[68] T. H. Pedersen and N. Zacharov, "The development of a sound wheel for reproduced sound," *138th Audio Eng. Soc. Conv. 2015*, vol. 2, pp. 765–777, 2015.

[69] F. E. Toole, *Sound reproduction - Loudspeakers and Rooms*. Elsevier Ltd, 2008.

[70] P. J. Chapman, "Programme material analysis," *Audio Eng. Soc.*, vol. 4277, pp. 1–20, 1996.

[71] Head Acoustics, "Conducting Listening Tests. HEAD acoustics Application Note Conducting Listening Tests." pp. 1–17, 2011.

[72] "Blocking Factor - Statistics How To." [Online]. Available: https://www.statisticshowto.com/blocking-factor/. [Accessed: 21-Apr-2020].

[73] "Latin Squares - Williams Design." [Online]. Available: https://statpages.info/latinsq.html. [Accessed: 21-Apr-2020].

[74] C. J. Struck and S. F. Temme, "Simulated free field measurements," *AES J. Audio Eng. Soc.*, vol. 42, no. 6, pp. 467–482, 1994.

[75] "Klippel dB-Lab Analyzer Software." [Online]. Available: https://www.klippel.de/dm/?page=details&pid=1804. [Accessed: 18-Apr-2020].

[76] J. Begin, "Working In Tandem: Combining Near And Far Field Measurements - ProSoundWeb," 2019. [Online]. Available: https://www.prosoundweb.com/working-in-tandem-combining-near-and-far-field-measurements/. [Accessed: 19-Apr-2020].

[77] W. Klippel, "Merging Near and Far Field Measurements," *Appl. Note to KLIPPEL R&D Syst.*, 2014.

[78] Mathworks, *MATLAB version 9.6.0,1047502 (R2019a)*. Natick, Massachusetts: The MathWorks Inc., 2019.

[79] P. Jumppanen, "Lumped Parameter Analysis of a Dynamic Loudspeaker," pp. 1–17, 2013.

[80] A. Farina, "Simultaneous Measurement of Impulse Response and Distortion With a Swept-Sine Technique," *Proc. AES 108th conv, Paris, Fr.*, vol. 5093, no. I, pp. 1–15, 2000.

[81] G. Peeters, "A large set of audio features for sound description (similarity and classification) in the CUIDADO project," no. 0, pp. 1–25, 2004.

[82] D.-N. Jiang, L. Lu, H.-J. Zhang, J.-H. Tao, and L.-H. Cai, "Music type classification by Spectral Contrast Feature," *Microsoft Res.*, pp. 0–3, 2002.

[83] P. Newell and K. Holland, *Loudspeakers for Music Recording and Reproduction*, 2nd ed. New York: Audio Engineering Society, 2019.

[84] E. Geddes and L. Lee, "Auditory Perception of Nonlinear Distortion," *Audit. Percept. Nonlinear Distortion*, 2003.

[85] E. Geddes and L. Lee, "Auditory Perception of Nonlinear Distortion - Theory," in *Audio Engineering Society*, 2002.

[86] Ronald M. Aarts, "A Comparison of Some Loudness Measures for Loudspeaker Listening Tests*," 1992.

[87] "REW - Room EQ Wizard Room Acoustics Software." [Online]. Available: https://www.roomeqwizard.com/. [Accessed: 21-Apr-2020].

[88] "REAPER | Audio Production Without Limits," *Cockos, Inc.* [Online]. Available: https://www.reaper.fm/. [Accessed: 22-Apr-2020].

[89] "SWS / S&M Extension," *Cockos, Inc.* [Online]. Available: http://www.sws-extension.org/. [Accessed: 22-Apr-2020].

[90] "Introduction to OSC | opensoundcontrol.org." [Online]. Available: http://opensoundcontrol.org/introduction-osc. [Accessed: 22-Apr-2020].

[91] Mark, "Send Open Sound Control (OSC) Messages - version 1.1.0.0," *File Exchange - MATLAB Central*, 2011. [Online]. Available: https://it.mathworks.com/matlabcentral/fileexchange/31400-send-open-sound-control-osc-messages. [Accessed: 23-Apr-2020].

[92] "Audio Toolbox Documentation," *MathWorks Italia*, 2020. [Online]. Available: https://it.mathworks.com/help/audio/index.html?s_tid=CRUX_lftnav. [Accessed: 23-Apr-2020].

[93] A. Mirhashemi, "Introducing spectral moment features in analyzing the SpecTex hyperspectral texture database," *Mach. Vis. Appl.*, vol. 29, no. 3, pp. 415–432, 2018.

[94] F. Antonacci, A. Canclini, and A. Sarti, "Sound Analysis Lecture Notes (Draft)," 2013.

[95] T. Giannakopoulos and A. Pikrakis, "Introduction to Audio Analysis: A MATLAB® Approach - Theodoros Giannakopoulos, Aggelos Pikrakis - Google Libri," *Academic Press*, 2014. [Online]. Available: https://books.google.it/books?id=zbHVAQAAQBAJ&pg=PA98&lpg=PA98&dq=MPEG7+spectral+envelope+matlab&source=bl&ots=a_UYO5iU66&sig=ACfU3U1U7FXToDMEm6pcwk-jmmGsKLgWww&hl=it&sa=X&ved=2ahUKEwj8mPKfzOrnAhUnSxUIHSdfBfEQ6AEwAnoECAoQAQ#v=onepage&q&f=false. [Accessed: 21-Apr-2020].

[96] O. Lartillot, "MIRtoolbox 1.7 User' s Manual," 2017.

[97] J. Kirchner, "Data Analysis Toolkit #10: Simple linear regression," *Univ. California, Berkely*, no. 6, p. 3, 2001.

[98] T. Naes, O. Tomic, and P. B. Brockhoff, *Statistics for Sensory and Consumer Science*. 2010.

[99] T. Naes and E. Risvik, *Multivariate analysis of data in sensory science*. 1996, pp. 307–344.

[100] P. H. Garthwaite, "An interpretation of partial least squares," *J. Am. Stat. Assoc.*, vol. 89, no. 425, pp. 122–127, 1994.

[101] A. Kondylis, "PLS methods in regression Model assessment and inference," Universitè de Neuchatel, 2006.

[102] ITU-R BS.1116-3, "Methods for the subjective assessment of small impairments in audio systems," *Int. Telecommun. Union, Geneva*, vol. 3, 2015.

[103] "TotalMix FX - Mixing/Routing with superior features for Studio and Live Work - RME Audio Interfaces | Format Converters | Preamps | Network Audio & MADI Solutions." [Online]. Available: https://www.rme-audio.de/totalmix-fx.html. [Accessed: 29-Apr-2020].

[104] "KA3 - Klippel Analyzer 3." [Online]. Available: https://www.klippel.de/products/rd-system/analyzer-hardware/ka3-klippel-analyzer-3.html. [Accessed: 29-Apr-2020].

[105] "Machine Learning and Deep Learning for Audio - MATLAB & Simulink - MathWorks Italia." [Online]. Available: https://it.mathworks.com/help/audio/feature-extraction-and-deep-learning.html. [Accessed: 22-Apr-2020].

[106] E. Zechmann, "Nth Octave Frequency Bands - File Exchange, Version 1.3.0.1," *Mathworks File Exchange*. [Online]. Available: https://it.mathworks.com/matlabcentral/fileexchange/19033-nth-octave-frequency-bands?focused=3842661&tab=function. [Accessed: 22-Apr-2020].

[107] "Multiple-sample tests for equal variances - MATLAB vartestn," *MathWorks Italia*. [Online]. Available: https://it.mathworks.com/help/stats/vartestn.html#btuo7w_. [Accessed: 23-Apr-

2020].

[108] Ipek, "Normality test package - version 1.2.0.0," *MATLAB Central File Exchange*. [Online]. Available: https://it.mathworks.com/matlabcentral/fileexchange/60147-normality-test-package. [Accessed: 23-Apr-2020].

[109] R. Bedre, "ANOVA using Python." [Online]. Available: https://reneshbedre.github.io/blog/anova.html. [Accessed: 25-Apr-2020].

[110] "Multiple comparison test - MATLAB multcompare," *MathWorks Italia*. [Online]. Available: https://it.mathworks.com/help/stats/multcompare.html. [Accessed: 23-Apr-2020].

[111] "Statistics and Machine Learning Toolbox Documentation," *MathWorks Italia*. [Online]. Available: https://it.mathworks.com/help/stats/index.html?s_tid=CRUX_lftnav. [Accessed: 03-May-2020].

[112] "Partial least-squares regression - MATLAB plsregress," *MathWorks Italia*. [Online]. Available: https://it.mathworks.com/help/stats/plsregress.html. [Accessed: 06-May-2020].

[113] N. Zacharov, "Assessor selection process for multisensory applications," *Conference Paper*, no. December, Munich, p. 4, 2009.

[114] J. D'Appolito, "Measuring Loudspeaker Low-Frequency Response," *audioXpress*, 2018.

[115] EBU R 128, "Loudness Normalization and permitted maximum level of audio signals," *EBU*, no. June, pp. 1–5, 2014.

[116] ITU-R BS.1770-4, "Algorithms to measure audio programme loudness and true-peak audio level," *Radiocommun. Sect. ITU*, vol. 4, 2015.

[117] Bob Katz, "An Integrated Approach to Metering, Monitoring, and Levelling Practices," vol. 65, no. 8, pp. 1141–1146, 1999.

[118] Chester.W.Rice, "LOUD SPEAKER," 1929.

[119] "Home | Inkscape 0.92." [Online]. Available: https://inkscape.org/it/. [Accessed: 23-Apr-2020].